Digital Contact Controller

ATA2508 User's Manual

Version 1.4

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1. General Description

1.1 Introduction

@Lab's second generation touch sensor IC, ATA2508, provides lots of new features. First of all it features programmability to increase flexibility thus gives better performance and stability to the specific applications and more opportunity to broader application areas. Its AIC™ (Automatic Impedance Calibration) feature may be easily configured to support different sensitivity between channels, or change parameters such as a calibration interval. AIC™ may be temporarily paused and resumed by a host. It provides a strong new feature APIS™ (Adjacent Pattern Interference Suppression) to eliminate adjacent key or pattern interference. It also supports touch-strength outputs in addition to touch outputs. Furthermore, twelve general purpose DIOs may be configured and programmed to meet customer's specific requirements which gives customers greater flexibility and values.

1.2 Feature

- I^2C for the host interface
- Twelve channels (40QFN) or nine channels (32QFN,24SSOP) available
- Configurable AIC™ (Automatic Impedance Calibration)
- Two kinds of interrupts: GINT for general purpose, TINT for touch detection
- De-bounced touch outputs
- Eight bit resolution of touch strength data (256 step)
- APIS™: Mode1, Mode2, Mode3 (Adjacent Pattern Interference Suppression)
- Configurable twelve DIO pins as direct touch outputs, extended GPIOs, or external interrupt inputs.
- Beep generation for tactile feeling
- Idle and Sleep modes for power saving

1.3 Package Types (Lead free packages) and Outline

Package Type	Product Code	Package Dimension	Pin Pitch	No. of Sensor Input	Digital Output
40QFN	ATA2508DA-40N	5mm X 5mm X 0.85mm	0.4mm	12ea	12ea
32QFN	ATA2508DA-32N	4mm X 4mm X 0.85mm	0.4mm	9ea	8ea
24QFN	ATA2508DC-24N	4mm X 4mm X 0.85mm	0.5mm	6ea	3ea
30SSOP	ATA2508DA-30S	12.74mm X 10.3mm X 2.5mm	0.8mm	12ea	6ea
24SSOP	ATA2508DA-24S	8.2mm X 7.8mm X 2.0mm	0.65mm	9ea	3ea
20SSOP	ATA2508DA-20S	6.5mm X 6.4mm X 1.85mm	0.65mm	6ea	2ea

1.4 Electrical Characteristics

Parameter	Symbol	Unit	MIN	TYP	MAX	Note
Absolute Maximum Ratings						
Storage Temperature	Tstg	℃	-45		95	
Operating Temperature	Topr	℃	-40		90	
Operating Humidity	Hopr	%	5		95	
Power Supply Voltage	VPH	V	2.3	3.3	5.5	VPH should be higher than 3V when using internal LDO
Power Supply Voltage(V25)	V25	V	2.3	2.5	2.7	
Input Voltage	Vin	V			VPH+0.3	
ESD (HBM)	HBM	V			8000	Sensor Input
Recommended Operating Conditions						
Power Supply Voltage(VPH)	Vddp	V	2.5		5	VPH should be higher than 3V when using internal LDO
Power Supply Voltage(V25)	Vddc	V	2.4	2.5	2.6	
Digital Input Rising Time	Tr_i	ns	-	-	5	
Digital Input Falling Time	Tf_i	ns	-	-	5	
AC Electrical Specifications (Typical values at Ta=25℃ and VPH=3.3V)						
Input frequency	fi	KHz	2.5	5	20	
Sample frequency	fsmp	Hz	10	500	5000	
Touch Sensitivity	Stch	pF		0.06		
Output Rising Time	Tr_o	ns	-	50	60	Load = 100pF
Output Falling Time	Tf_o	ns	-	50	60	Load = 100pF
DC Electrical Specifications (Typical values at Ta=25℃ and VPH=3.3V , using external 2.5V LDO)						
Supply Current (Active mode)	Idd_o	μA		98		
Supply Current (Idle mode)	Idd_i	μA		60		
Supply Current (Sleep mode)	Idd_s	μA		0.1		
Digital Input Low Voltage	Vil	V			0.7	
Digital Input High Voltage	Vih	V	0.8xVPH			
Digital Output Low Voltage	Vol	V			0.6	
Digital Output High Voltage	Voh	V	VPH-0.5			
GPIO Driving Current	Idr	mA			2	

1.5 Functional Characteristics

Clock Frequency	Active to Idle	Idle to Active	Active to Sleep	Idle to Sleep	Sleep to Active
System Clock 1.6MHz Sensor Clock 20KHz	0.25 X **A** sec	Min: 2ns Max: 10ms	1ns	1ns	10us
System Clock 1.6MHz Sensor Clock 10KHz	0.5 X **A** sec	Min: 2ns Max: 20ms	1ns	1ns	10us
System Clock 800KHz Sensor Clock 10KHz	0.5 X **A** sec	Min: 2ns Max: 20ms	1ns	1ns	10us
System Clock 800KHz Sensor Clock 5KHz	1 X **A** sec	Min: 2ns Max: 40ms	1ns	1ns	10us
System Clock 400KHz Sensor Clock 5KHz	1 X **A** sec	Min: 2ns Max: 40ms	1ns	1ns	10us
System Clock 400KHz Sensor Clock 2.5KHz	2 X **A** sec	Min: 2ns Max: 80ms	1ns	1ns	10us
System Clock 200KHz Sensor Clock 2.5KHz	2 X **A** sec	Min: 2ns Max: 80ms	1ns	1ns	10us
System Clock 200KHz Sensor Clock 1.25KHz	4 X **A** sec	Min: 2ns Max: 160ms	1ns	1ns	10us

A: IDLE TIME Register Value

1.6 Communication Specification for I^2C

Table. 1. Electrical DC Specification for I^2C bus

PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
LOW level input voltage: Fixed input levels V_{DD}-related input levels	V_{IL}	 -0.5 -0.5	 1.5 0.3VDD	 n/a -0.5	 n/a 0.3 V_{DD} (1)	 V V
HIGH level input voltage: Fixed input levels V_{DD}-related input levels	V_{IH}	 3.0 0.7 V_{DD}	 (2) (2)	 n/a 0.7 V_{DD}	 n/a (2)	 V V
Hysteresis of Schmitt trigger inputs: V_{DD} > 2V V_{DD} < 2V	V_{hys}	 n/a n/a	 n/a n/a	 0.05 V_{DD} 0.1 V_{DD}	 - -	 V V
LOW level output voltage(open drain or collector) at 3mA sink current: V_{DD} > 2V V_{DD} < 2V	 V_{OL1} V_{OL3}	 0 n/a	 0.4 n/a	 0 0	 0.4 0.2 V_{DD}	 V V
Output fall time from V_{IHmin} to V_{ILmax} with a bus capacitance from 10pF to 400pF	t_{of}	-	250(4)	20+0.1Cb(3)		ns
Pulse width of spike which must be Suppressed by the input filter	t_{sp}	n/a	n/a	0	50	ns
Input current each I/O pin with an input Voltage between 0.1VDD and 0.9V V_{DDmax}	I_i	-10	10	-10(5)	10(5)	uA
Capacitance for each I/O Pin	C_i	-	10	-	10	pF

Note

1. Devices that use non-standard supply voltages which do not conform to the intended I^2C bus system levels must relate their input levels to the V_{DD} voltage to which the pull-up resistors R_p are connected.
2. Maximum VIH = V_{DDmax} + 0.5V.
3. C_b = capacitance of one bus line in pF.
4. The maximum t_f for the SDA and SCL bus lines quoted in table 2(300ns) is longer than the specified maximum t_{of} for the output stages (250ns). This allows series protection resistors (R_s) to be connected between the SDA/SCL Pins and the SDA/SCL bus lines as shown in Fig.36 without exceeding the maximum specified t_f.
5. I/O pins of Fast-mode devices must not obstruct the SDA and SCL lines if V_{DD} is switched off.

n/a = not applicable

Table. 2. Electrical AC Specification for I^2C bus

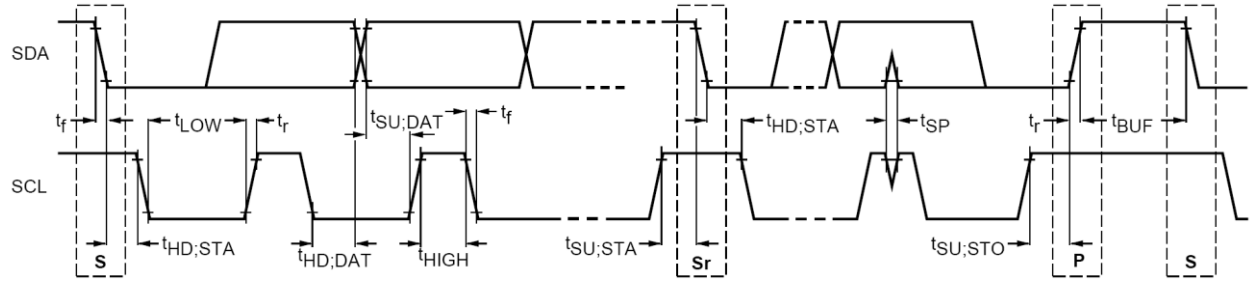
PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
SCL clock frequency	f_{SCL}	0	100	0	400	KHz
Hold time (repeated) START condition. After this period, the first clock pulse is generated	$t_{HD;STA}$	4.0	-	0.6	-	us
LOW period of the SCL clock	t_{LOW}	4.7	-	1.3	-	us
HIGH period of the SCL clock	t_{HIGH}	4.0	-	0.6	-	us
Set-up time for a repeated START condition	$t_{SU;STA}$	4.7	-	0.6		us
Data hold time: For CBUS compatible master For I^2C bus devices	$T_{HD;DAT}$	5.0 $2^{(2)}$	- $3.45^{(3)}$	- $0^{(2)}$	- $0.9^{(3)}$	us us
Data set-up time	$t_{SU;DAT}$	250	-	100(4)	-	ns
Rise time of both SDA and SCL signals	t_r	-	1000	$20+0.1C_b$(5)	300	ns
Fall time of both SDA and SCL signals	t_f	-	300	$20+0.1C_b$(5)	300	ns
Set-up time for STOP condition	$t_{SU;STO}$	4.0	-	0.6	-	us
Bus free time between a STOP and START condition	f_{BUF}	4.7	-	1.3	-	us
Capacitive load for each bus line	C_b	-	400	-	400	pF
Noise margin at the low level for each connected device(including Hysteresis)	V_{nL}	$0.1V_{DD}$	-	$0.1\ V_{DD}$	-	V
Noise margin at the HIGH level for each connected device(including Hysteresis)	V_{nH}	$0.2\ V_{DD}$	-	$0.2\ V_{DD}$	-	V

Notes

1. All values referred to V_{IHmin} and V_{ILmax} levels (See Table 1)
2. A device must internally provide a hold time of at least 300ns for the SDA signal (referred to the V_{IHmin} of the SCL signal) to bridge the undefined region of the falling edge of SCL
3. The maximum $t_{HD;DAT}$ has only to be met if the device dose not stretch the Low period(t_{LOW})of the SCL signal.
4. A Fast-mode I^2C-bus device can be used in a Standard-mode I^2C-bus system, but the requirement $t_{SU;DAT} \geq 250$ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line $t_{max} + t_{SU;DAT} = 1000 + 250 = 1250$ns(according to the Standard-mode I^2C bus specification) before the SCL line is released.
5. C_b = total capacitance of one bus line in pF. If mixed with Hs-mode devices, faster fall-times according to Table are allowed.

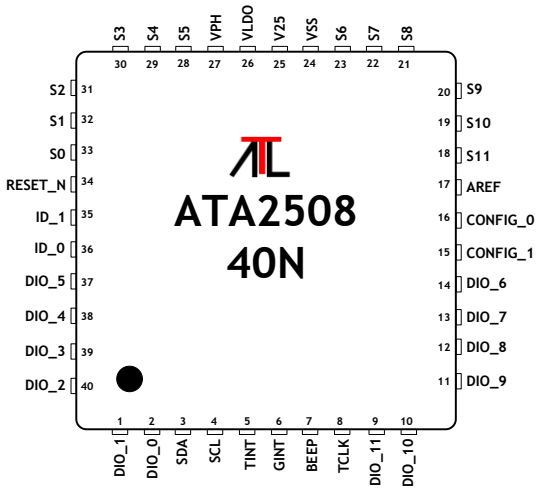
n/a = not applicable

Fig. 1. Definition of timing for F/S-mode devices on the I^2C-bus

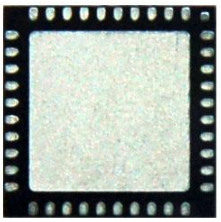


1.7 Top View and Pin Descriptions

1.7.1 40QFN Pin descriptions

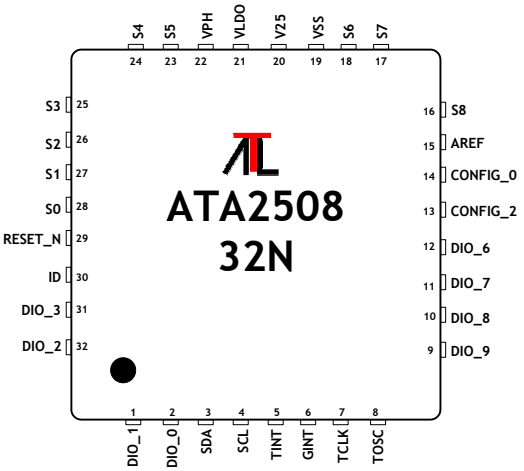


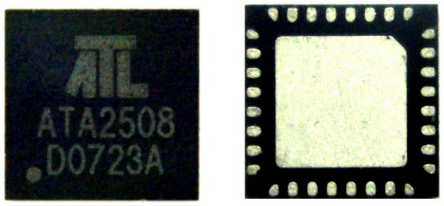




Name	IO	Pin #	Description
RESET_N	I	34	Reset, active LOW
TCLK	I	8	Test Clock Input
S	I	18~23 28~33	Twelve Sensor Inputs from external Touch Pads.
AREF	I	17	Reference Input.
DIO	IO	1,2,9~14 38~40	Configured by HOST as below: - extended GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	3	Bidirectional I^2C Data from/to Host
SCL	I	4	I^2C CLK from Host
TINT	O	5	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	6	General Interrupts including touch interrupt, and they can be masked.
BEEP	O	7	Beep Output.
ID	I	35,36	I^2C Chip ID Select(00:0x58, 01:0x59, 10:0x5A, 11:0x5B)
CONFIG	I	15,16	MCU Control Mode or Fixed Register Mode (00:MCU Control Mode, 01: Fixed 1 Mode, 10: Fixed 2 Mode, 11:Fixed 3 Mode)
VPH	P	27	Power (2.5V ~ 5.5V)
VLDO	O	26	2.5V Regulator Power Output
V25	P	25	2.5V Power Input
VSS	P	24	Ground

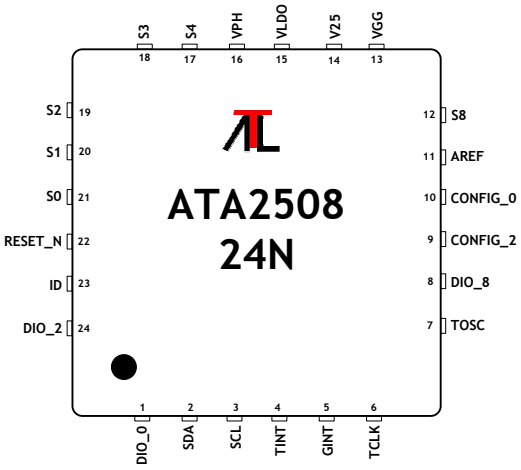
1.7.2 32QFN Pin descriptions



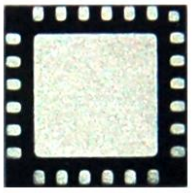


Name	IO	Pin #	Description
RESET_N	I	29	Reset, active LOW
TCLK	I	7	Test Clock Input
S	I	16~18 23~28	Nine Sensor Inputs from external Touch Pads.
AREF	I	15	Reference Input.
DIO	IO	1,2,9~12 31,32	Configured by HOST as below: - extended GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	3	Bidirectional I^2C Data from/to Host
SCL	I	4	I^2C CLK from Host
TINT	O	5	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	6	General Interrupts including touch interrupt, and they can be masked.
TOSC	I	8	The port for test
ID	I	30	I^2C Chip ID Select(0:0x58, 1:0x5B)
CONFIG	I	13,14	MCU Control Mode or Fixed Register Mode (00:MCU Control Mode, 01: Fixed 1 Mode, 10: Fixed 2 Mode, 11:Fixed 3 Mode)
VPH	P	22	Power (2.5V ~ 5.5V)
VLDO	O	21	2.5V Regulator Power Output
V25	P	20	2.5V Power Input
VSS	P	19	Ground

1.7.3 24QFN Pin descriptions

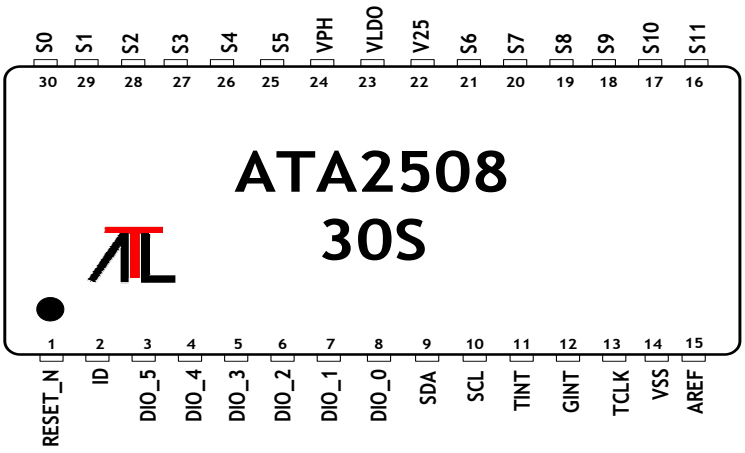






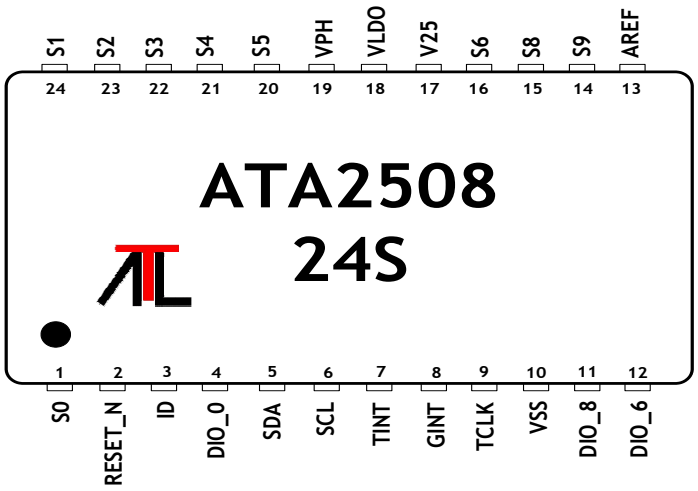
Name	IO	Pin #	Description
RESET_N	I	22	Reset, active LOW
TCLK	I	6	Test Clock Input
S	I	12 17~21	Six Sensor Inputs from external Touch PAD.
AREF	I	11	Sensor Reference Input.
GPIO	IO	1, 8, 24	Configured by HOST as Input of Output - GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	2	Bidirectional I^2C Data from/to Host
SCL	I	3	I^2C CLK from Host
TINT	O	4	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	5	General Interrupts including touch interrupt, also they can be masked.
ID	I	23	I2C Chip ID Select(0:0x58 1:0x5B)
CONFIG	I	9, 10	MCU Control Mode or Fixed Register Mode (00:MCU Control Mode, 01: Fixed 1 Mode, 10: Fixed 2 Mode, 11:Fixed 3 Mode)
VPH	P	16	Power (2.5V ~ 5.5V)
VLDO	O	15	2.5V Regulator Power Output
V25	P	14	2.5V Power Input
VSS	P	13	Ground

1.7.3 30SSOP Pin descriptions



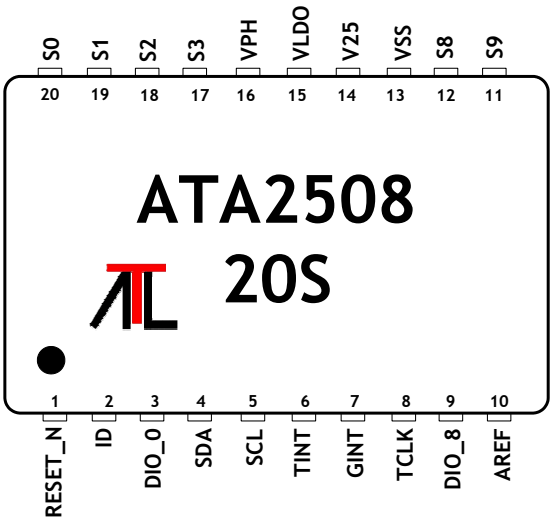
Name	IO	Pin #	Description
RESET_N	I	1	Reset, active LOW
TCLK	I	13	Test Clock Input
S	I	16~21 25~30	Nine Sensor Inputs from external Touch Pads.
AREF	I	15	Reference Input.
DIO	IO	3~8	Configured by HOST as below: - extended GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	9	Bidirectional I^2C Data from/to Host
SCL	I	10	I^2C CLK from Host
TINT	O	11	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	12	General Interrupts including touch interrupt, and they can be masked.
ID	I	2	I^2C Chip ID Select(0:0x58, 1:0x5B)
VPH	P	24	Power (2.5V ~ 5.5V)
VLDO	O	23	2.5V Regulator Power Output
V25	P	22	2.5V Power Input
VSS	P	14	Ground

1.7.4 24SSOP Pin descriptions



Name	IO	Pin #	Description
RESET_N	I	2	Reset, active LOW
TCLK	I	9	Test Clock Input
S	I	1,14~16,20~24	Nine Sensor Inputs from external Touch Pads.
AREF	I	13	Reference Input.
DIO	IO	4,11,12	Configured by HOST as below: - extended GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	5	Bidirectional I^2C Data from/to Host
SCL	I	6	I^2C CLK from Host
TINT	O	7	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	8	General Interrupts including touch interrupt, and they can be masked.
ID	I	3	I^2C Chip ID Select(0:0x58, 1:0x5B)
VPH	P	19	Power (2.5V ~ 5.5V)
VLDO	O	18	2.5V Regulator Power Output
V25	P	17	2.5V Power Input
VSS	P	10	Ground

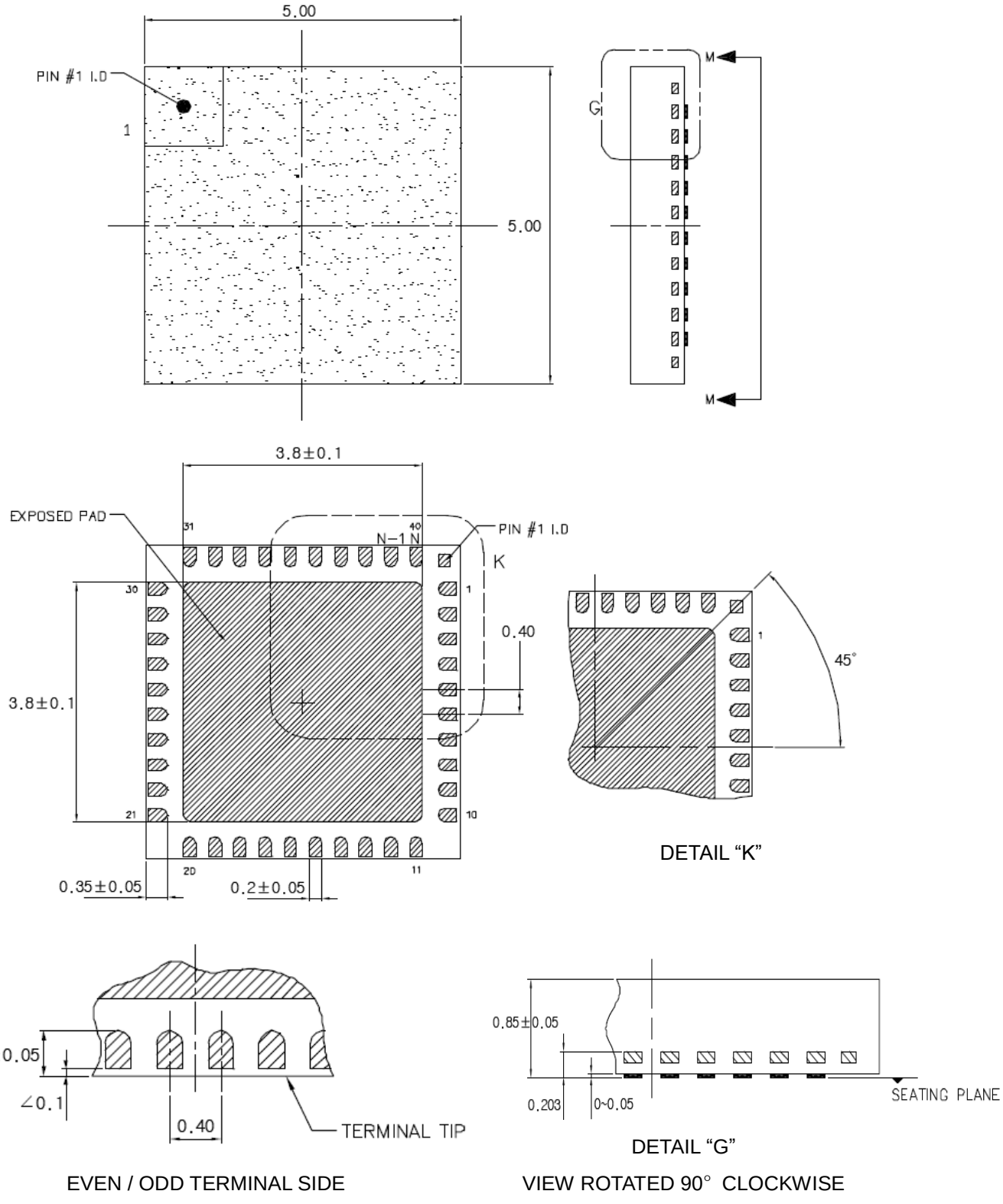
1.7.5 20SSOP Pin descriptions



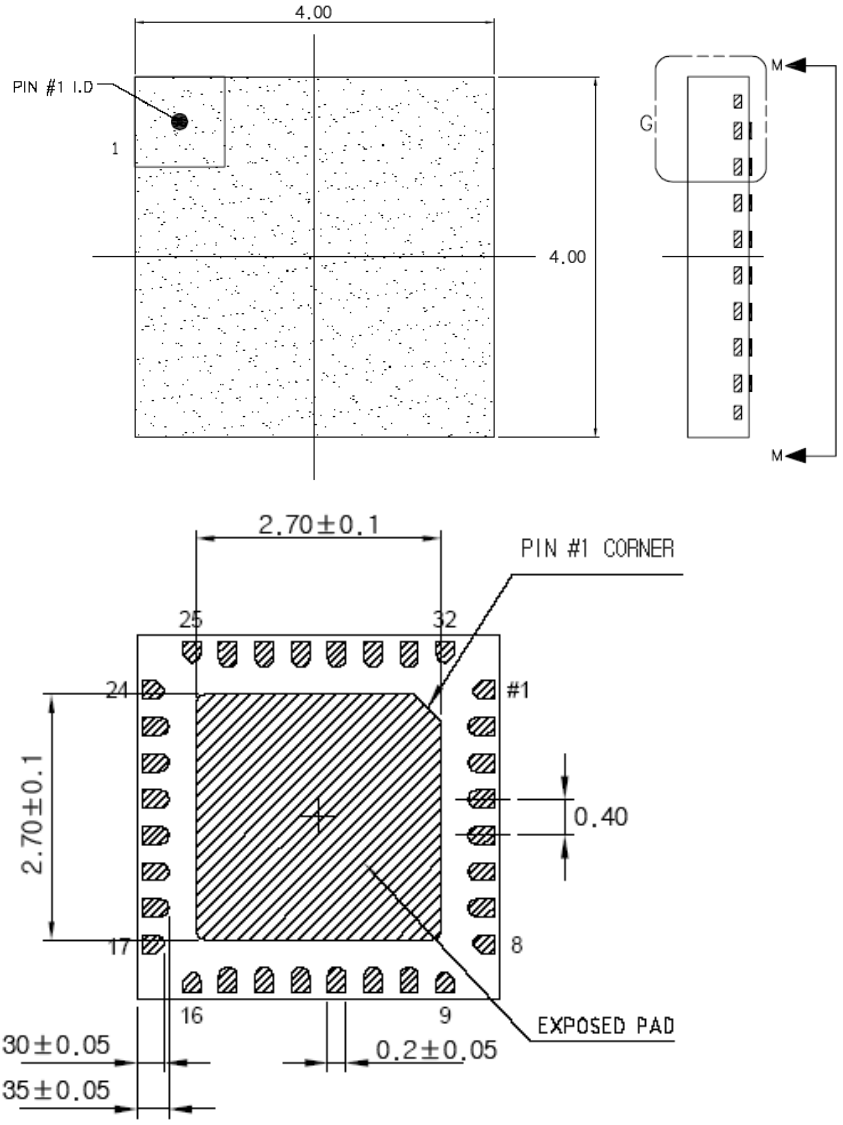
Name	IO	Pin #	Description
RESET_N	I	1	Reset, active LOW
TCLK	I	8	Test Clock Input
S	I	11~12,17~20	Six Sensor Inputs from external Touch Pads.
AREF	I	10	Reference Input.
DIO	IO	3,9	Configured by HOST as below: - extended GPIOs, Direct Button Outputs or External Interrupt inputs
SDA	IO	4	Bidirectional I^2C Data from/to Host
SCL	I	5	I^2C CLK from Host
TINT	O	6	Touch Interrupt, it can be generated when touch status is changed.
GINT	O	7	General Interrupts including touch interrupt, and they can be masked.
ID	I	2	I^2C Chip ID Select(0:0x58, 1:0x5B)
VPH	P	16	Power (2.5V ~ 5.5V)
VLDO	O	15	2.5V Regulator Power Output
V25	P	14	2.5V Power Input
VSS	P	13	Ground

1.7.6 Package Dimension

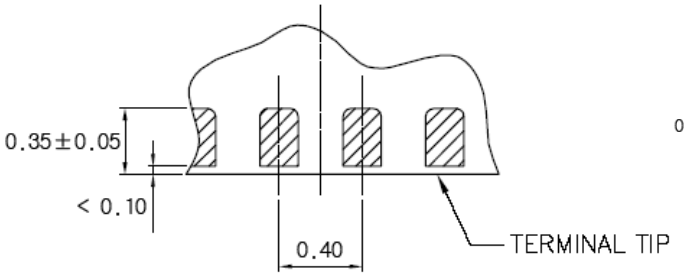
40QFN (unit: mm)



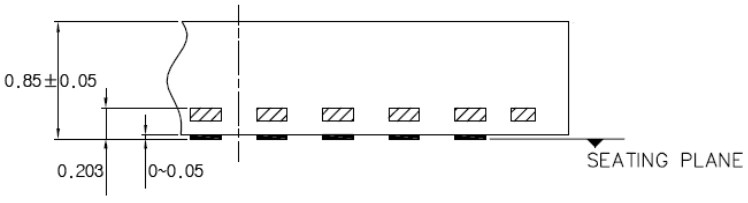
32QFN (unit: mm)



VIEW M-M



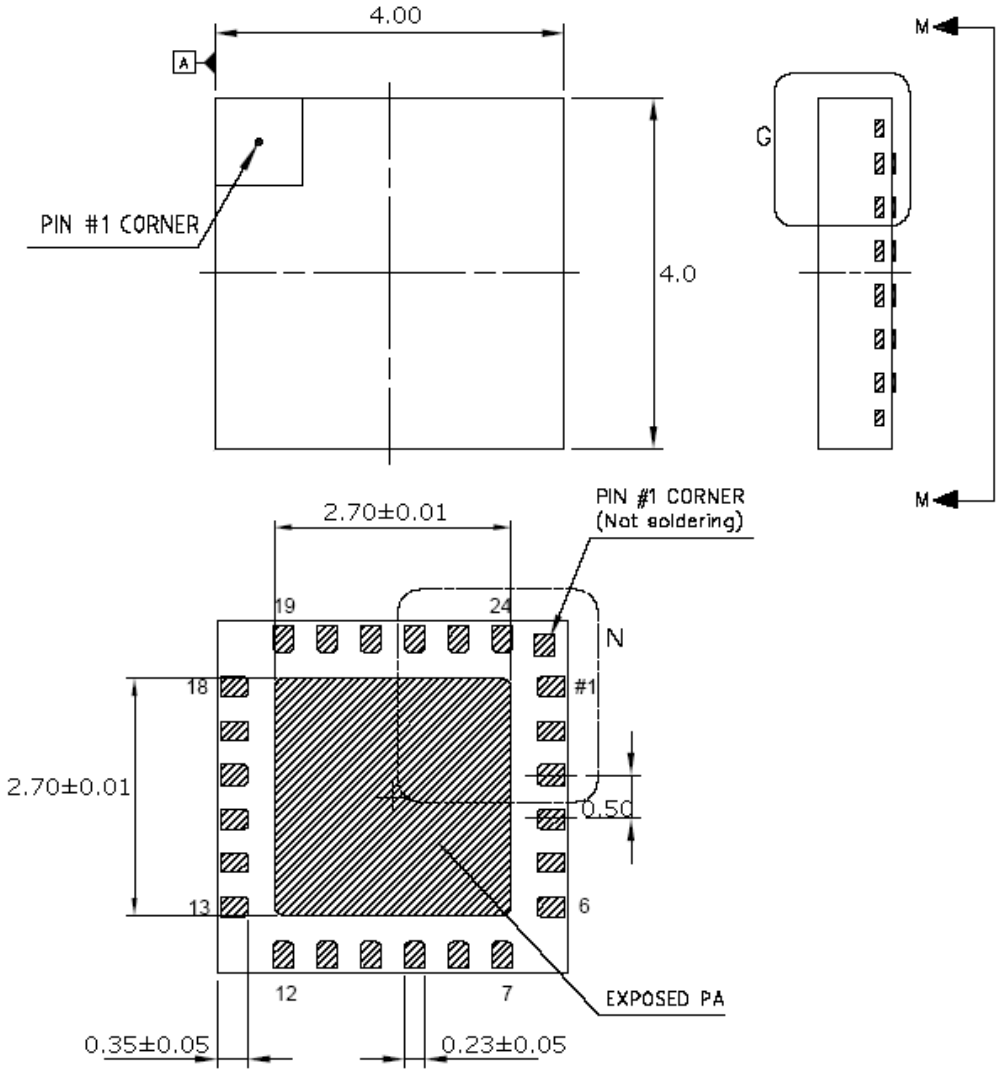
EVEN / ODD TERMINAL SIDE



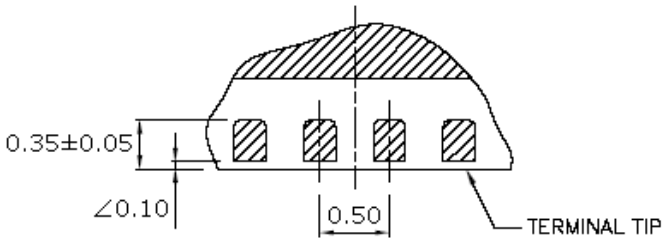
DETAIL G

VIEW ROTATED 90° CLOCKWISE

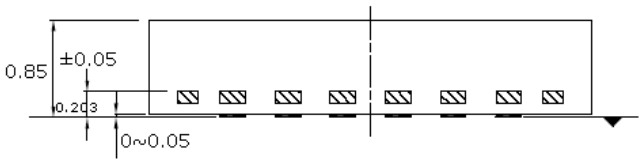
24 QFN (unit: mm)



VIEW M-M



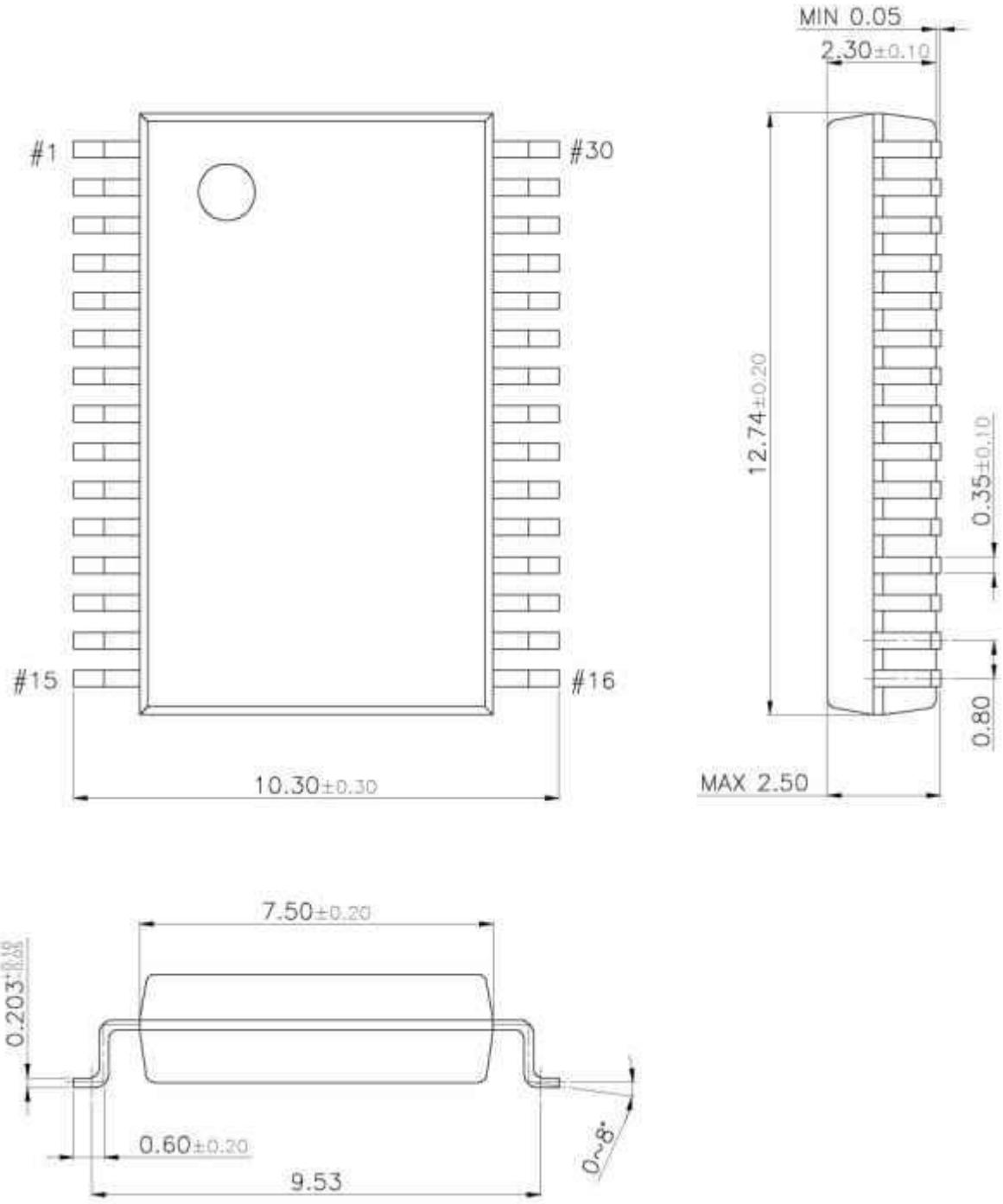
EVEN / ODD TERMINAL SIDE



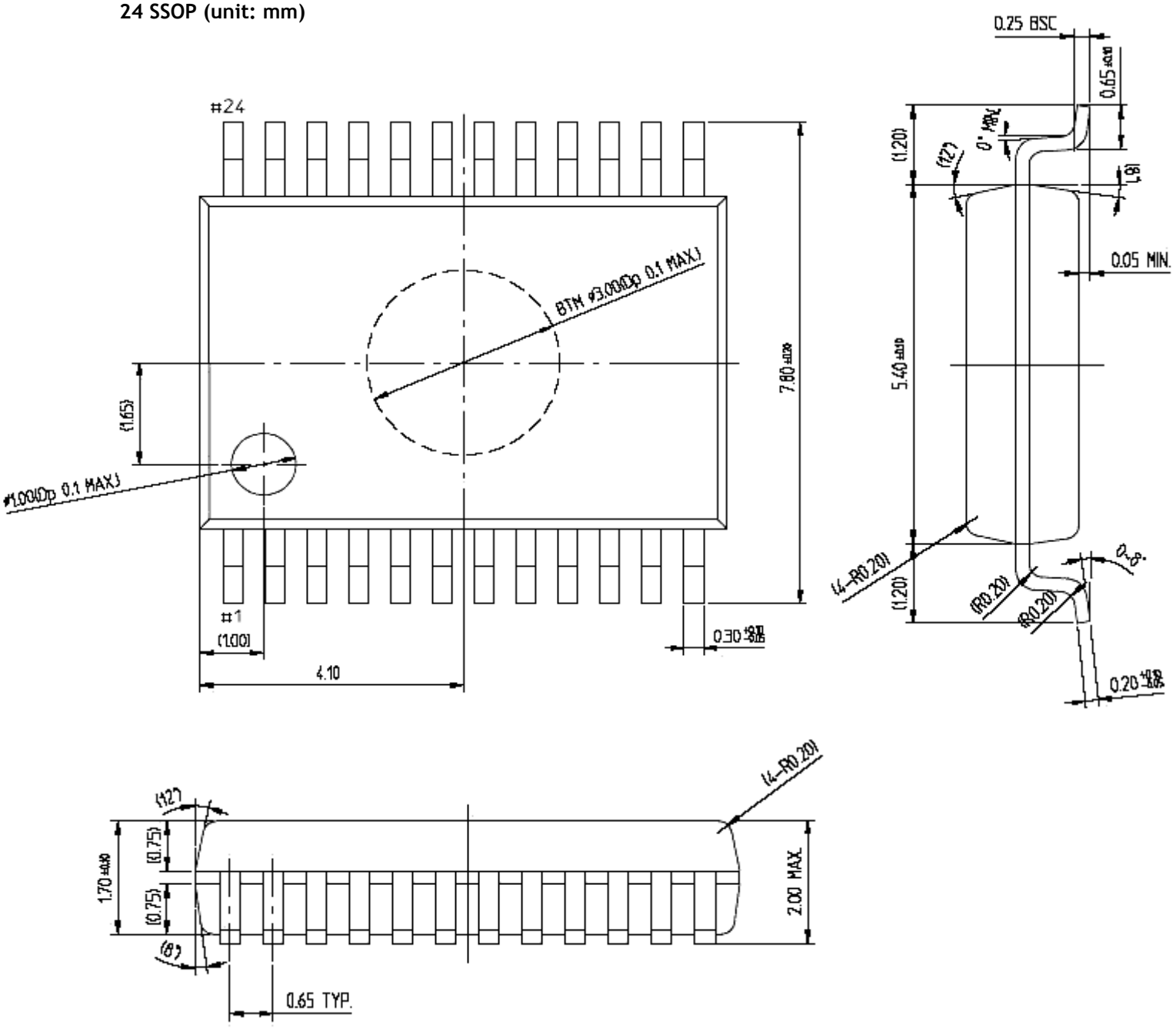
DETAIL G

VIEW ROTATED 90° CLOCKWISE

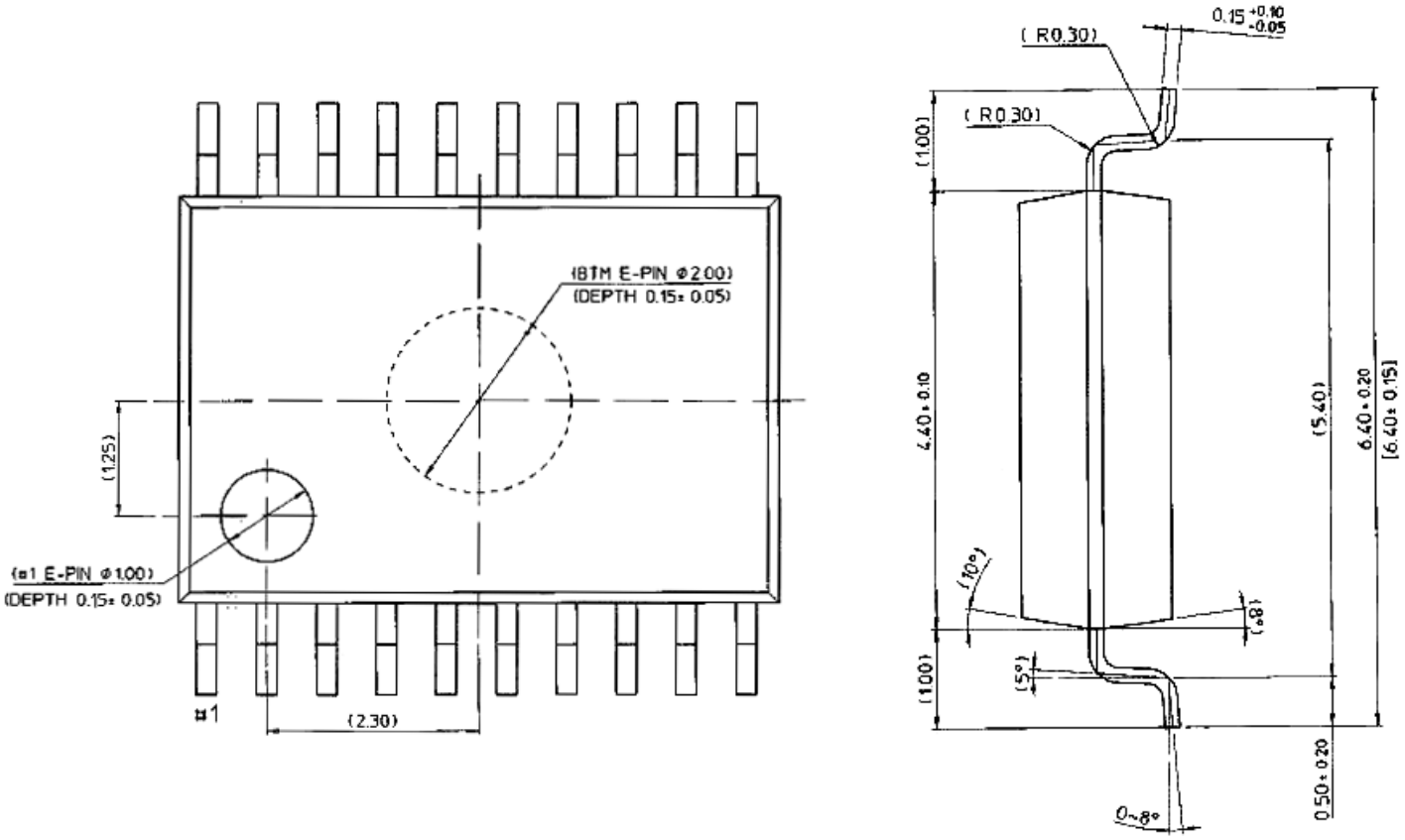
30 SSOP (unit: mm)

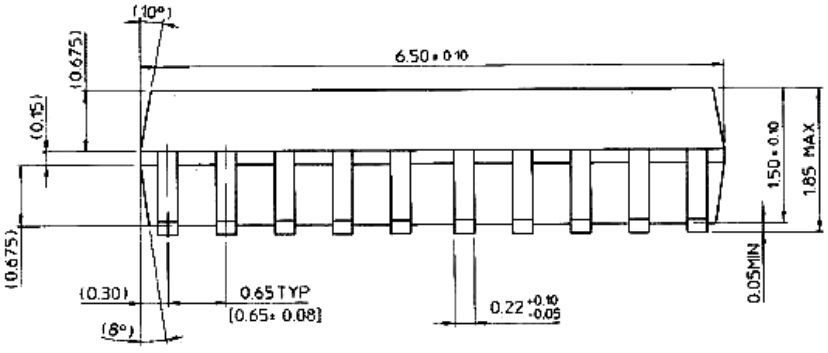


24 SSOP (unit: mm)



20 SSOP (unit: mm)





1.8 Operation Principles

Touch Detection

ATA2508 contains ATLab's proprietary technology of Impedance Change Detection engine within the device. It detects impedance difference between reference and sensor input.

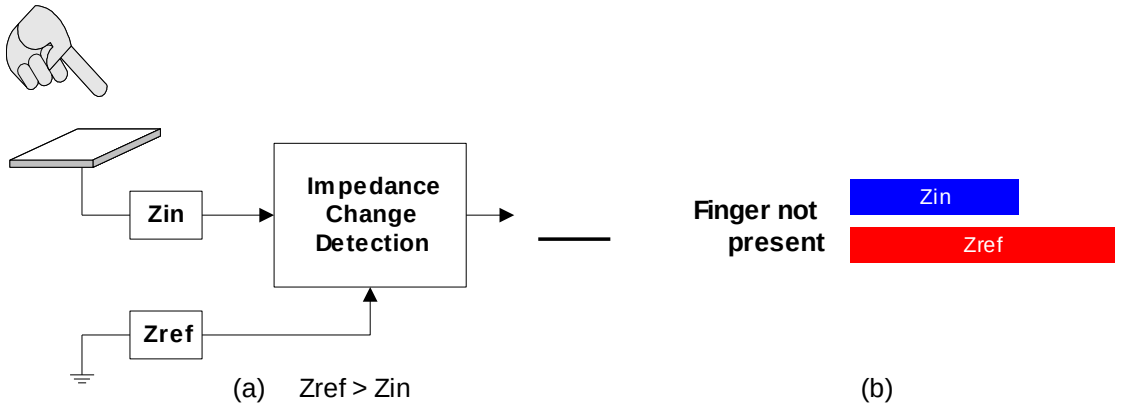


Figure 1-1 Case of none touch: (a) Touch PAD System Model, (b) Impedance Status

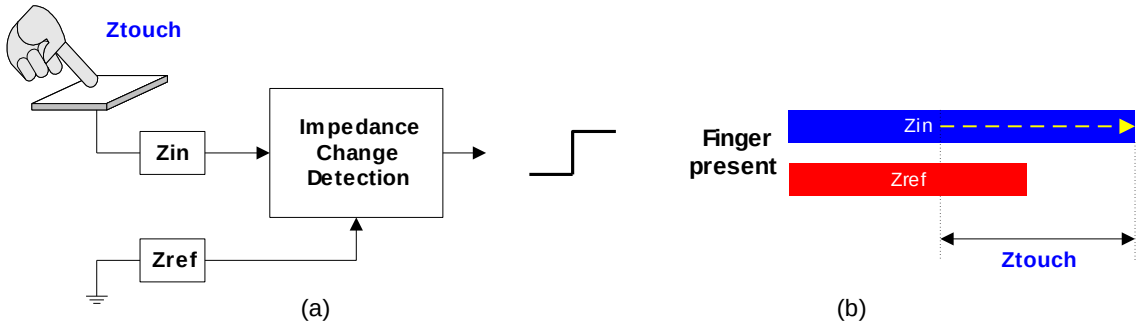


Figure 1-2 : Case of touch: (a) Touch PAD System Model, (b) Impedance Status

As shown in Figures 1-1 and 1-2, if the pad is not touched, the impedance of the sensor input Zin should be less than the impedance of the reference Zref, that is, Zref > Zin condition should be sustained, whereas if the pad is touched, Zin is increased by Ztouch. When Ztouch becomes greater than the difference of Zin and Zref in none touch state by touching the pad, the Impedance Change Detection (ICD) engine within the ATA2508 generates the acknowledged output signal.

$$ICD = \begin{cases} 1, \text{ if } Zin - Zref > 0.1\ pF \\ 0, \text{ otherwise} \end{cases}$$

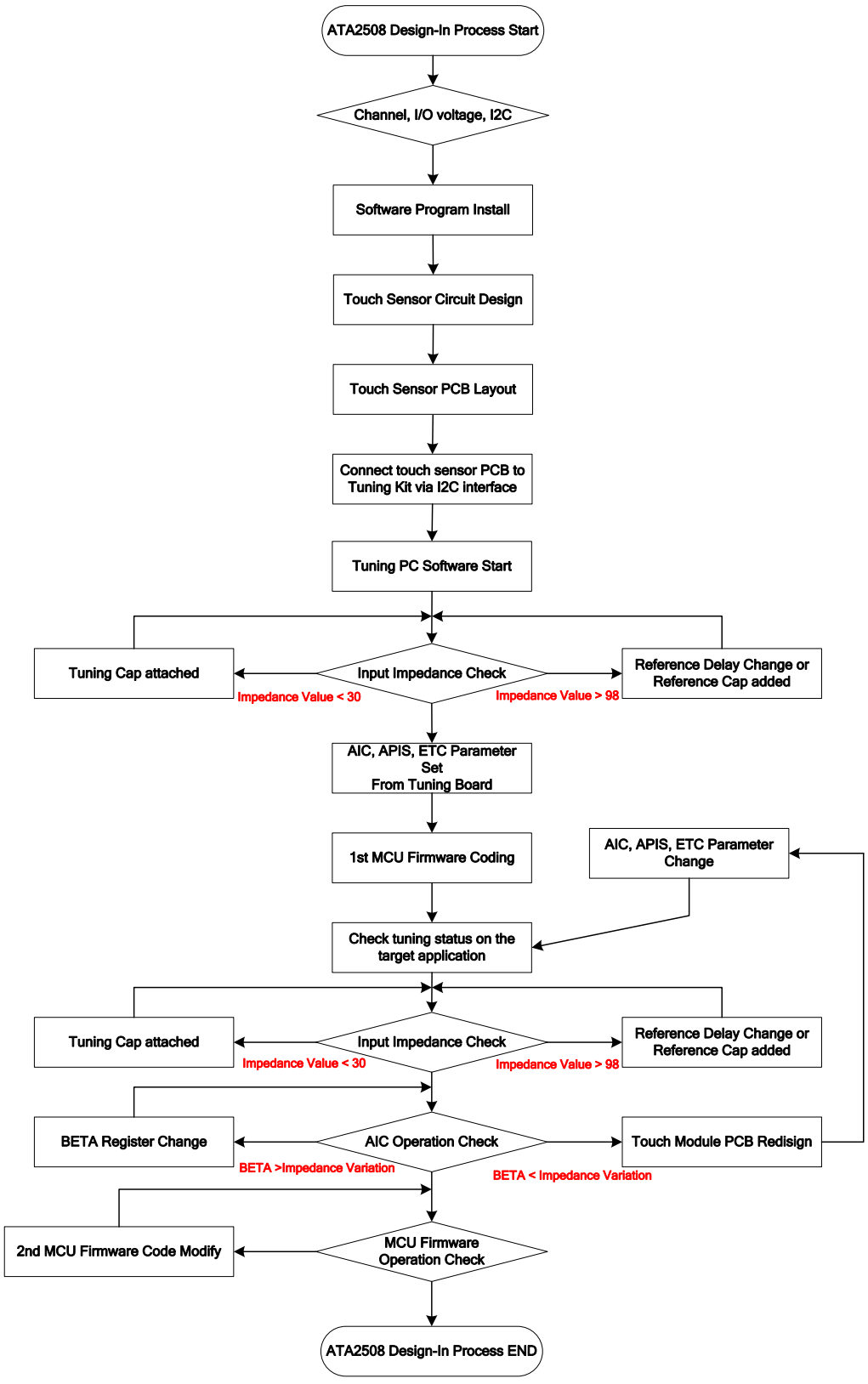
Notice that the ICD in the ATA2508 generates touch detection signal when Zin becomes greater than Zref as the amount of 0.1pF or above in order to maintain stable output against various noises. The sensor input impedance, Zin, includes parasitic capacitance of the input line, tuning capacitance of input pin and on-chip input impedance, while Zref includes on-chip impedance, AIC control values and external tuning capacitance if necessary. The AIC control values will be explained later.

AIC (Automatic Impedance Calibration)

AIC function, one of ATLab's patented technologies, is to maintain uniform sensitivity against external environmental changes such as temperature, humidity, supply voltage and current, and system level variations. This helps users to develop their applications more conveniently by providing actual impedance value of each sensor input. For developers, ATLab provides Tuning Viewer program that helps to optimize PCB design and to decide AIC input parameters. More detail will be explained in the chapter 3, Software.

The ICD engine residing in the ATA2508 controls reference impedance value for each sensor input pin by acquiring each input impedance data. It periodically updates all reference impedance values under the condition that all twelve touch pads remain no-touch status. This auto-calibration function absorbs environmental changes and guarantees product stability.

1.9 Tuning Process Flowchart



2. Part Ⅰ: Hardware

2.1 Tuning System

The conceptual tuning system is shown in the following Figure 2-1.

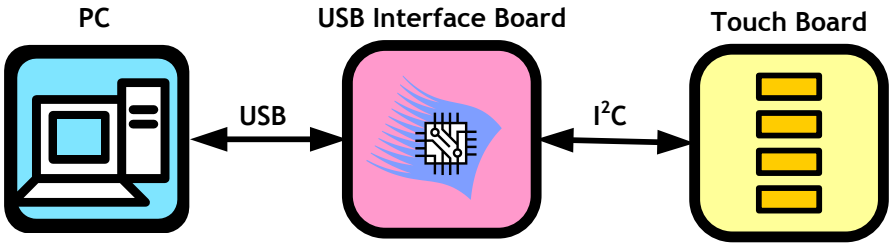


Figure 2-1 Conceptual Tuning System

2.1.1 Composition of Tuning System

The tuning kit version 3.0 for ATA2508 consists of one USB interface board and several touch boards. You can send commands to the MCU in the USB interface board or receive touch data from the MCU through USB interface on PC. The MCU in USB interface board will control ATA2508 in the touch board via I²C interface by reading/writing data to access internal registers in ATA2508.

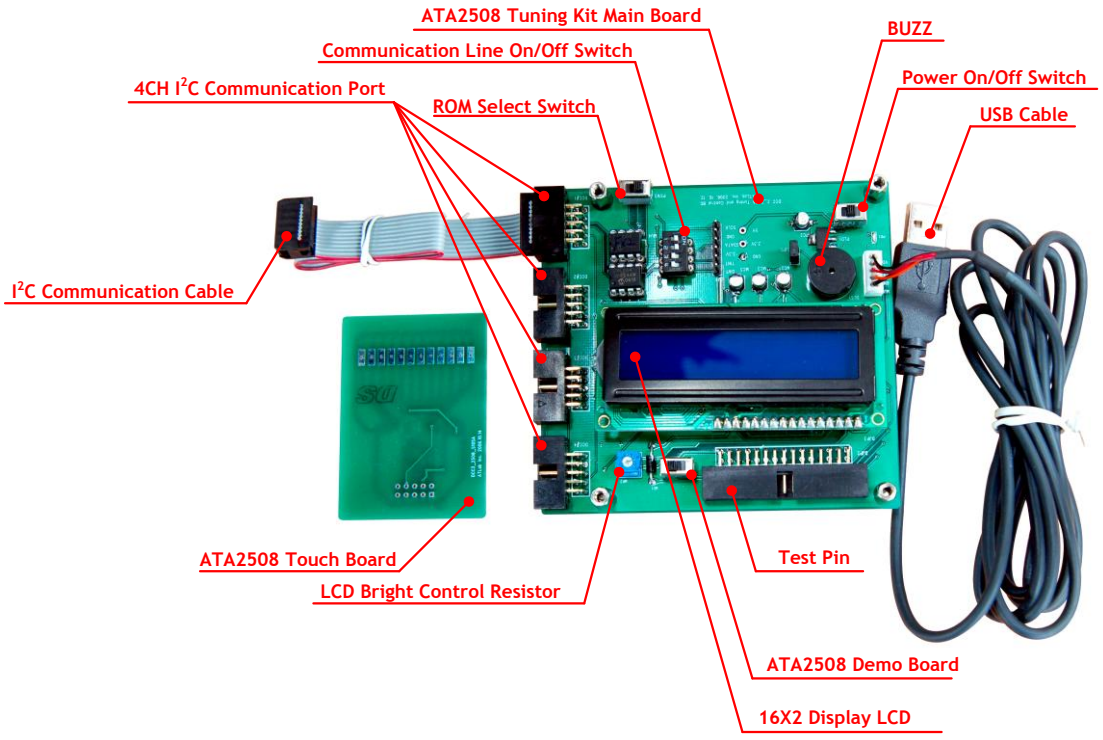


Figure 2-2 The Composition of Tuning System

2.1.2 Connecting USB Interface Board and Touch Board

2.1.2.1 Connecting USB Interface Board to PC

Before starting tuning with the ATA2508, you need to connect USB interface board (hereafter called USB B/D) to PC. PC also requires device driver program to control the tuning system. Once you install setup program included in the tuning package, .inf file is created in the \Inf folder. After finishing this installation procedure, PC automatically detects corresponding device driver program when USB B/D is connected, or you can assign it manually. After the installation was completed and if PC recognized the tuning system correctly, you can see the message, "ATA2508 Tuning B/D" on the monitor. From hence you can control ATA2508 through the tuning program installed on PC.

2.1.2.2 Connecting ATA2508 Touch Board to USB Interface Board

ATA2508 Touch Board (hereafter called Touch B/D) can be connected to USB B/D via I^2C bus cable. This cable is composed of 10 pins and pin assignments are described below in Figure 2-3.

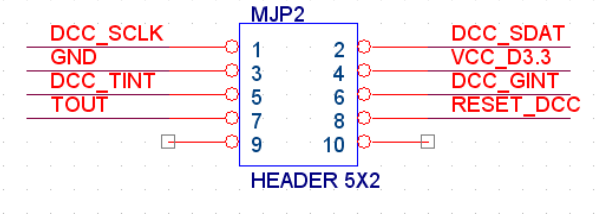


Figure 2-3 Pin Assignment of I^2C Bus Cable

There is a connector in USB B/D to configure 4 different types of I^2C address. Therefore, if you assign different I^2C address for each ATA2508 Touch B/D, you can connect 4 ATA2508s simultaneously. I^2C addresses start 0x58 through 0x5B (7 bit), and the assigned ID depends on electric status of ID0 and ID1 pins of ATA2508 shown in Table 2-1.

I^2C ID (Hex)	ID1	ID0
0x58	0	0
0x59	0	1
0x5A	1	0
0x5B	1	1

Table 2-1 I^2C ID Configuration Table

In order to access the ATA2508 you want, tuning program must select corresponding I^2C ID first by sending the command to USB B/D. The detail instruction will be described in chapter 3, Software Part later.

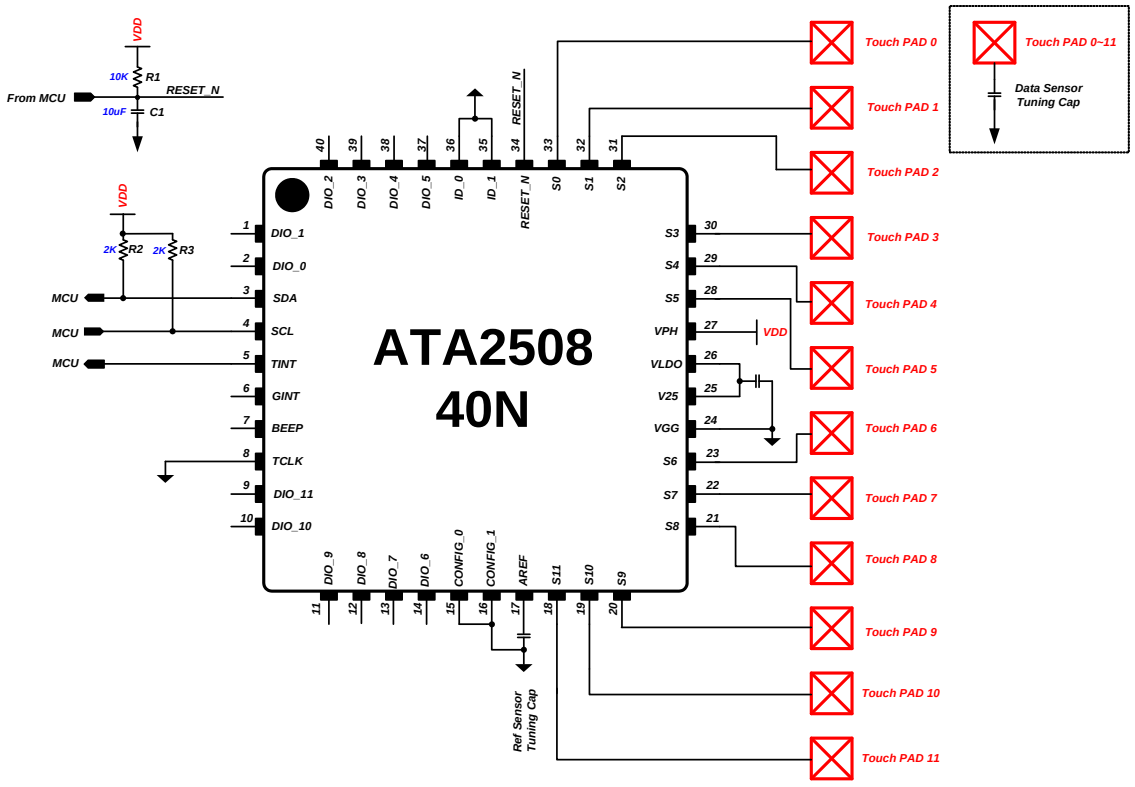
The explanation of each pin in Figure 2-3 follows. Pin 1 and pin 2 are for I^2C communication, which are clock and data line respectively. Pin 3 is ground pin and pin 4 is Vdd pin for power supply. Pin 5 and 6 deliver 2 interrupt signals generated by ATA2508 to MCU in USB B/D. Pin 5 delivers touch interrupts and pin 6 delivers general interrupts. The characteristics of interrupts will be described in chapter 3, Software Part later. Pin 7 is for buzzer whose frequency and period are predefined by users. Pin 8 is used to give cold reset to ATA2508 when it seems to be unstable through MCU control menu of tuning program in PC. "DCC Ext. Reset" button In MCU control menu activates this function.

2.1.3 Composition of Touch Board

There are 3 kinds of touch boards in the tuning package, linear scroll board, circular scroll board and button board. Even if the shape of each touch board is different, the circuit is almost identical. However, MCU program or internal registers' values of ATA2508 in each touch board should be different according to the applications you implement. Each touch board has a pin option to assign unique I^2C address and option pad to select internal LDO or external power for supplying voltage to I/O and core part. The voltage of I/O is acceptable from 3.3V to 5V.

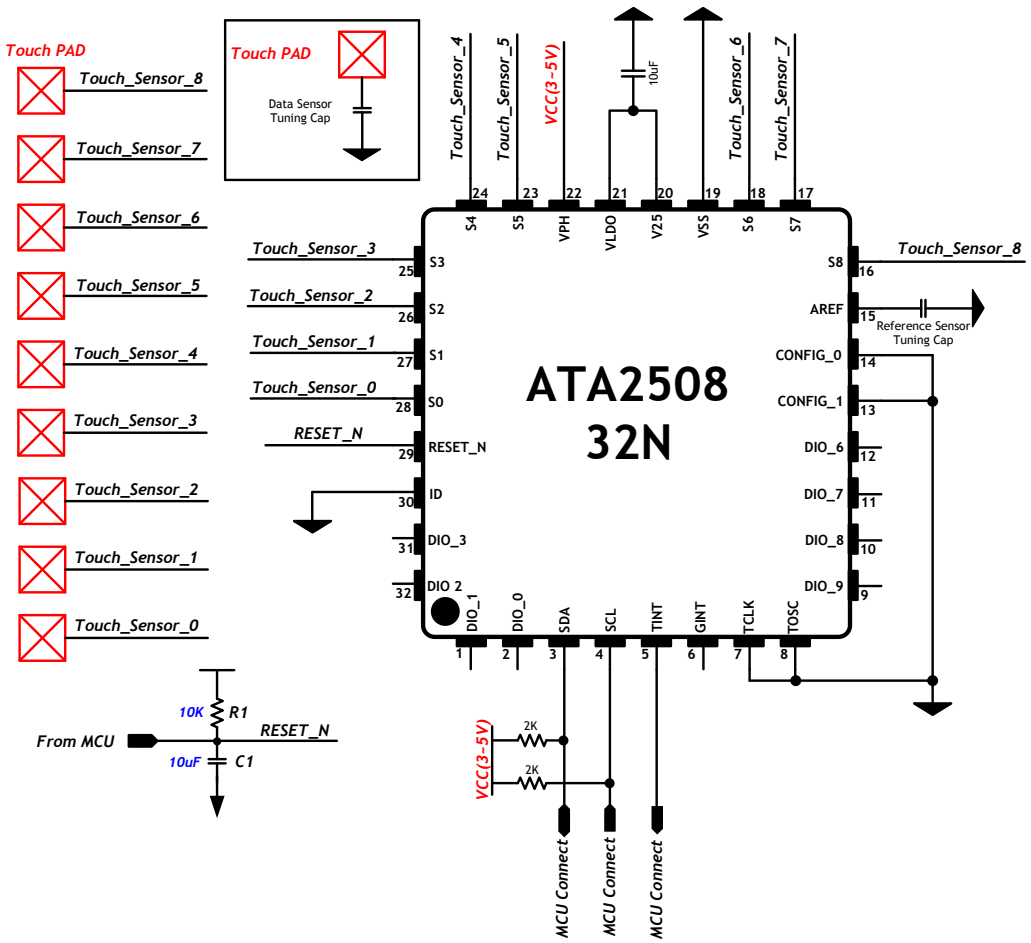
2.2 Typical Application Circuit

2.2.1 40QFN Application Circuit



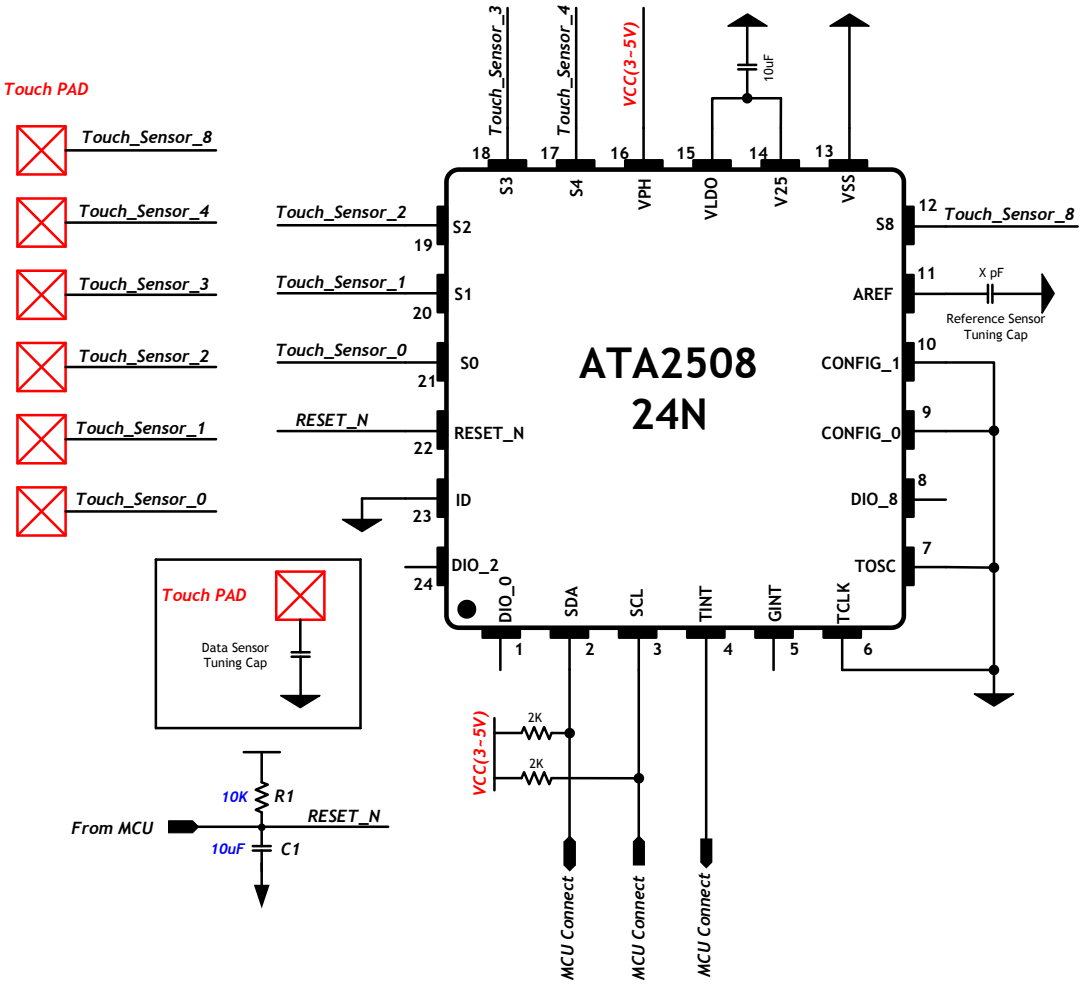
- The voltage of Vdd can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 12 channels. Each DIO port can drive 2mA current.

2.2.2 32QFN Application Circuit



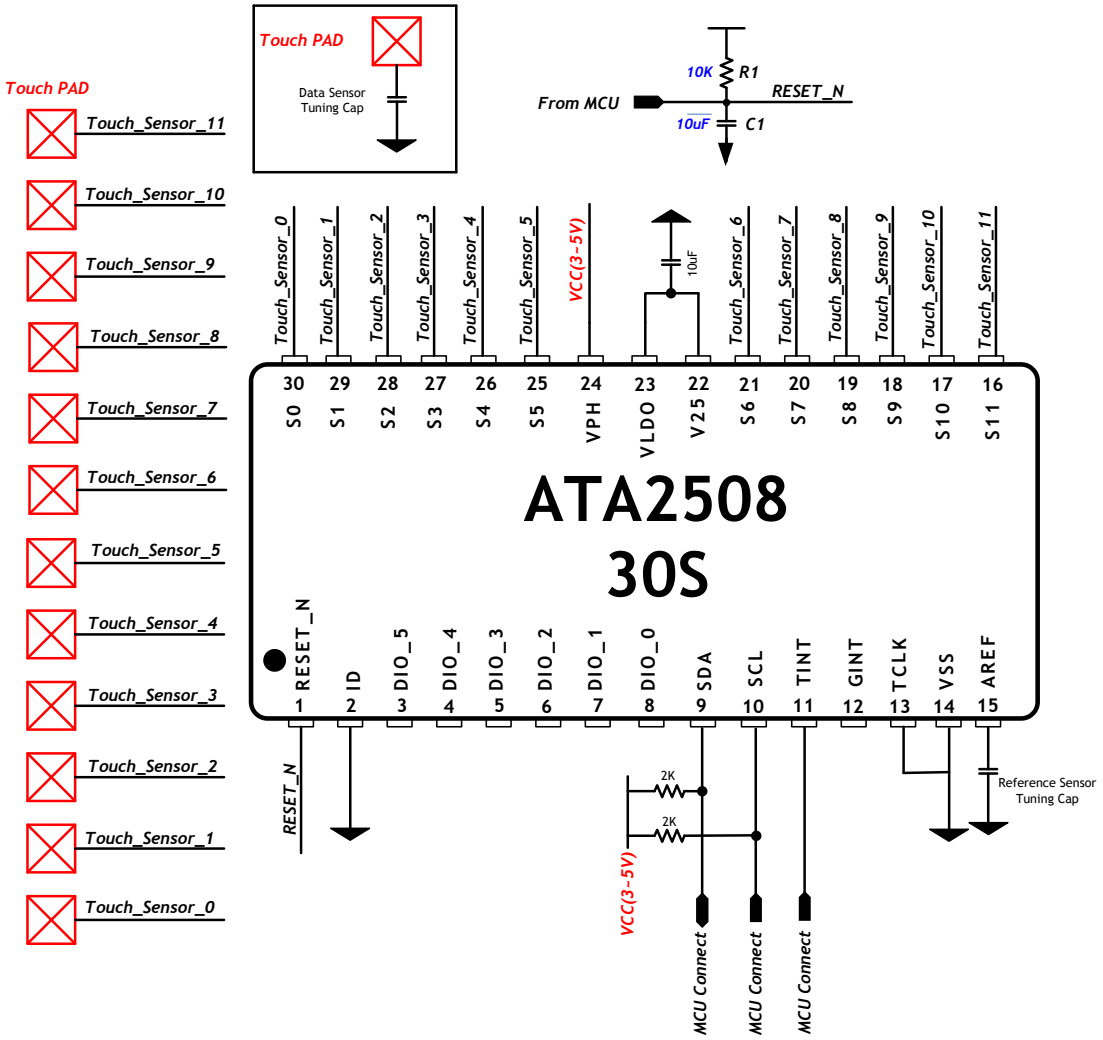
- The voltage of VCC can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 8 channels. Each DIO port can drive 2mA current.

2.2.3 24QFN Application Circuit



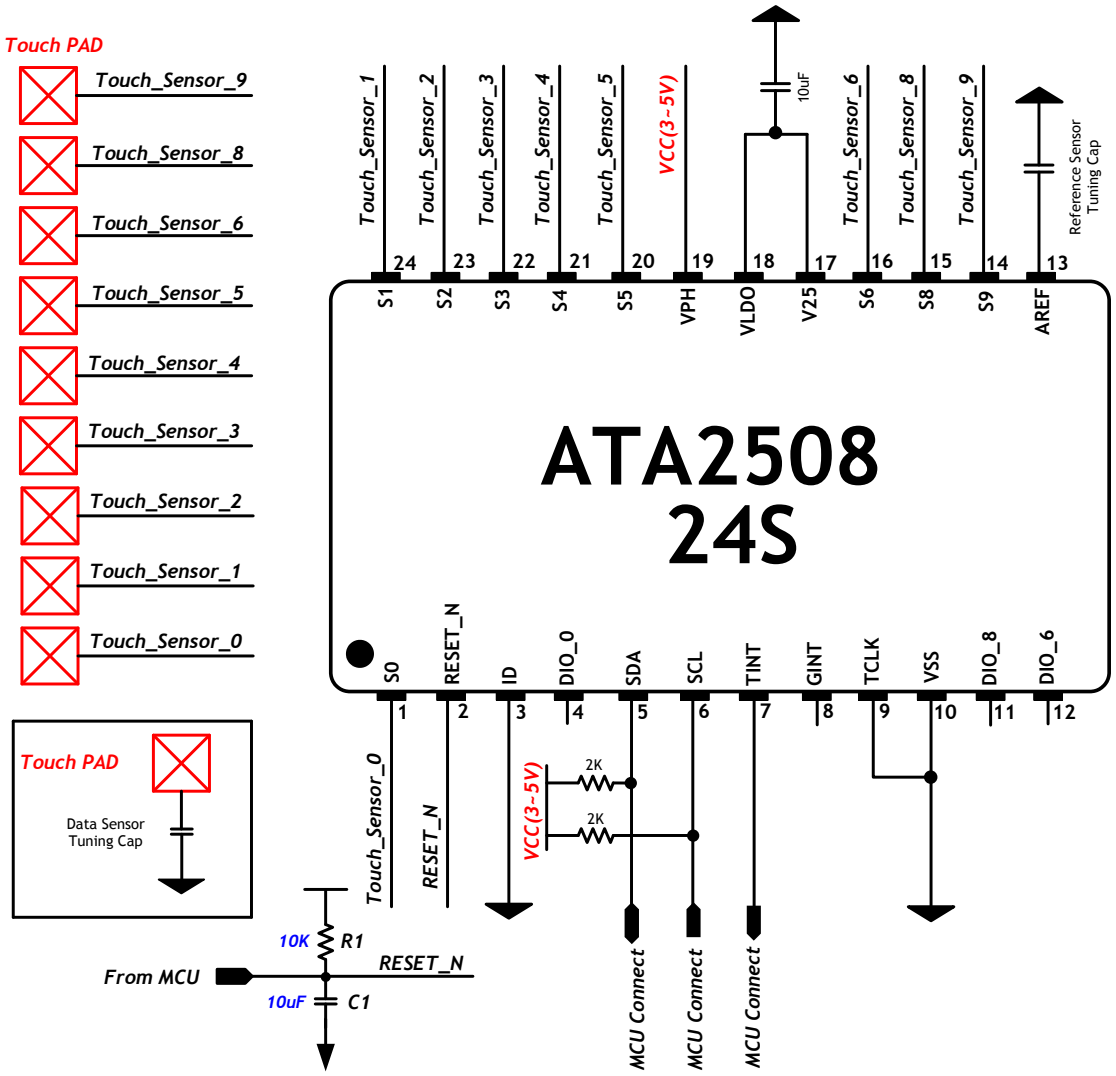
- The voltage of VCC can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 8 channels. Each DIO port can drive 2mA current.

2.2.4 30SSOP Application Circuit



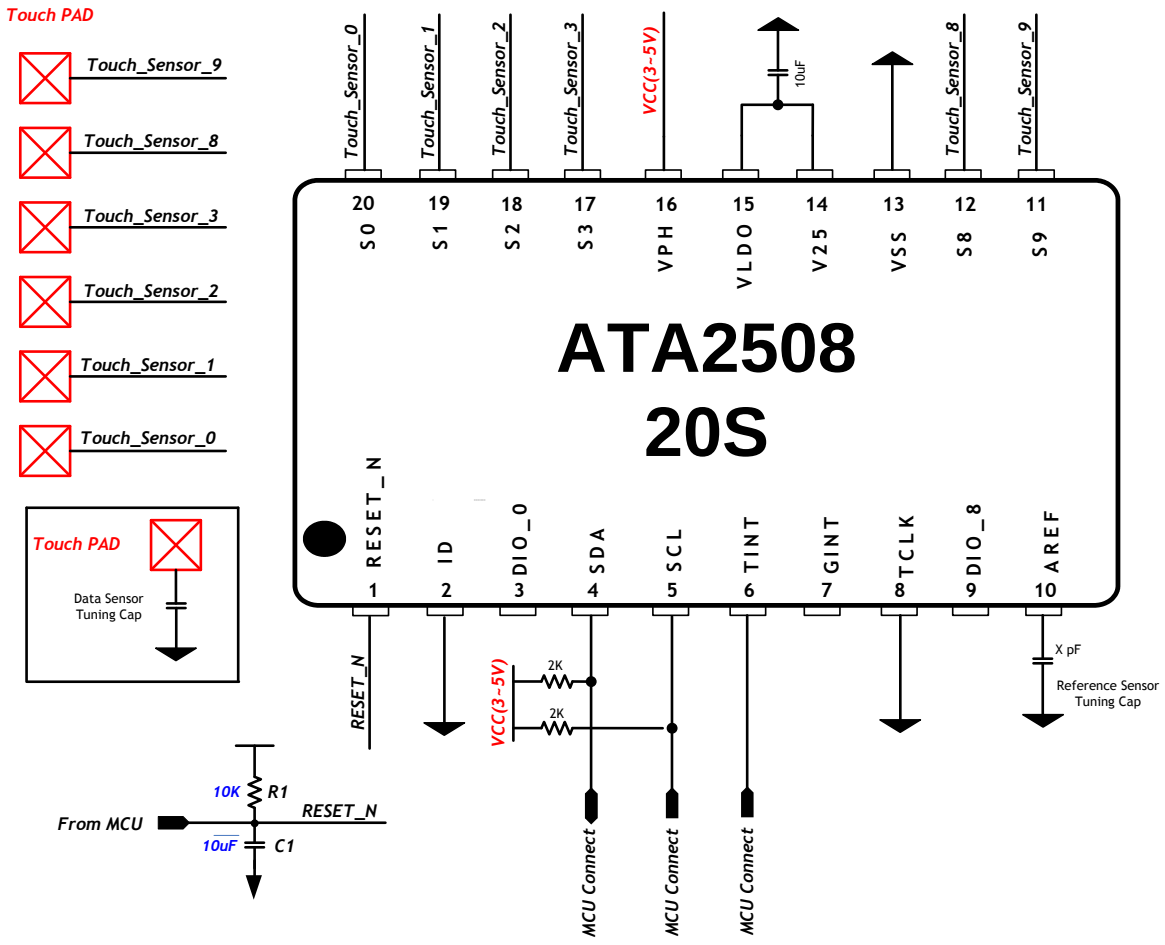
- The voltage of VCC can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 8 channels. Each DIO port can drive 2mA current.

2.2.5 24SSOP Application Circuit



- The voltage of VCC can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 8 channels. Each DIO port can drive 2mA current.

2.2.6 20SSOP Application Circuit



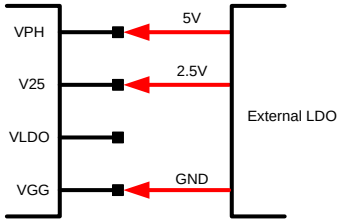
- The voltage of VCC can be 3V through 5V.
- Each tuning capacitor is optional component depends on PCB layout environment.
- The circuit above is a typical application circuit using internal LDO.
- R1 and C1 are not necessary if MCU provides reset signal.
- 2K pull-up resistor for I^2C communication can be eliminated if other device already used it for I^2C communication.
- LEDs can be directly attached to DIO ports as indication of touch status of 8 channels. Each DIO port can drive 2mA current.

2.2.7 Power Connection

There are two methods to supply power to ATA2508. One is to receive V25 core voltage from internal LDO, and the other is to receive core voltage from external power supply. In the case of using internal LDO, LDO should be turned on in Sleep mode. Therefore, it will cause slightly more power consumption than using external power supply for V25 core voltage.

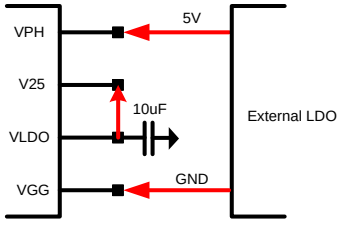
►Power Connection Example

Case A. VPH: External 5V, VLDO: External 2.5V (Internal LDO Off: Register Control)



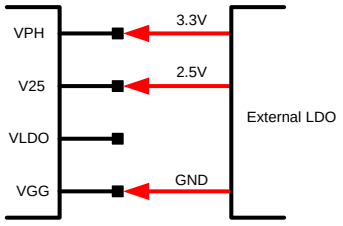
IO interface to other chip is 5V.

Case B. VPH: External 5V, VLDO: Internal LDO 2.5V



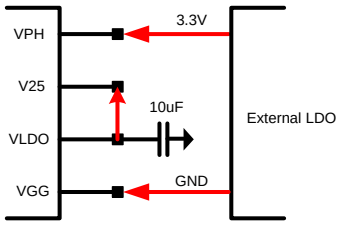
IO interface to other chip is 5V.

Case C. VPH: External 3.3V, VLDO: External 2.5V (Internal LDO Off: Register Control)



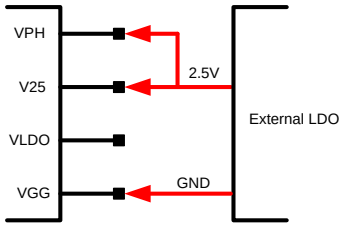
IO interface to other chip is 3.3V.

Case D. VPH: External 3.3V, VLDO: Internal LDO 2.5V



IO interface to other chip is 3.3V.

Case E. VPH: External 2.5V, VLDO: External 2.5V



IO interface to other chip is 2.5V.

If VPH receives 2.5V, internal LDO can not be used because VLDO can not output 2.5V when VPH receives 2.5V from external LDO.

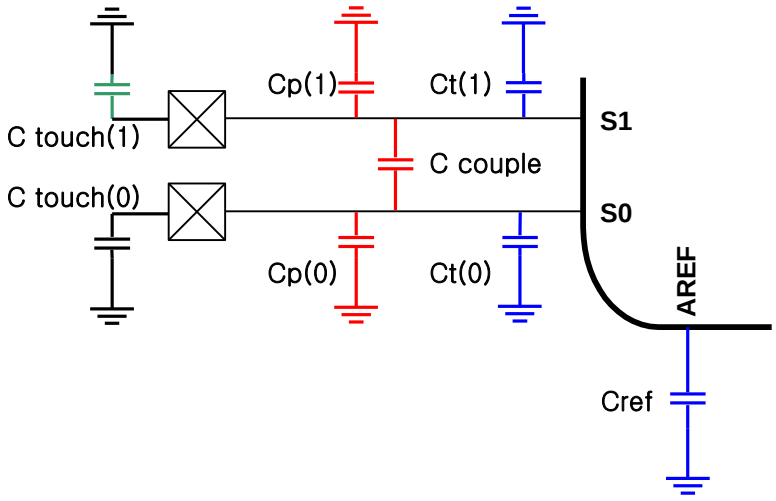
2.3 Guidelines for Touch Pad and PCB Designs

2.3.1 Impedance Details

Sensor Input Impedance (Zin) is determined by three major external factors. They are Parasitic Capacitance (Cp) between input line and GND, Cross-coupling Capacitance (Ccouple) between input lines, and Tuning Capacitance (Ct). To keep these values lower than Reference Capacitance (Cref), input sensitivity should be controlled by Tuning Capacitance (Ct) as

$$Zin = Ct(n) + Cp(n) + Ccouple$$
$$Zref = Cref$$

In particular, Cross-coupling Capacitance (Ccouple) is a critical factor to deteriorate operation so it should be minimized. Since cross-coupling capacitance existing on sensor input patterns can cause crosstalk, one should carefully design sensor layout to minimize this crosstalk by placing enough space or ground pattern between sensor input lines. Inevitable crosstalk should be compensated by software. But, smaller Cross-coupling Capacitance can minimize software load.



Cp : Parastic capacitance by input line layout
Ct : Tuning capacitance for sensitivity adjustment
Cref : Reference capacitance value
C couple : Coupling capacitance among input lines
C touch : Capacitance between human body and touch pad

Figure 2-4 : Impedance Details

2.3.2 PCB Design

The important thing in PCB design is to draw sensor lines to reduce influence caused by internal or external noise sources. The types of noise sources and their design guides will be described.

- The noise influence by other chips

In touch module, we recommend that only ATA2508 be mounted without any other chips because other chips can cause noise signals by controlling several types of components such as LCD or Buzzer, etc.

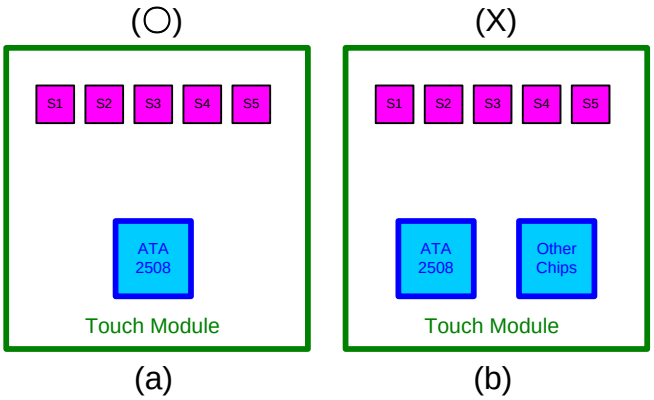


Figure 2-5 : A layout example of noise influence

- Cross coupling capacitance

Noise signal can be generated between sensor input lines. If sensor lines are close and parallel long, they could be noise sources. In order to avoid this, we recommend to design like figure 2-6(b) to enlarge line space and to make lines short in parallel.

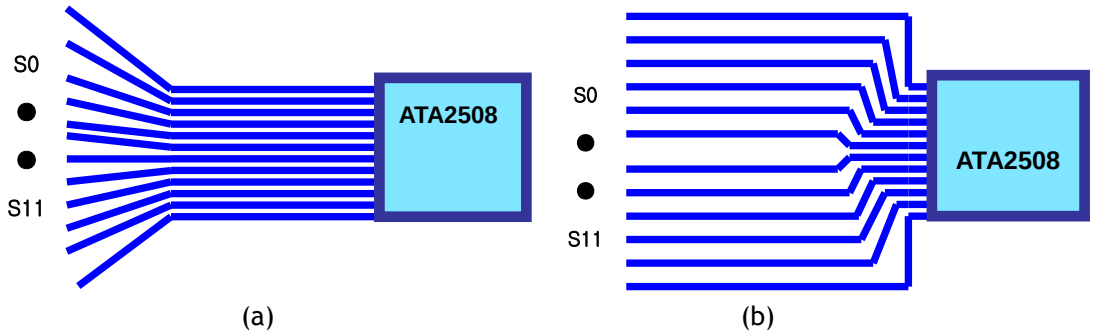


Figure 2-6 : A layout example of sensor input lines

- Disposition of data lines and sensor input lines

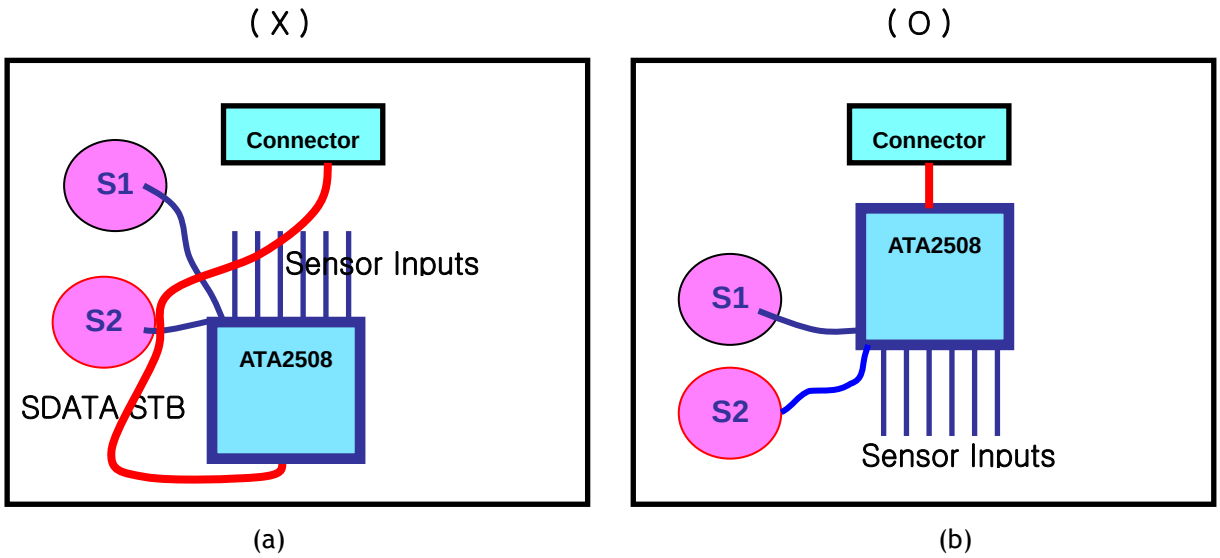


Figure 2-7 : A layout example of sensor input lines

Figure 2-7(a) shows a problem caused by overlapping sensor input lines with data lines. For example, the capacitance generated by power lines with characteristics of consistent voltage output will not deeply affect sensor input lines. However, the capacitance generated by data lines fluctuating high and low voltage output will make sensor input lines unstable. Therefore, the data lines in the front panel application should be placed closer to the connector in order to avoid bad influence on sensor input lines. Another important thing in layout design is that sensor input lines should be placed on the opposite side of data output lines. Finally, since overlapping data lines with sensor pads will be worse than overlapping data lines with sensor input lines, it is recommended that sensor pads should be apart from data lines.

- Several ATA2508 Sensor Line Noise

If 2 ATA2508's are mounted on the same PCB, they can be noise sources to each other. Therefore, in the application using 2 chips you need to design touch pad area like Figure 2-8(a).

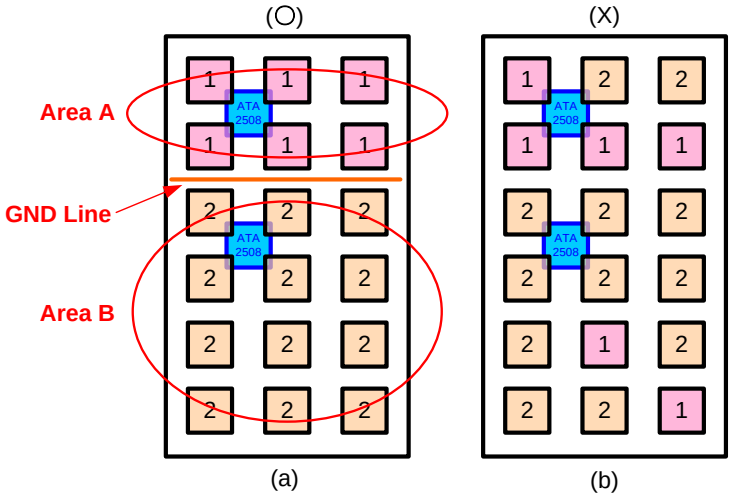


Figure 2-8 : A Layout example of several ATA2508 using

- LCD Control Signal Noise

If PCB which includes LCD control lines is located near by touch PCB, it could be a noise source even if it is not on same PCB. Therefore, you need to design PCB like Figure 2-9(b) less affected by LCD control signal. All kinds of pulse type signals could be better apart from the ATA2508.

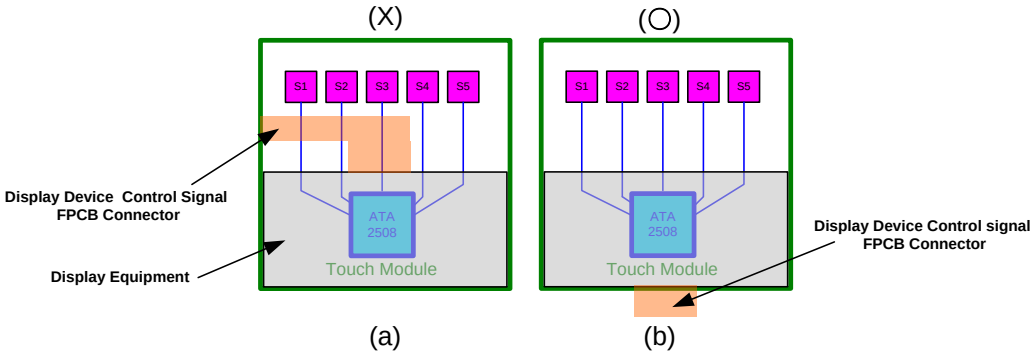


Figure 2-9 : Example of LCD control signal noise

- Charge sharing

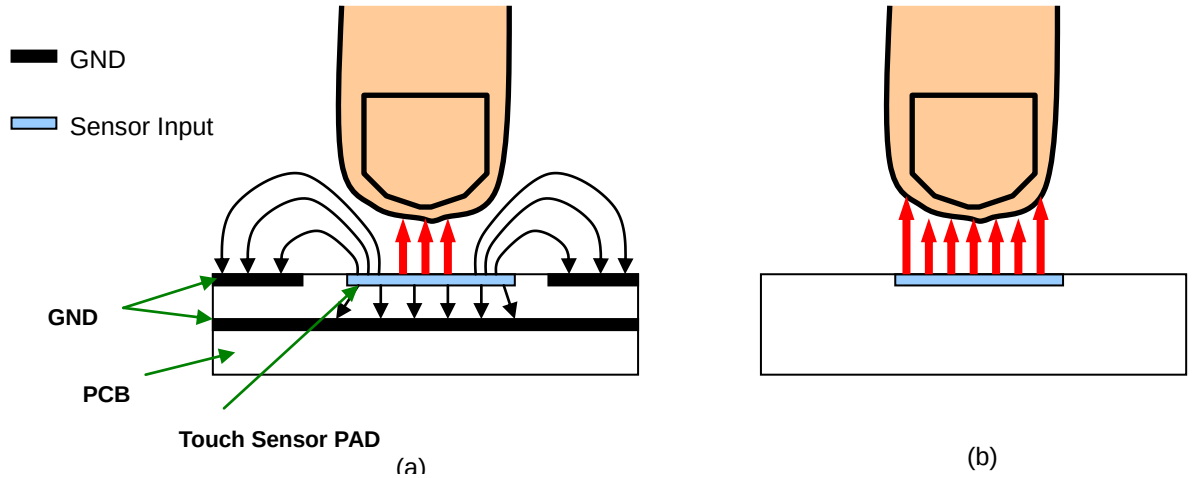


Figure 2-10 : Charge sharing by GND pattern

The sensitivity of the ATA2508 will be decreased if GND pattern is located at nearby sensor input pads and lines because electric field generated by GND patterns will attenuate the strength of capacitance generated by the finger touch. This will decrease the sensitivity of the sensor input shown in Figure 2-10(a). Even if GND Pattern is used for reducing the interference of the lines, make sure to keep enough distance from the sensor input pad.

- Mismatch in Each Sensor Input

For normal AIC function, each sensor input capacitances of the system should be within 6pF.

- Large sensor input pad

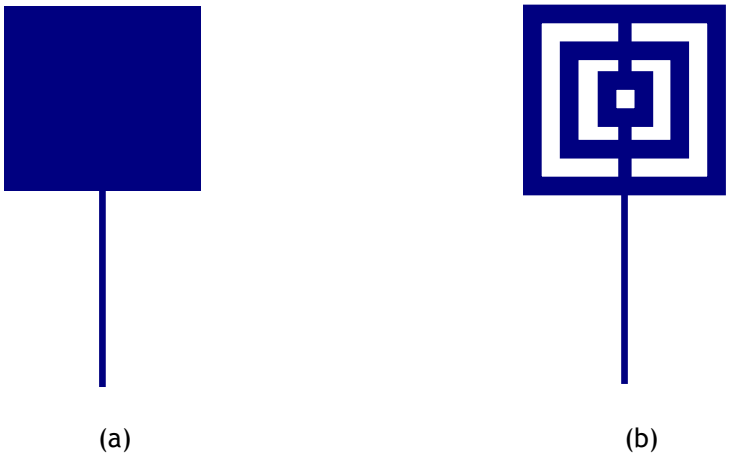


Figure 2-11 : Substitute Pad Pattern for Large Size Sensor Pad

If the pad is bigger than 10mmX10mm, it will become very sensitive by external environmental change.

As the result, input impedance during no touch could be unstable. In order to avoid this case, we recommend to make pad layout shown in Figure 2-11(b) which is exactly the same pad size as shown in Figure 2-11(a), but it will eliminate the problem by reducing real surface area.

3. Part Ⅱ: Software

3.1 Installation Procedures

Tuning program should be installed first to use ATA2508 tuning viewer. Program installation starts by double click of "setup.exe" file shown in Figure 3-1 in CD included in tuning package.

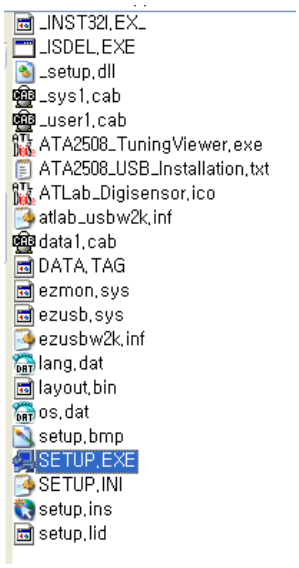


Figure 3-1 Files for Setup

Once installation starts, ATLab's touch sensor logo, DigiSensor™ is displayed shown in Figure 3-2. You need to click logo to approach next installation process or a few moments later it goes to next step by itself.



Figure 3-2 ATLab touch sensor logo during installation

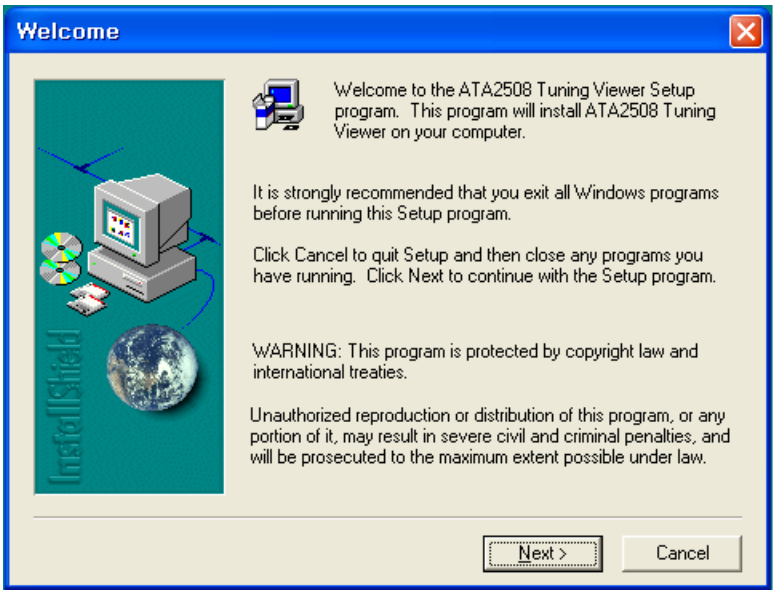


Figure 3-3 Installation Guide Window

Figure 3-3 is the guide message to show basic warnings before starting installation. By clicking "Next" button it goes to next step.

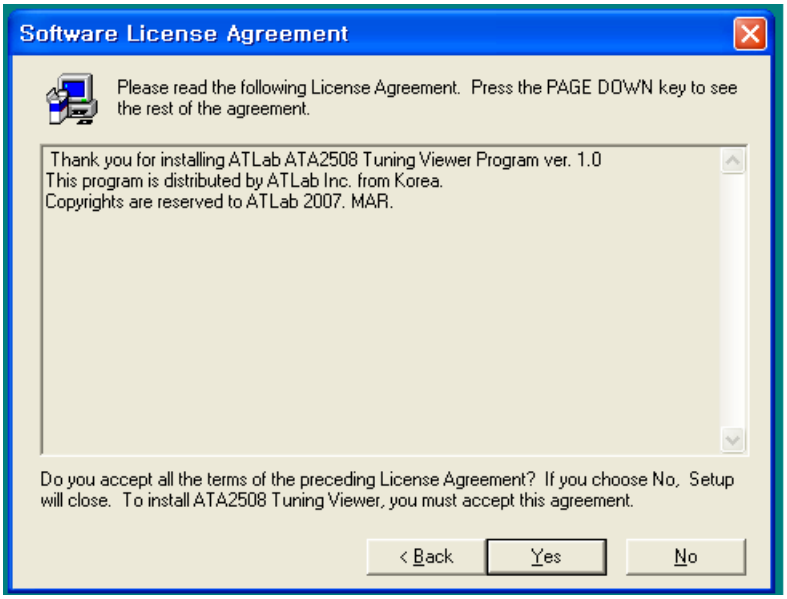


Figure 3-4 Copyright Window

Figure 3-4 shows copyright message of ATA2508 tuning program. Select "Yes" to go next step.

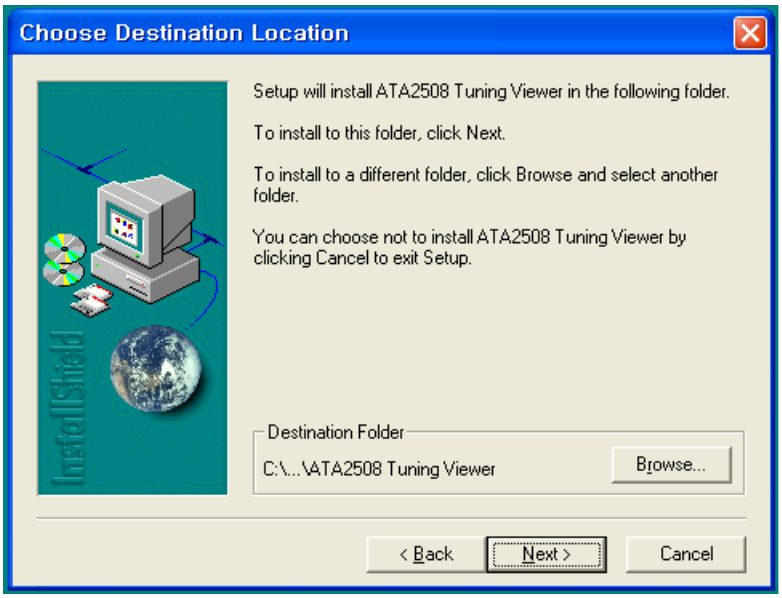


Figure 3-5 Destination Folder Window

Figure 3-5 displays to choose destination location. You can change different folder as recommended or tuning program is installed at "C:\Program Files\ATLab\ATA2508 Tuning Viewer". After selection of destination folder, click "Next" button to approach next step.

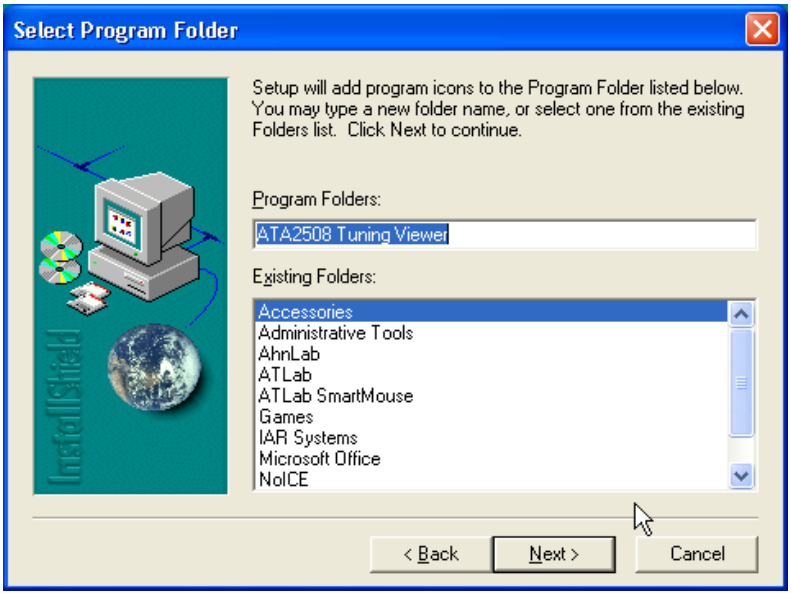


Figure 3-6 Program Folder Window

You can select program folder name at Figure 3-6. Default name is "ATA2508 Tuning Viewer" and you can change it if you want to. Then, by clicking "Next" button, it starts to copy all

necessary files to program folder.

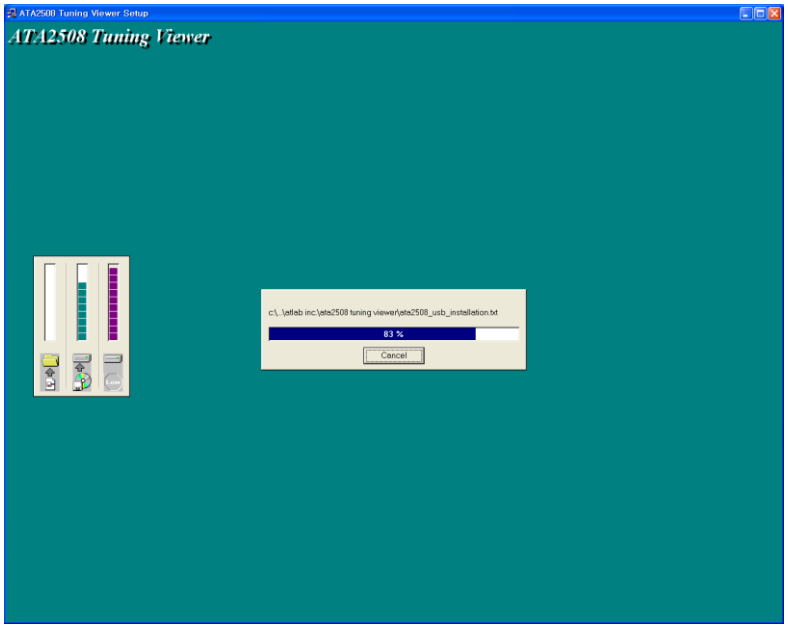


Figure 3-7 Copy Process Window

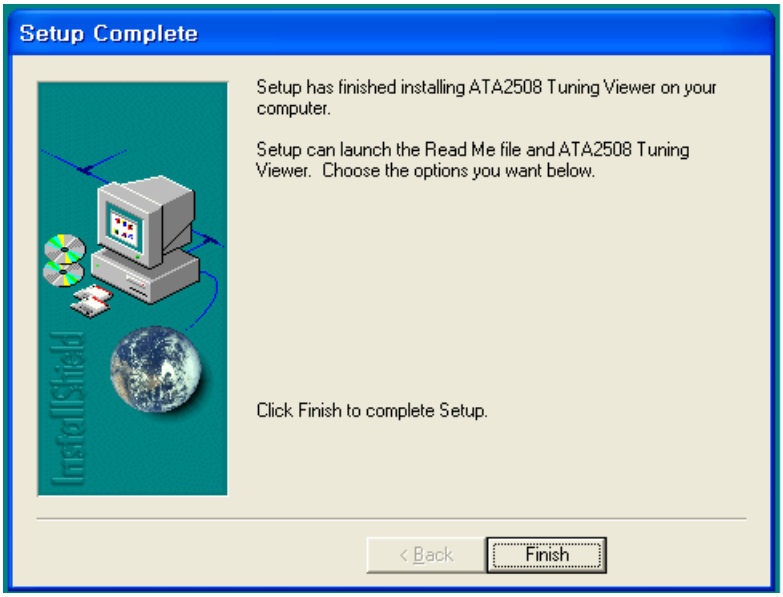


Figure 3-8 Setup Complete Window

Figure 3-8 is displayed after completion of file copy. By clicking "Finish" button the installation ends.

From now on, I will explain how to install ATA2508 USB interface board on PC. When you

connect USB interface board to PC USB port, the window, "Found New Hardware Wizard" is displayed. This window is displayed only when the first connection is made.

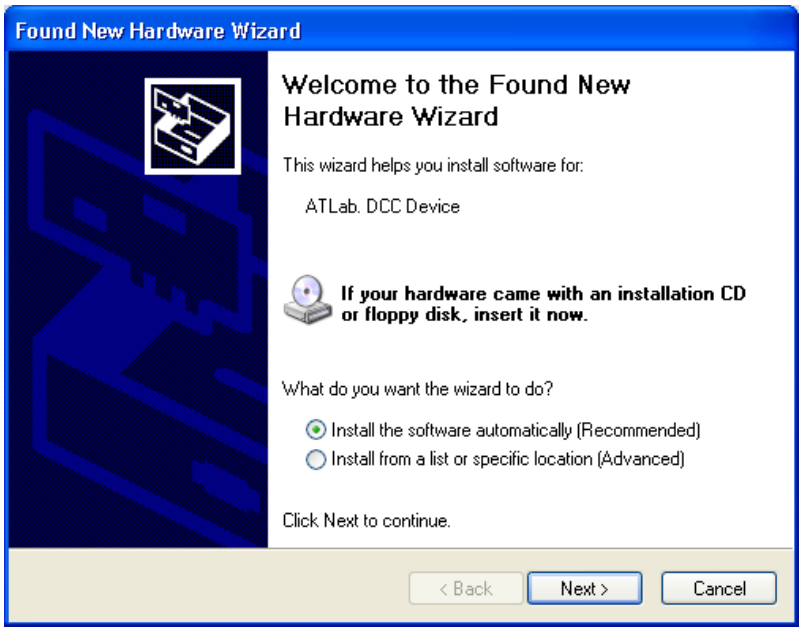


Figure 3-9 Window of New Hardware Wizard

Select "Install from a list or specific location" shown in Figure 3-10 then click "Next" button to go into next step.

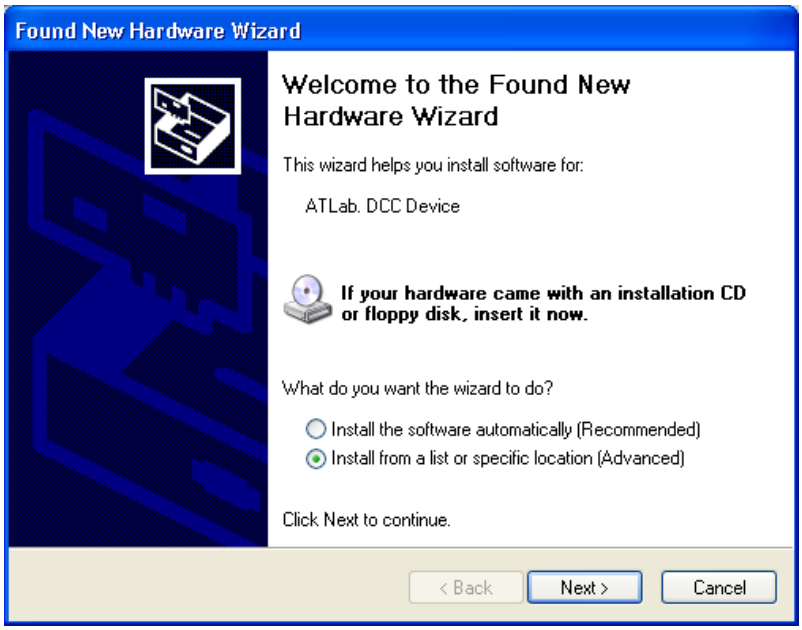


Figure 3-10 Selection Installation Method

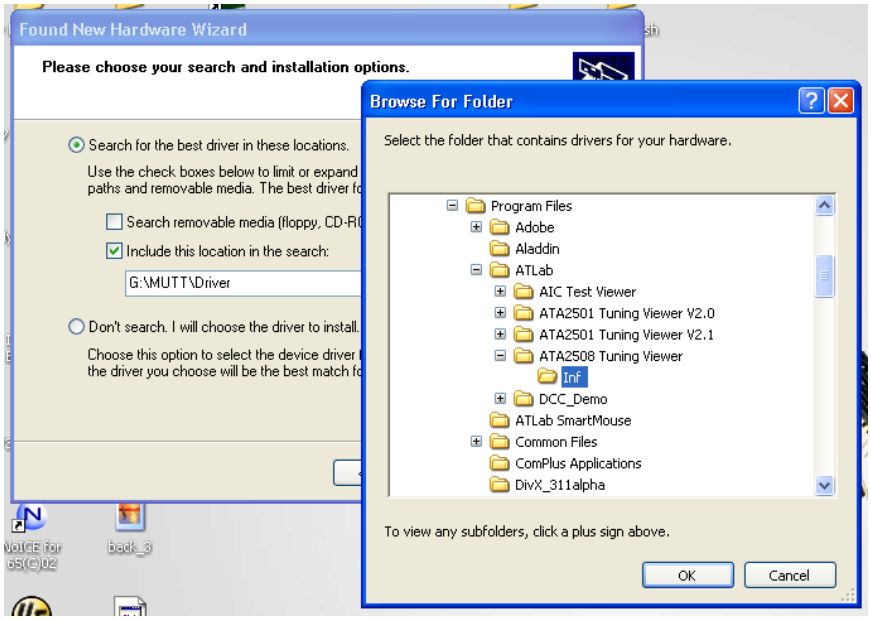


Figure 3-11 Selection of Inf Folder

Check "Include this location in the search". And select "Inf" folder in ATA2508 Tuning Viewer shown in Figure 3-11. If you have difficulty to find destination folder, Select first option, "Install the software automatically" in Figure 3-9. Figure 3-12 shows the window after selection of "Inf" folder.

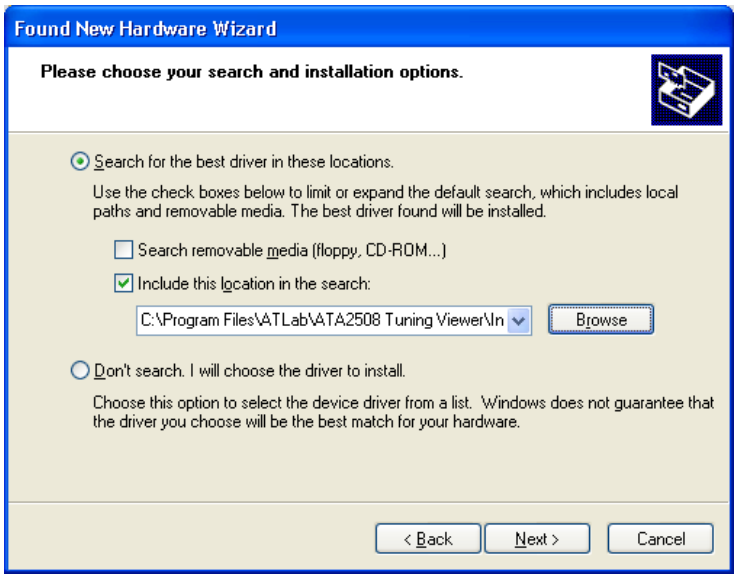


Figure 3-12 Window after Selection of "Inf" Folder

By clicking "Next" button, it starts to search driver programs and copy them to window system folder shown in Figure 3-13.

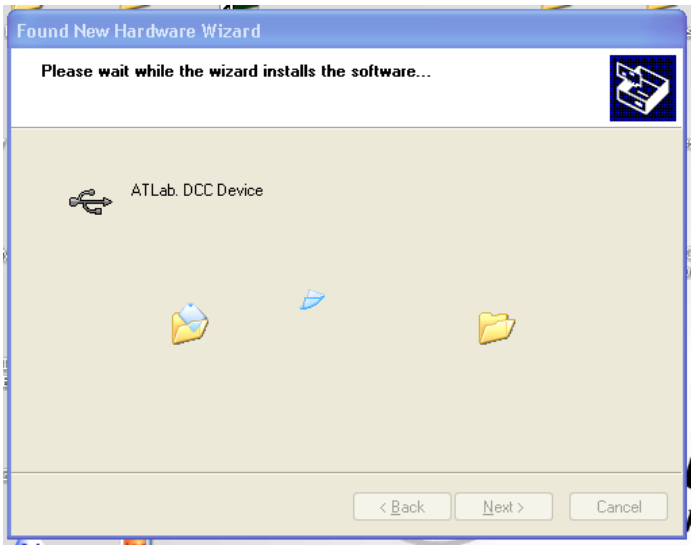


Figure 3-13 Searching Inf file

During copy process, you can find warning messages shown in Figure 3-14, but ignore these messages by clicking "Continue Anyway" button to complete installation successfully.

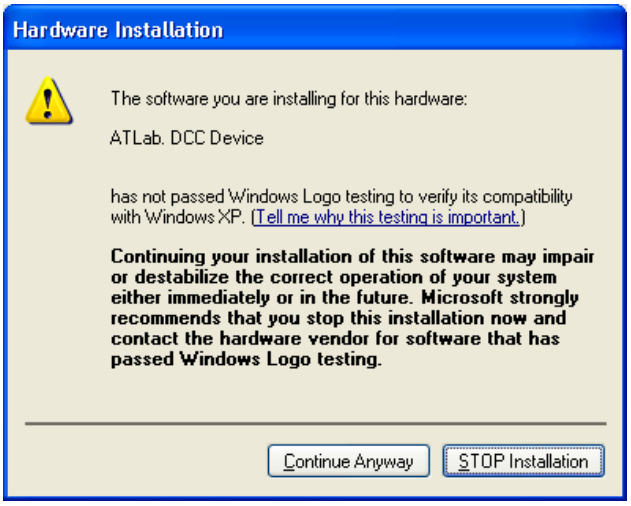


Figure 3-14 Warning Message about WHQL compatibility Check

Figure 3-15 will be displayed after finishing driver program installation successfully. By clicking "Finish" button, the installation process is terminated.

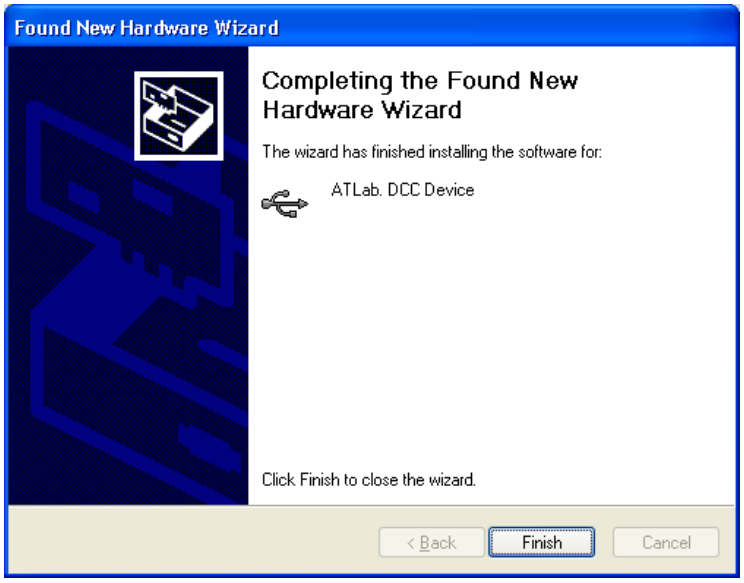


Figure 3-15 End message of installation

Since software installation and hardware connection were done, we are ready to use tuning program. It will be explained in next chapter, "3.2 How to use Tuning Viewer".

3.2 How to use Tuning Viewer

3.2.1 Main Window of Tuning Viewer

In order to run tuning viewer program on PC, USB B/D must be connected to PC first and acknowledged by PC. Its procedure is explained in chapter 3.1. After USB B/D is connected and acknowledged by PC, you can start it by clicking in the order named "Start" -> "All Programs" -> "ATLab" -> "ATA2508 Tuning Viewer" -> "ATA2508 Tuning Viewer". After launching the tuning viewer, Figure 3-16 which is main window of tuning program will be displayed.



Figure 3-16 Main Window of Tuning Viewer

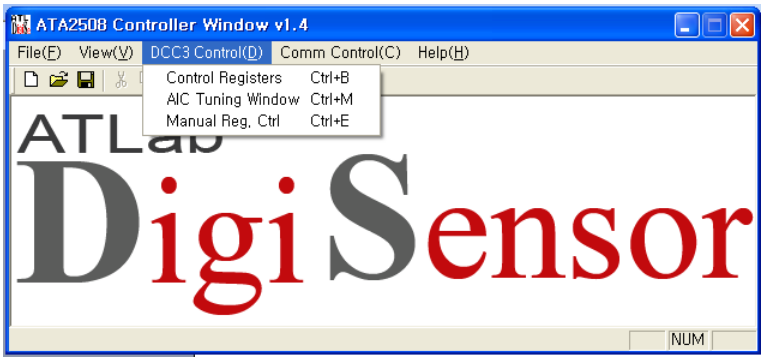


Figure 3-17 DCC3 Control Menu List

3.2.2 Reading and Writing Registers

If you select "DCC3 control" in the menu bar, three sub commands will pop up as shown in Figure 3-17. If you choose "Control Registers" in the sub commands, the dialog window with four tabs will appear as shown in Figure 3-18. Each page has registers with similar functions or

contiguous addresses for the convenience. All commands in the menu are connected to dialog boxes as described before.

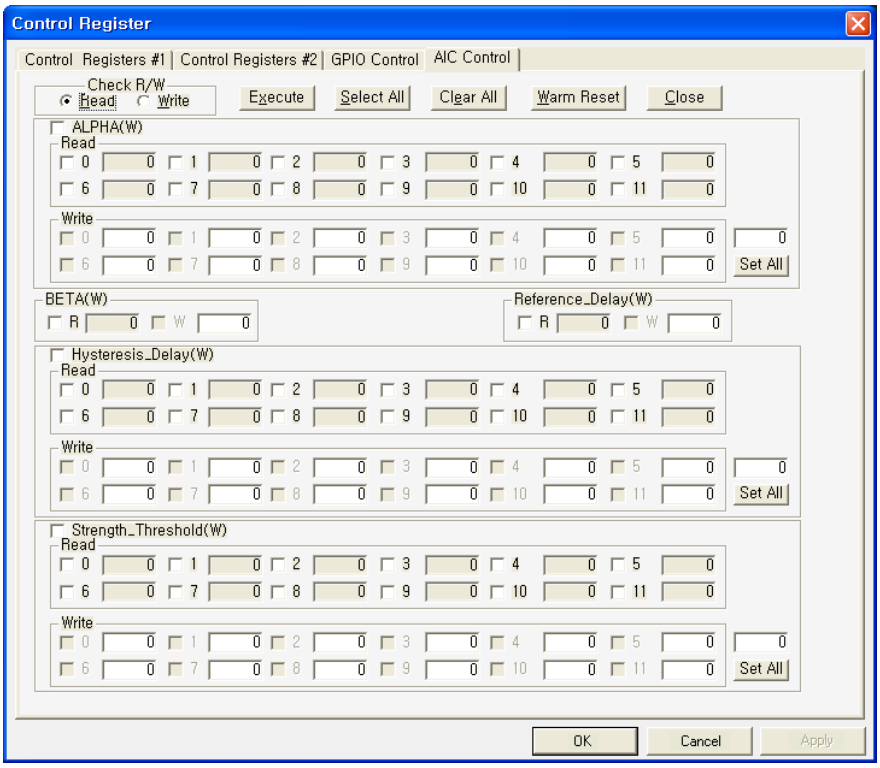


Figure 3-18 AIC Control Tab in Control Register Dialog Box

For example, Figure 3-18 shows registers regarding AIC parameter setup such as ALPHA, BETA and Reference Delay, etc.



Figure 3-19 Control Buttons of Registers

When you need to read the values of registers, select "Read" in "Check R/W" as shown in Figure 3-19. After that, activate check boxes of registers you want to read. For example, if you want to read ALPHA registers, check the rectangular box near by ALPHA. You can recognize V mark in the box as shown in Figure 3-20.

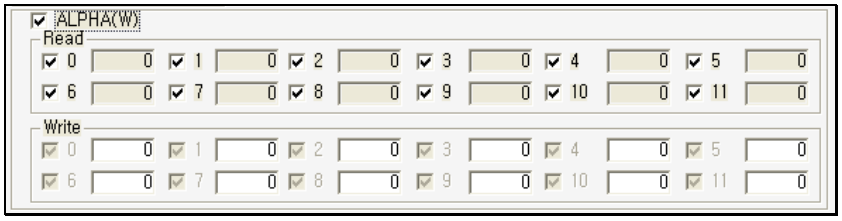


Figure 3-20 V mark after checking registers

After finishing register selection, read the values of registers by pressing "Execute" button. The values to be read appear in the read area below ALPHA.

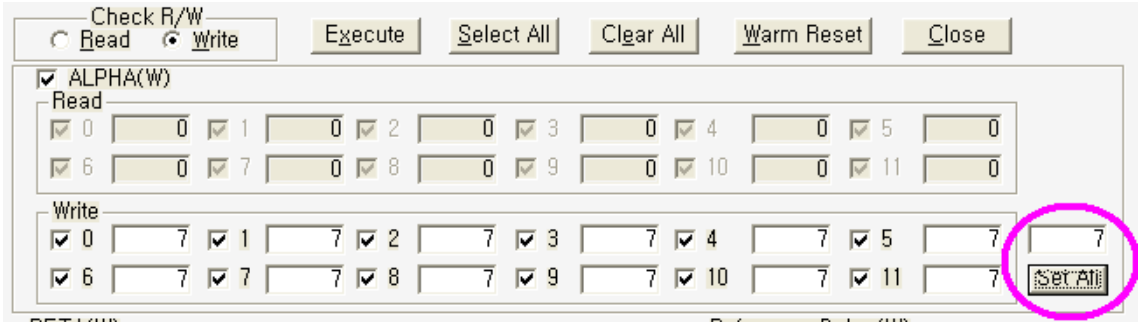


Figure 3-21 Writing values to registers

In the case of write action, check "Write" in "Check R/W", and select registers to write by checking the check boxes near by the registers, then write appropriate values in white boxes in write area of registers. You can enter either hex or decimal values according to notation of HEX or DEC near by write area. Important tip before writing registers is to read them first when you are not sure about the current status of registers to be written. For your convenience, to enter same values into the registers, the edit box and "Set All" button are located in the right corner as shown in Fig 3-21.

For example, when set alpha values as the value of 7 for all twelve channels, simply enter 7 and press 'Set All' button. Then all twelve ALPHA write registers have the same value of 7 and check boxes nearby are also checked and Write window is highlighted too. You can press 'Execute' button to transmit these data to the ATA2508.

Sometimes you need to try two or three times to read registers due to USB communication failure. If you still keep reading unexpected values from registers, check I^2C cable connection between USB B/D and Touch B/D. If I^2C cable connection is fine, check USB cable connection between USB B/D and PC. If you run "Execute" when USB cable connection is wrong or PC does not acknowledge USB B/D properly, the error message shown in Figure 3-22 appears. The way to solve this problem is to close tuning viewer program, and unplug USB cable from USB B/D. Then, run the program again after reconnecting USB cable.



Figure 3-22 Error Message caused by USB Failure.

The other control buttons shown in Figure 3-23 are described hereafter. "Select All" and "Clear All" buttons are used to select or to clear all registers in the dialog box, respectively. "Cold reset" and "Warm reset" are used to initialize ATA2508. The main difference between them is that "Cold reset" initializes all registers with default values before turning on the power, otherwise "Warm reset" maintains the current values of registers. Therefore, you can use "Warm reset" when you need to change register values and apply them right away because some registers need "Reset" for new register value updates. "Close" button is used to close current dialog box as implied.

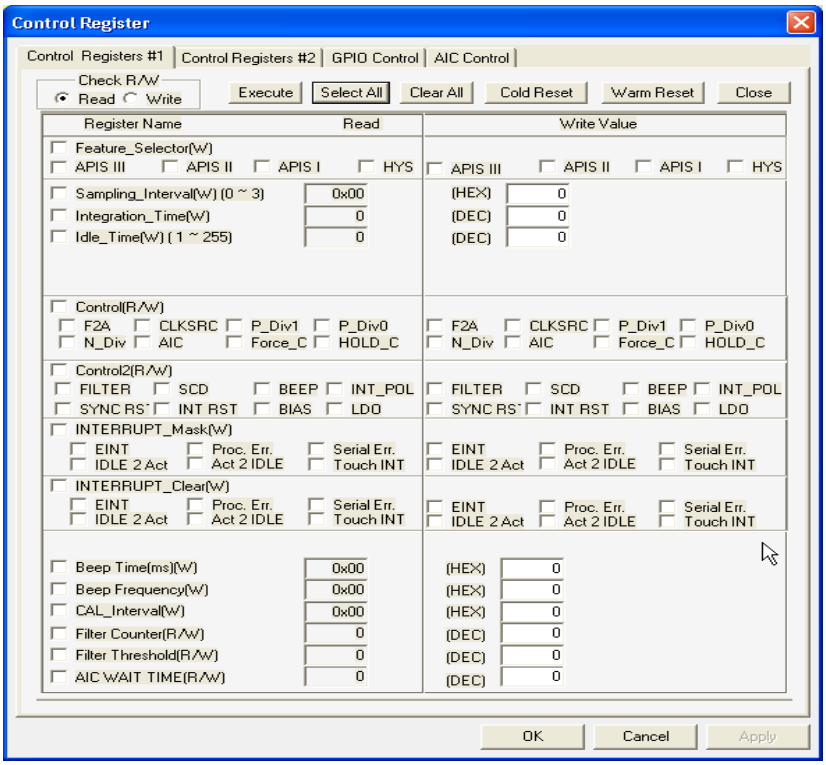


Figure 3-23 the Window of Control Registers, Page 1

The functions of all registers are explained more detail in chapter "3.5 Register Table." This chapter briefly describes each page view. Figure 3-23 gathers registers regarding Feature Selection, Control of Features, Interrupt Mask, Interrupt Clear and other important control features.

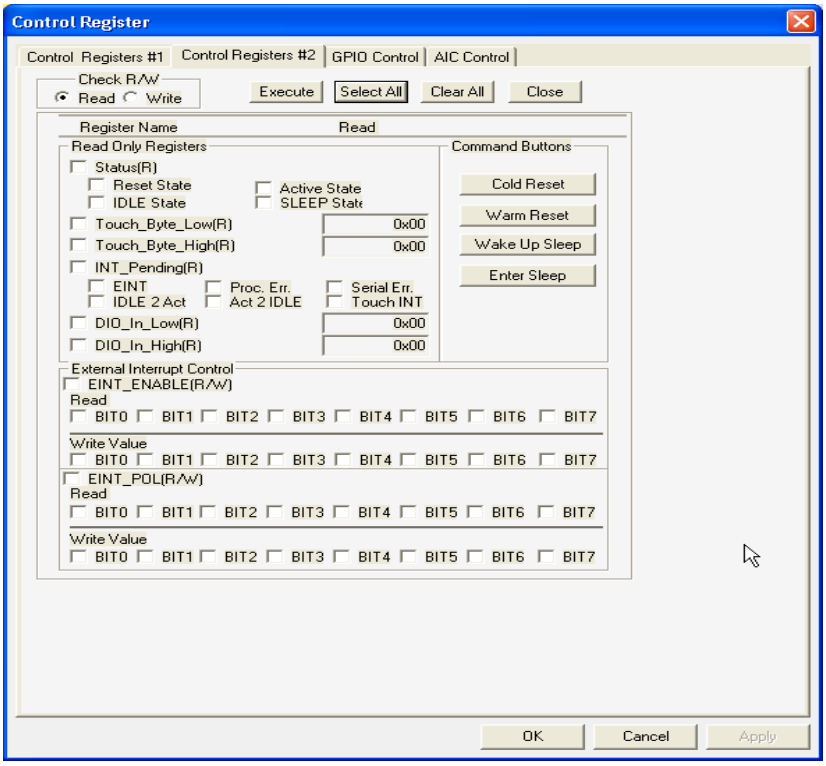


Figure 3-24 The Window of Control Registers, Page 2

Figure 3-24 includes registers regarding Status of ATA2508, Contact of Touch Pad, Interrupt Pending, External Interrupt Control, and External Interrupt Polarity.

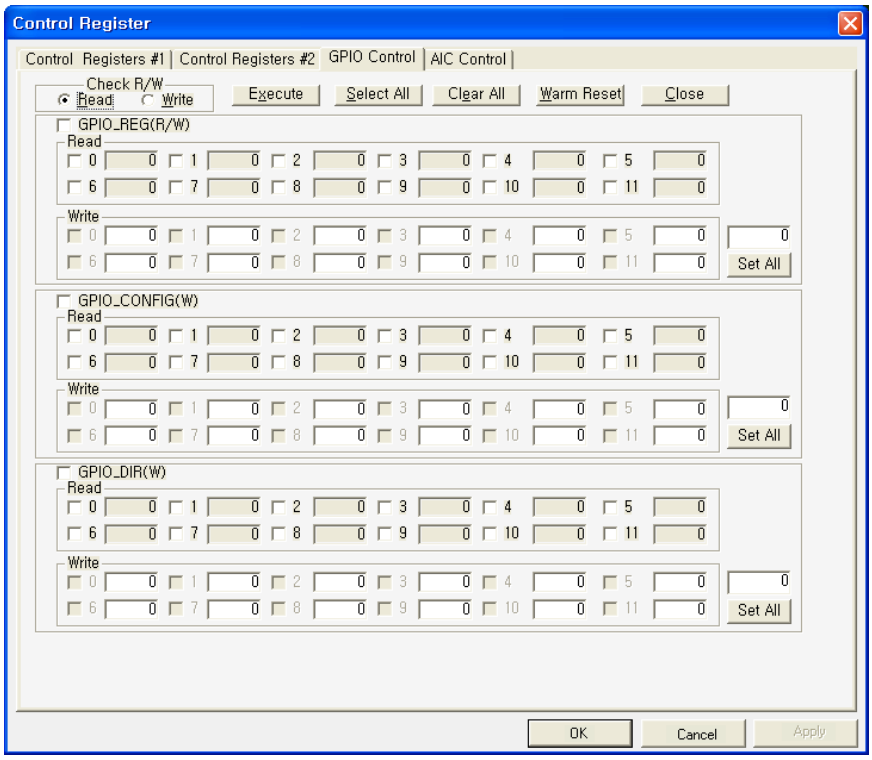


Figure 3-25 The Window of GPIO Control

Figure 3-25 contains registers regarding GPIO control when DIO is configured as GPIO.

3.2.3 Monitoring and Tuning Touch Module

When new PCB design is required for an ATA2508 application, sensitivity tuning is fundamental to achieve stable operation. This tuning activity is accomplished by AIC Tuning window as shown in Figure 3-26. This window includes ALPHA, BETA, and Reference Delay registers for AIC operation. It is also composed of touch view to detect human touch and strength view to monitor accumulated touch data for a certain period of time. If you click "Monitor" button to watch operating condition, it turns to "Stop" button indicating that monitoring is activated. Once monitoring is activated, DN_TH and SET_DN_TH display the graph of impedance variation of each channel from 00 to 11. Blue colored bar and red colored bar signify sensor input impedance and reference input impedance respectively. When you want to stop monitoring, click "Stop" button, then it goes back to "Monitor" button indicating that it was stopped. Once you start monitoring, you can't access registers, hence need to stop it before accessing registers.

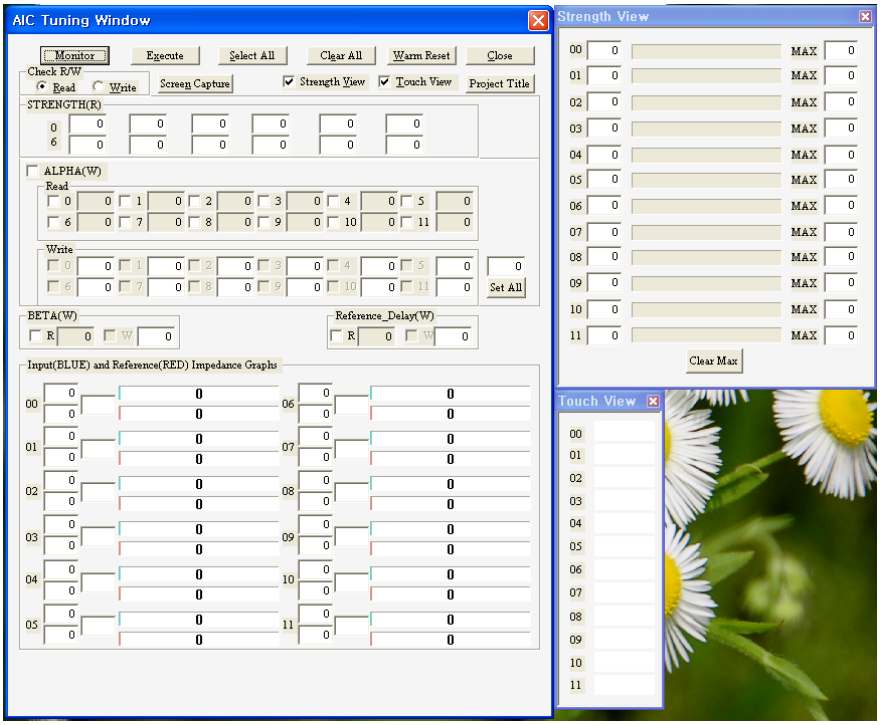


Figure 3-26 AIC Tuning Window

If you press "Screen Capture" button, then you can store current AIC Tuning Window Screen as a bitmap file where you want to save. The file name will be automatically generated with present date and time.

Figure 3-27 shows project information and the result is in next picture.

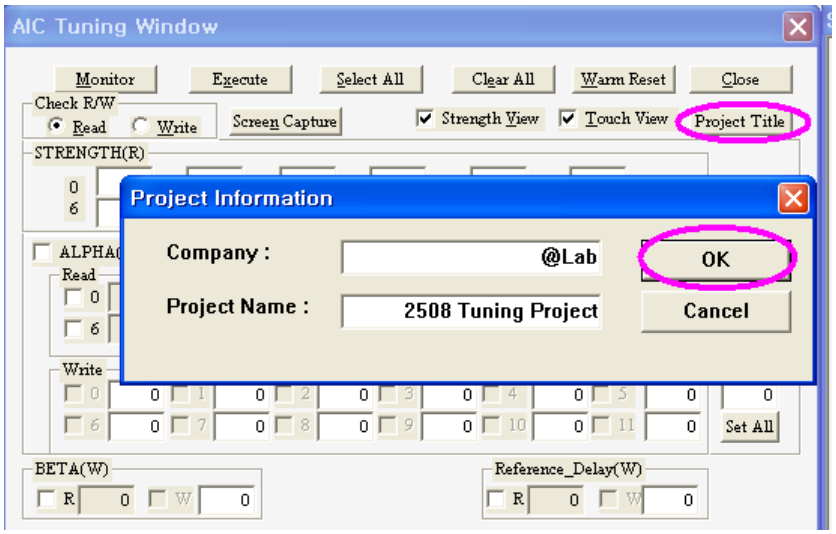


Figure 3-27 Project and Company name window

Figure 3-28 shows the company and project names in the title bar of the dialog window.

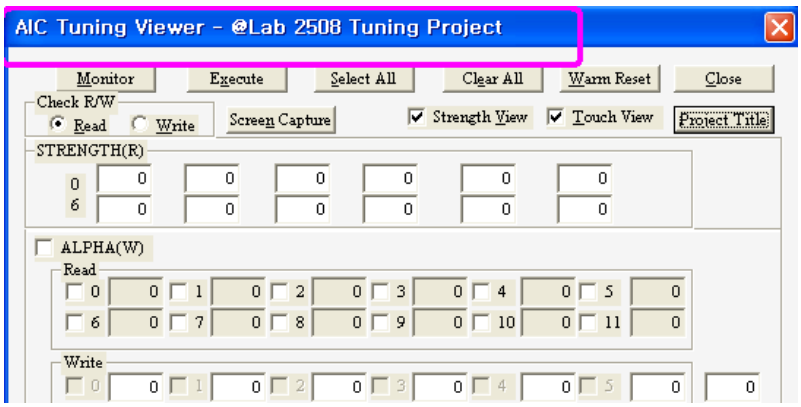


Figure 3-28 Project Name set finished

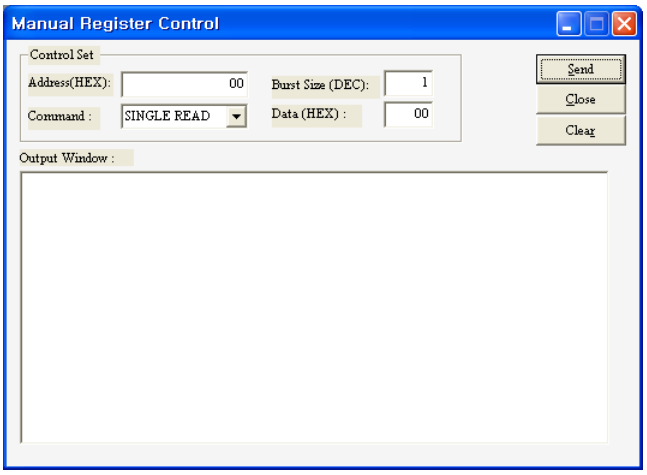


Figure 3-29 Manual Register Control Window

You can access registers directly without using control windows through Manual Register Control Window. It supports I^2C single and burst read and write. The data size must be bigger than 1 byte to use burst mode. You can also access other I^2C device through this window by connecting it to USB B/D. Before accessing other I^2C device, you need to assign its I^2C address (denoted as chip ID hereafter) in MCU Configuration Window as shown in Figure 3-30.

3.2.4 MCU Configuration Window

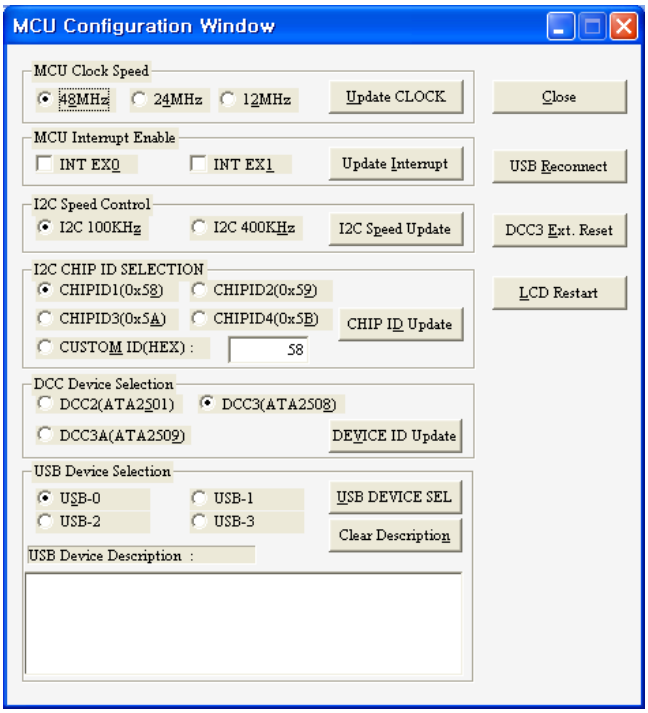


Figure 3-30 MCU Configuration Window

You can control MCU in USB B/D through MCU Configuration Window as shown in Figure 3-30. It is located at the menu, Comm_Control -> MCU Control. As stated before, since MCU in USB B/D is able to control four ATA2508 chips simultaneously, it could be manipulated through this window. The result of control by this window is reflected to LCD panel in USB B/D. Four ATA2508 chips may exist in one Touch B/D or one USB B/D can accommodate four Touch B/Ds containing each ATA2508.

Let's assume that two ATA2508's are connected to the Touch B/D. You need to memorize their pre-assigned Chip IDs described in the chapter 2.1.2.2 to select the corresponding chip you want to access at the area of I^2C CHIP ID SELECTION as shown in Figure 3-31. After choosing the chip, click "CHIP ID Update" button to inform MCU in USB B/D. From now on, the chip which has same chip ID reacts from MCU commands.

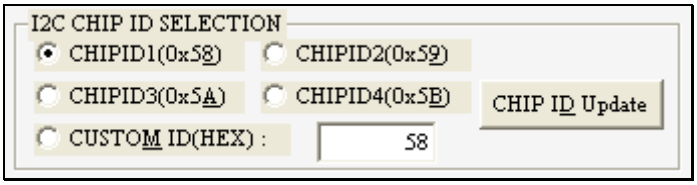


Figure 3-31 I^2C CHIP ID Selection

If you want to control other chip, you need to follow the procedure again described above. When you select different chip ID, it is shown on the LCD of USB B/D.



Figure 3-32 MCU Interrupt Enable

Another important function in MCU Configuration Window is to change MCU clock speed from 12MHz to 48MHz. It also contains interrupt control function as shown in Figure 3-32. TINT and GINT are handled by INT EX0 and INT EX1 respectively and the polarity of MCU interrupt pins is negative edge triggered. So, the polarity of ATA2508 interrupts must correspond with that of MCU interrupt pins. This can be done by checking INT_POL bit of Control2 register described at the chapter 3.5.22.

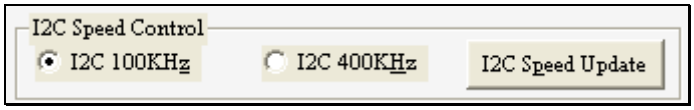


Figure 3-33 I^2C Speed Control

Figure 3-33 shows the area of I^2C Speed Control where you can select 100 KHz or high speed 400 KHz.

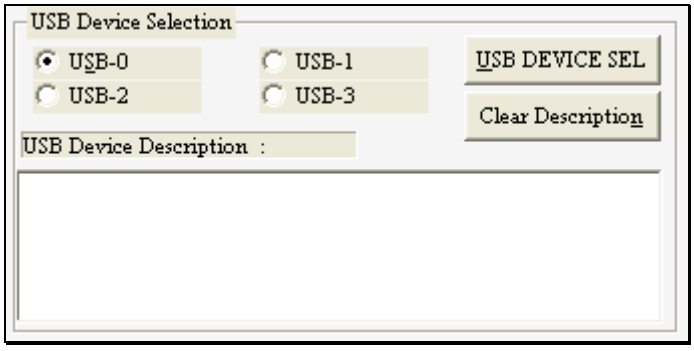


Figure3-34 USB Device Selection

In the area of USB Device Selection, you can control up to four USB B/Ds at the same time. It means that you can control up to sixteen ATA2508s simultaneously because each USB B/D can handle four ATA2508s. In order to use multiple USB B/Ds, you also need to open multiple

tuning viewer programs. Connect USB B/Ds to PC first and open tuning viewer program one by one. Whenever you open a tuning viewer program, you need to assign USB device ID from 0 to 3 and click "USB DEVICE SEL" button to inform MCU of corresponding USB B/D. The result will appear on USB Device Description area and also Chip ID (I^2C address) will be shown in the LCD of USB B/D. But, in the case of having same USB descriptions, you can hardly distinguish which one is under control. The easy way to avoid this is that you can activate "Monitor" button in AIC Tuning Window by watching the change of impedance or touch information as shown in Figure 3-25. If USB B/D is not connected to any USB ID from 0 to 3, error can be occurred. When it happens, you should close the tuning viewer program and re-start.

Four USB B/Ds can be connected to PC as shown in Figure 3-35.

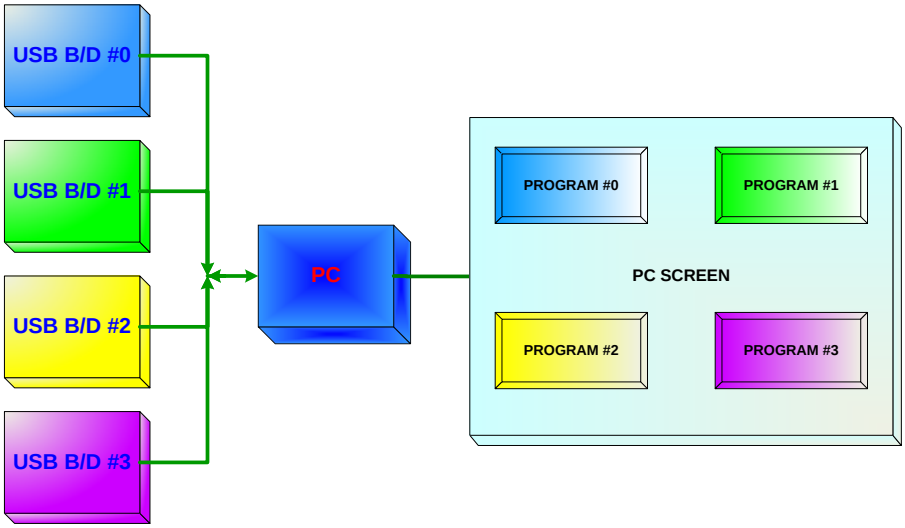


Figure 3-35 Example of Multiple Connections of USB B/Ds

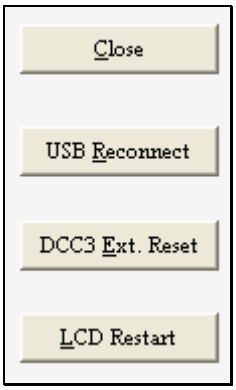


Figure 3-36 Side Buttons in MCU Configuration Window

Figure 3-36 shows the buttons located at right side area in MCU Configuration Window. "Close" button is used to close this window. "USB Reconnect" button is used when the status

of USB communication is bad. At this time you don't have to unplug the USB cable. After about 5 seconds, USB connection is re-established. If it is still in bad condition, close all tuning viewer programs and unplug USB B/D and try again. "DCC3 Ext. Reset" button enables reset to ATA2508 if its reset pin is connected to MCU. Please refer to Figure 2-3, Pin Assignment of I^2C Bus Cable in chapter 2.1.2.2 for reset connection. According to the Figure 2-3, you can connect the reset pin of ATA2508 to the eighth pin of this connector. Even if you can reset ATA2508 by reset commands, Cold Reset or Warm Reset via I^2C bus, this reset configuration is useful when I^2C is not properly controlled. "LCD Restart" button may initialize LCD when LCD operation is abnormal such as no characters shown.

3.2.5 Data Saving Window

Figure 3-37 and Figure 3-38 show the location of Data Save Window and how to save strength data of sensor input channel (blue bar) and reference channel (red bar) respectively.

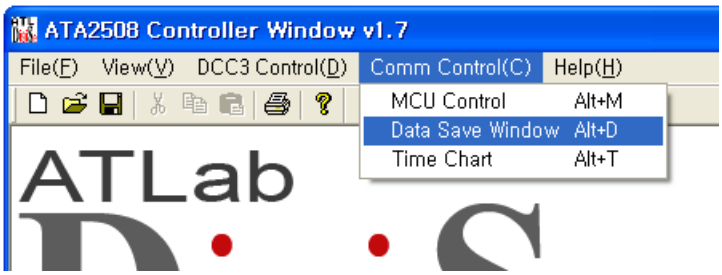


Figure 3-37 The Location of Data Save Window

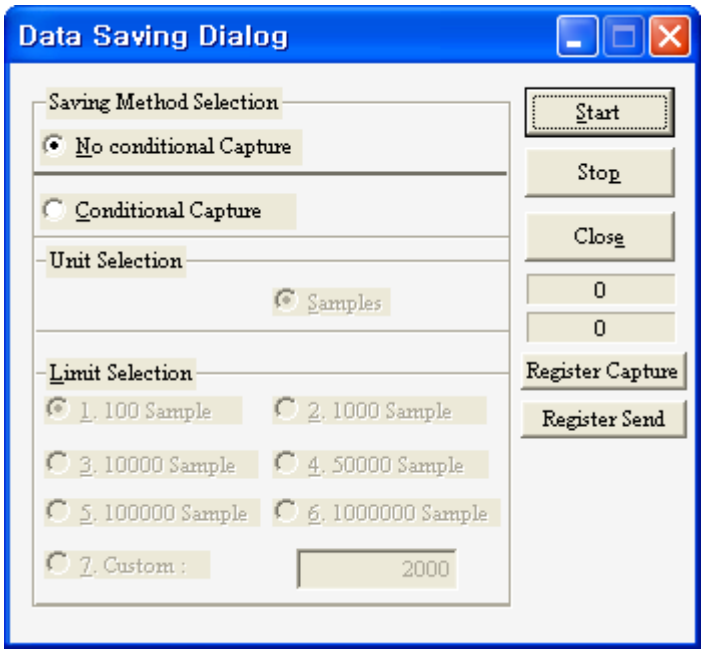


Figure 3-38 Data Save Window

There are two methods to save data. One is to save data without timing constraints until user stops saving operation by activating "No conditional Capture". The other is to save data with timing constraints by activating "Conditional Capture". Once you choose "Conditional Capture", the rest areas are activated. In the area of Limit Selection, you can select the number of samples to save in a given period. For example, if you choose "100 Samples", it only saves 100 samples and stops. Also you can designate the number of samples to save in the option 7, "Custom". After finishing save condition, save operation starts by pressing "Start" button. When you press "Start" button, AIC Tuning Window and File Save Window appear. In File Save Window, you can enter the location and file name as you want. Default name includes date, time and the number of samples to save.

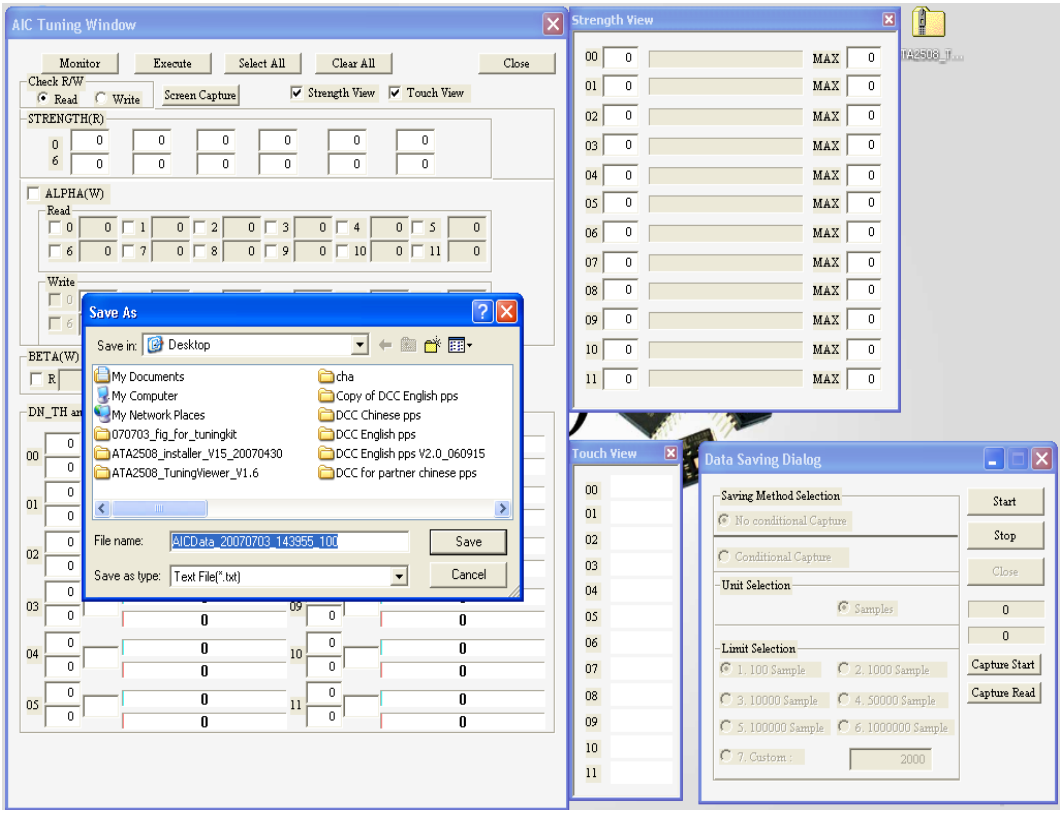


Figure 3-39 Data Save Window Start

Another function to save register data is 'Register Capture'. This function enables us to save the current values of 63 registers and to make a file. Company name and project title can be included using the functions of AIC Tuning Window.

Figure 3-40 below shows the picture just before pressing Register Capture button in Data Saving dialog.

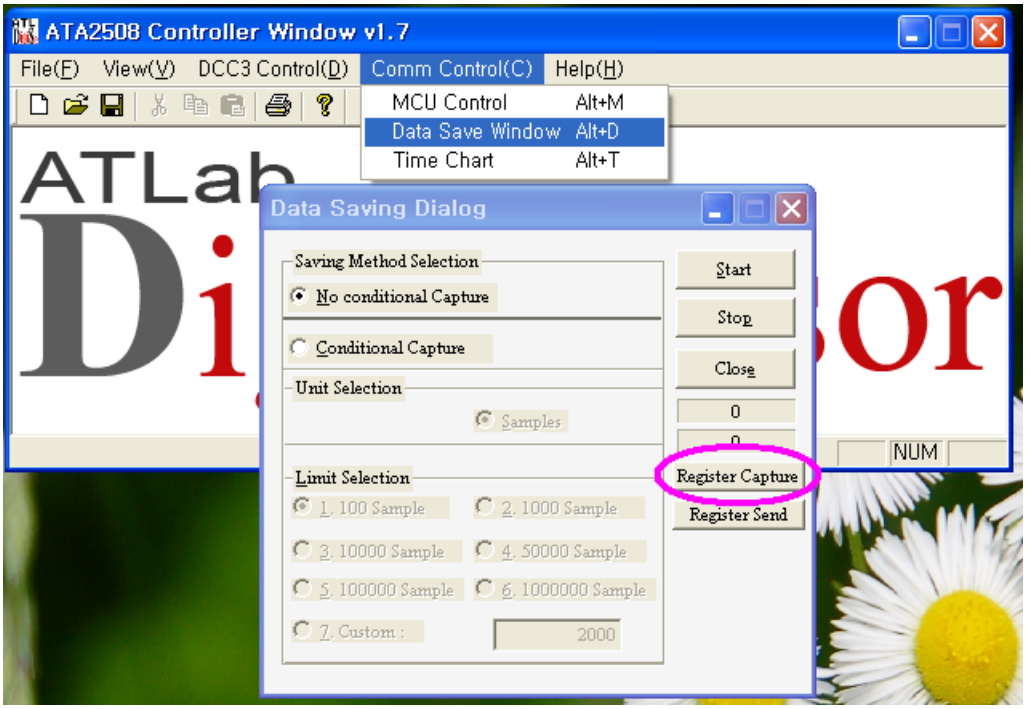


Figure 3-40 Register Capture window

Figure 3-41 will be shown when 'Register Capture' is pressed. You can confirm data saving before start it by pressing 'OK'. If I^2C communications are being processed, stop all communications and then use this function.

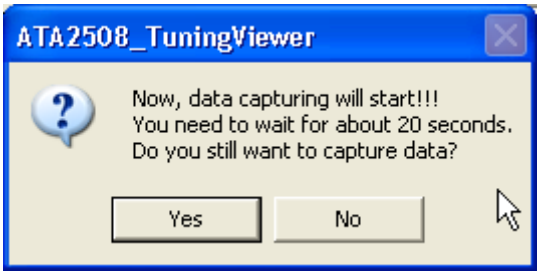


Figure 3-41 Confirm again

As soon as you select 'OK', all objects in dialog window are deactivated as shown in Figure 3-42 and a few seconds (about 10 ~20 seconds) later it will become active again.

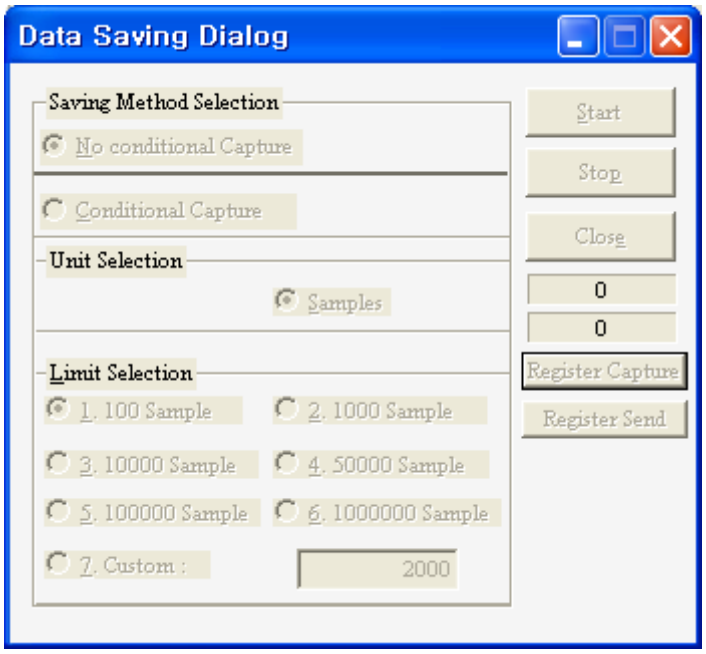


Figure 3-42 all objects are disabled while data capturing.

The saving result is displayed as shown in Figure 3-43.

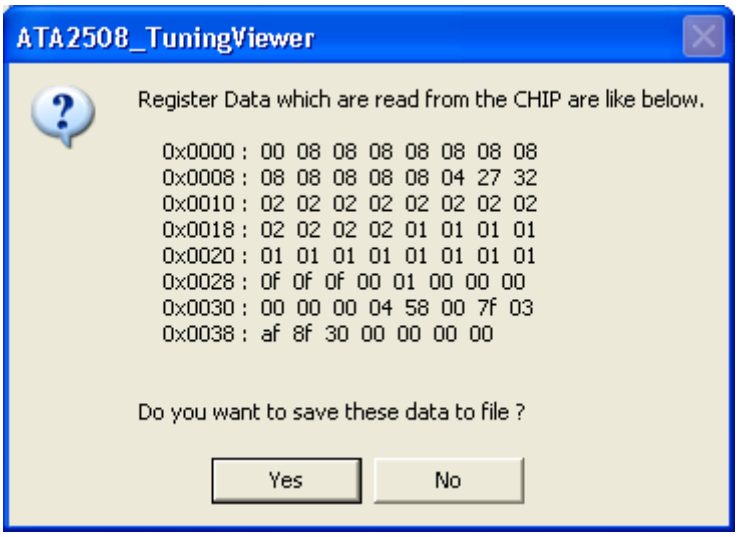


Figure 3-43 the window after capture is done.

Four-digit hexadecimal number is the start address of the register and the value after ':' is the current data stored in the corresponding register. One line displays 8 registers. It saves total 63 registers and the registers whose addresses are beyond 0x50 are read only or write only registers. Therefore, those registers are not included for saving operation. In this dialog

window you can select 'YES' to save these data as a file. AT this time the project name in AIC Tuning Window is also saved in the file.

Figure 3-44 shows where and how the file is created. The default file name automatically includes date and time to avoid unexpected duplication.

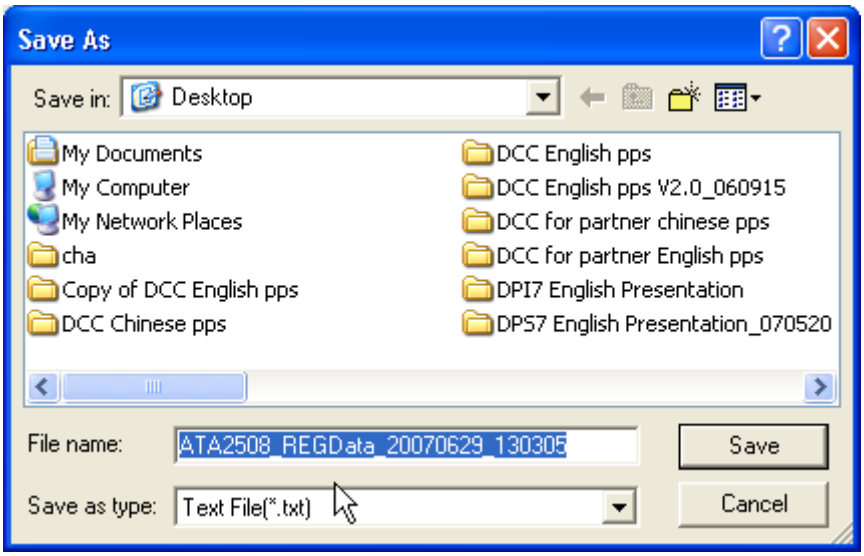


Figure 3-44 Set file path and file name

When you press 'Save', it starts saving and finishes.

And we will explain how to retrieve data stored in the register file. You can retrieve a saved file by selecting 'Register Send' button.

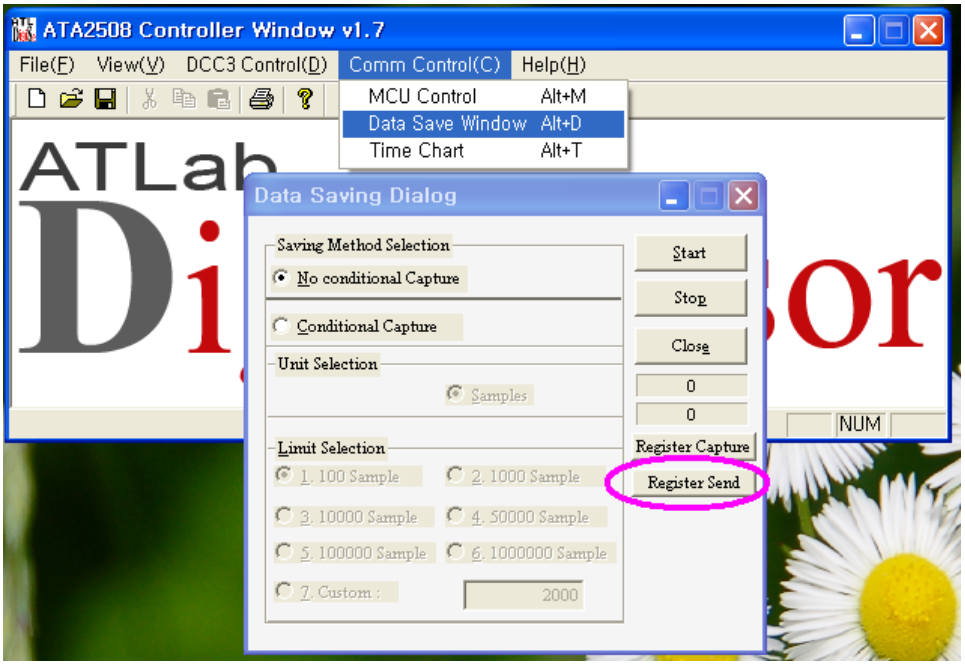


Figure 3-45 Register Send button

You can select a file you want to open through the file open window.

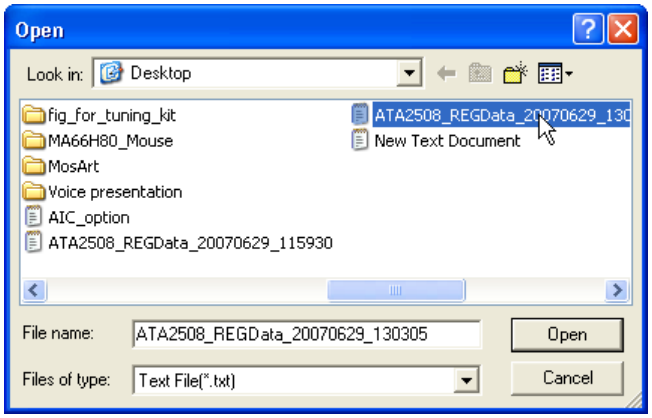


Figure 3-46 Select a file to load

After you select a file, press 'OPEN'. Then a dialog pops up and asks to send data to the chip.

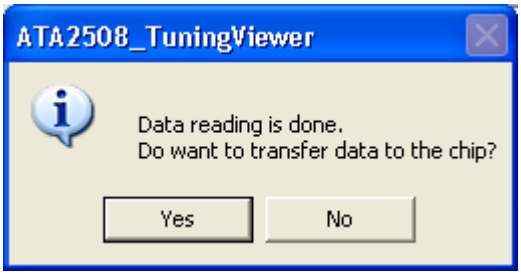


Figure 3-47 Confirm to upload register data to the chip

If you select "YES," it sends the data to the chip through I^2C. It will take about 2 ~ 5 seconds.

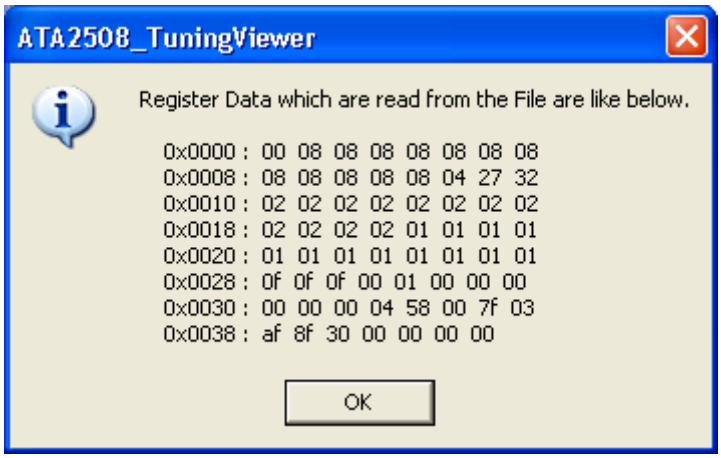


Figure 3-48 the window to show register values after data transfer is done

Figure 3-48 will be displayed after data transmission is done. You can confirm whether data transmission was successful by reading registers in the Register Control Window.

Lastly, we have time chart window that helps to calculate some AIC time related registers. As you see in the figure 3-49, the INTERNAL OSC is 1.6MHz (= 1600 KHz). And there are AIC timing related register below the chart. If you want to see the actual timing, you can type in the values on every edit boxes. And press "Update" button and you will see the changes.

In the left side of the chart, "P_Div" has 4 cases by 2 bit information, and "N_Div" has 1 bit, so you can see all combinations at once. And you can change the Register Set values below the chart. All the values are decimal. For example, if you want to see the actual IDLE TIME(= 15), drag the scroll bar of the chart, then you can see the idle time.

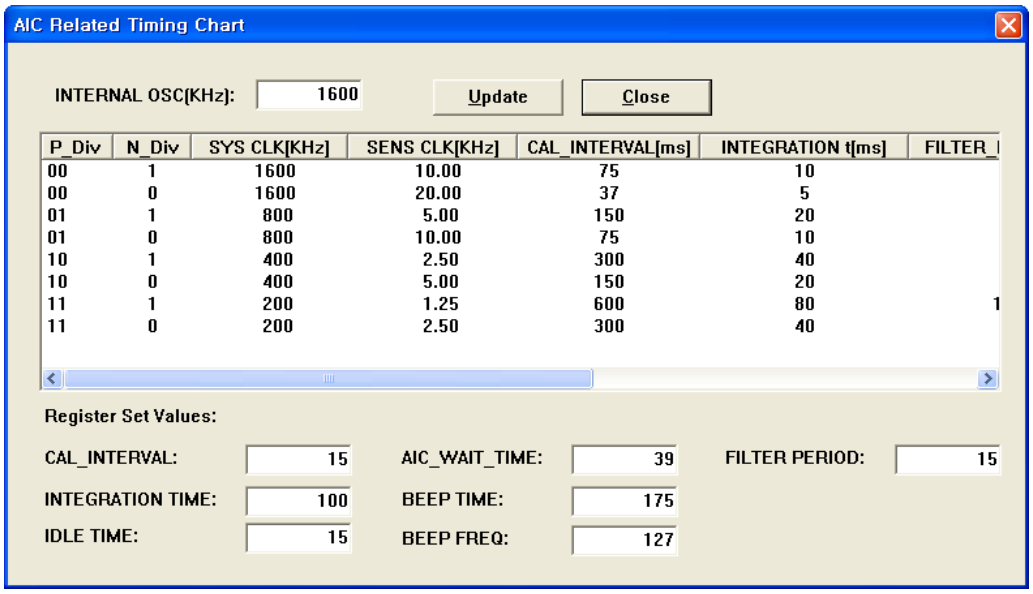


Figure 3-49 AIC Timing Chart Window

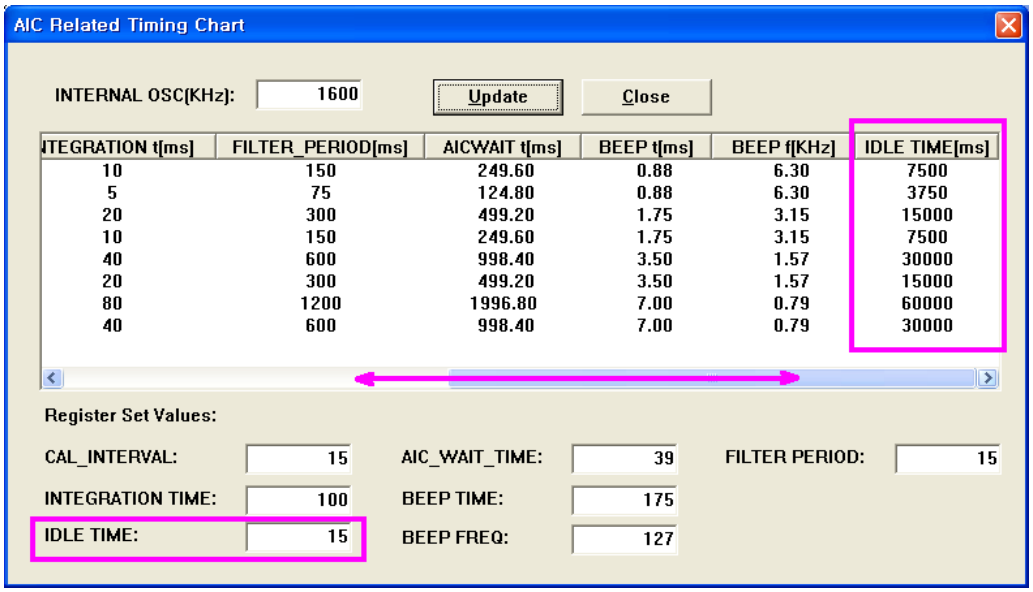


Figure 3-50 AIC Timing Chart Window

Basic operation about Tuning Viewer Program was explained so far. When you are not clear on these explanations, please contact ATLab Call Center, dcc@atlab.co.kr for further assistance.

3.3 Firmware Coding Guide

3.3.1 Initialization

In order to utilize ATA2508 (denoted as "sensor" hereafter) in mobile phone, MP3 or other I^2C Host device its registers must be initialized through I^2C interface as soon as power is supplied. For example, if the system needs to initialize the sensor as APIS mode, the register whose name and address are Feature Select and 0x00 (hex-decimal) must be initialized with an appropriate value. The registers of the sensor are explained more detail in the next chapter. After initialization, warm reset must follow to apply current setting right away and you can confirm its operation result by observing LCD or Tuning Viewer Program. Do not apply cold reset which will initialize all registers of the sensor with default values.

The registers which have similar characteristics have consecutive addresses to use I^2C burst mode. It will reduce I^2C access time significantly. For example, ALPHA registers0~11 starts the address 0x01 and ends 0x1C. Let's see the difference between single mode and burst mode by programming ALPHA registers.

Example Code 1 using ANSI-C code:

I2C_Write(BYTE chip_id, BYTE address, BYTE data, BYTE size); // I2C Write function

Let's assume that I^2C write function is above. Then we can program two ways.

Case 1: Single mode writing to ALPHA0 (0x01) through ALPHA11 (0x0C)

```
int chip_id = 0x58;
int i;
for (i=1; i<=12; i++)
{
I2C_Write(chip_id, i, 10, 1);          // i = address, 10 = data, 1 = data size
}
```

For the code example above, single I^2C write function is executed 12 times. After this program execution, all ALPHA registers have same value of 10. In this case data size is defined as one.

Case 2: Burst mode writing to ALPHA0 (0x01) through ALPHA11 (0x0C)

```
Int chip_id = 0x58;
I2C_Write(chip_id, 1,10,12);          // 1 = start address, 10=data, 12=data size
```

This code is an example of burst write. All ALPHA registers have same value of 10, but data size is defined as 12.

From the code examples above, if you can use burst mode operation on the consecutive addressed registers of ATA2508 as many as possible, it will reduce the access time of I^2C. In chapter 4, Register Map Summary, all registers are listed to review and Figure 3-51 is a flowchart of I^2C communication when host MCU initializes ATA2508 during boot up.

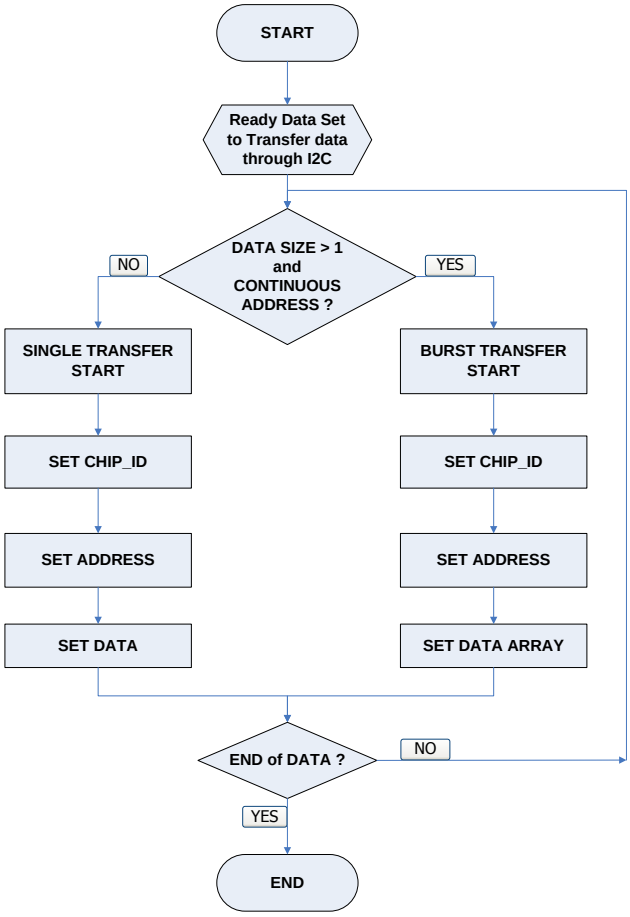


Figure 3-51 Initializations of registers through I^2C by Host MCU during boot up

3.3.2 APIS Touch Output

In order to read touch output of ATA2508 you need to access Touch_Byte_L (0x75) and Touch_Byte_H(0x76). Touch_Byte_L stores touch data of S0 through S7 in the order of LSB first, it means Touch_Byte_L[0] stores S0 and Touch_Byte_L[7] stores S7 respectively. Touch_Byte_H stores touch data of S8 through S11. Upper 4 bits of Touch_Byte_H are reserved and always 0.

If APIS mode is not defined, all touch data without APIS filtering are transmitted to MCU. But if your application is numeric keypad, you can use APIS mode 1 to get the strongest output and filter out all week touch inputs. It will reduce the burden of host computing time.

You can adjust APIS filtering capability at Integration Time Register and Strength Threshold Register. If you increase integration time, you can get higher quality of APIS but response time will be slower. Similarly, if you assign higher value to Strength Threshold Register, you can filter out week inputs more seriously but sometimes you can not detect touch inputs. Therefore, you need to compromise the level of APIS filtering.

Another important thing is you should be very cautious when you define the values of Integration Time Register and Strength Threshold Register. The value of Integration Time must be greater than the value of Strength Threshold. For example, if you assign the value of Integration time to 100, the value of Strength Threshold can never reach more than 100. So, you have to assign Strength Threshold value to less than 100. To define the value of Strength Threshold depends on the condition of applications such as cover material, cover thickness and pad size etc.

There are three kinds of APIS mode.

a. APIS mode I: reports the strongest output only.
b. APIS mode II: reports all outputs to overcome the value of Strength Threshold Register
c. APIS mode III: reports two strongest outputs for multi-touch application

All three cases are described by pictures below. The red colored bars are the output.

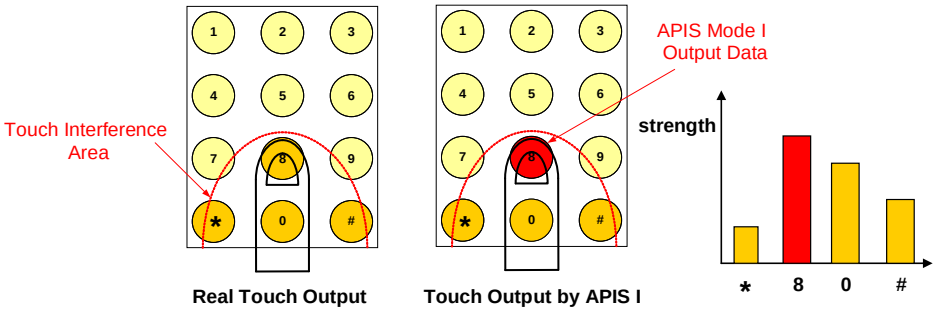


Figure 3-52 Operation of APIS mode I

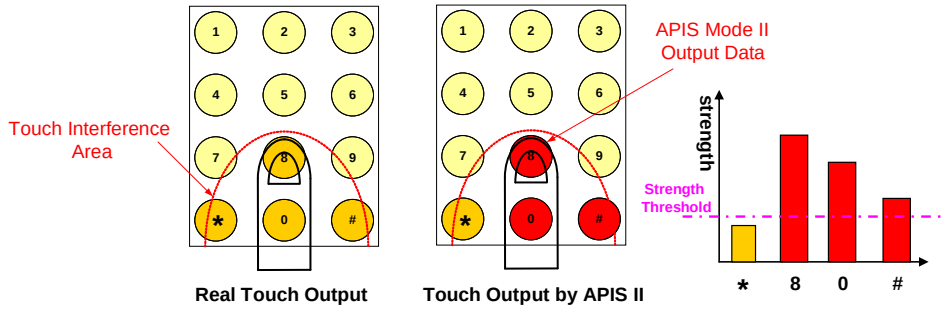


Figure 3-53 Operation of APIS mode II

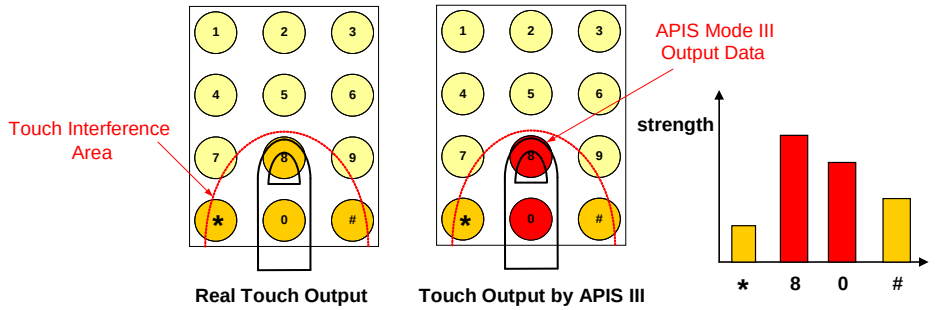


Figure 3-54 Operation of APIS mode III

3.3.3 Non-APIS Touch Output

When you are not satisfied with on-chip filtering function, you can turn off APIS filtering to get un-filtered sensor data. With these data you can apply your own filtering algorithms or functions by MCU programming. You can adjust data rate by changing Sampling Clock Speed. But, our recommendation is not to use Non-APIS mode.

3.4 Communication Protocol of I²C bus

3.4.1 Basic Transfer Form of I²C bus

ATA2508 follows all regulations defined by I²C specification and it works as slave mode. Basic I²C communication protocol is composed of as follows; START->CHIP_ID->DATA->STOP and other various formats explained later. Since I²C is base on 8 bit serial communication, host sends 8 clocks to slave device and additional 9th clock is sent as ACK/NAK for the purpose of checking communication error. If ACK/NAK bit is "L", it means data transition was normal. After checking ACK is "L", host can send/receive another data or terminate data transition by STOP signal. On the contrary, if ACK/NAK bit is "H", abnormal data transition was occurred. Then, host must terminate transition immediately by sending STOP signal.

Generally host initiates data transition by sending START signal to make SDA "L" and sends 7 bit CHIP_ID with clocks. CHIP_ID is used to designate which device must react to current host command. 8th bit data indicates Read/Write operation and 9th bit is ACK/NAK. Finally, host will terminate data transition by sending STOP signal to make both SDA and SCL "H". Therefore, SCL (clock line) and SDA (data line) remain in HIGH during none data transition.

The basic I²C data transition is shown in the Figure 3-55, 3-56, and 3-57.

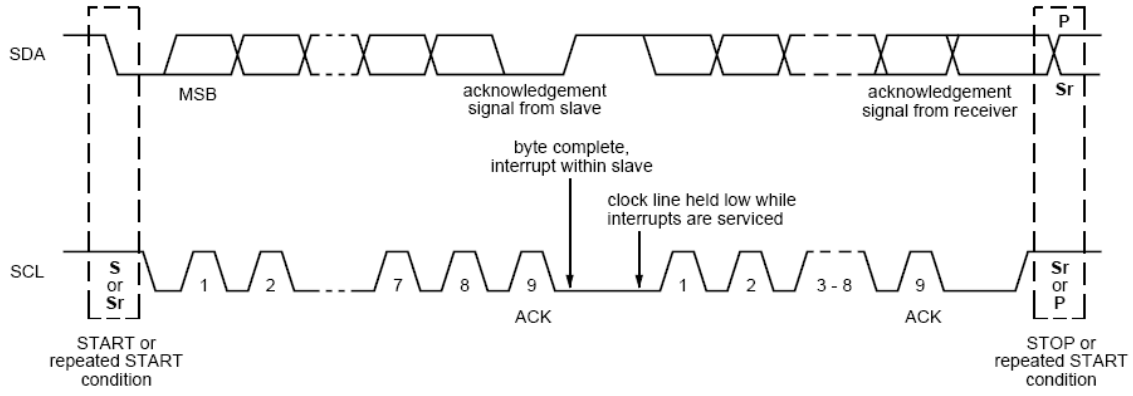


Figure 3-55 Basic Transfer form of I²C Bus

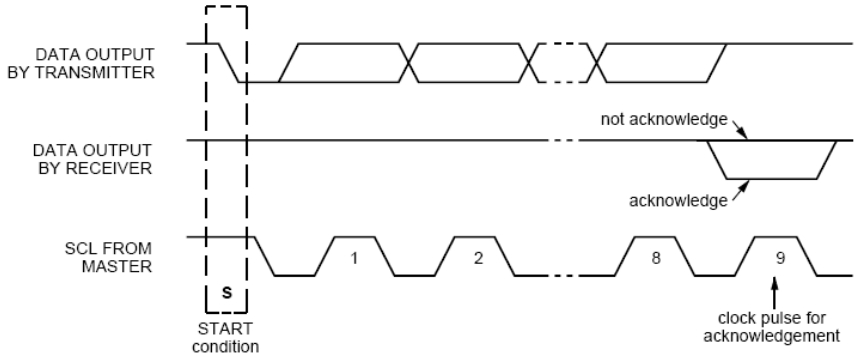


Figure 3-56 9th bit ACK/NAK waveform

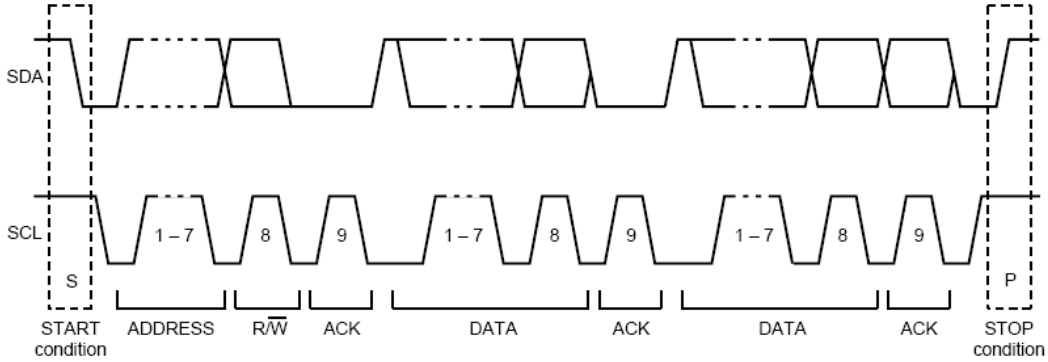


Figure 3-57 Continuous Data Transfer

3.4.2 Single Read Mode

ATA2508 has more than 100 internal registers which consist of read/write registers, read only registers and write only registers. All these registers can be accessed by I^2C interface. Since all registers have unique addresses, host must send write operation first to designate which registers are read. Therefore, data format is as follows;

START -> CHIP ID (7) ->WRITE (1) ->ACK (1) ->Register Address (8) -> ACK (1) -> Repeated START -> CHIP ID (7) -> READ (1) -> ACK (1) -> DATA (8) ->ACK (1) -> STOP where black colored is driven by host, red colored is driven by ATA2508 as slave device, and (n) means number of bits.

Figure 3-58 describes the operation of single read mode.

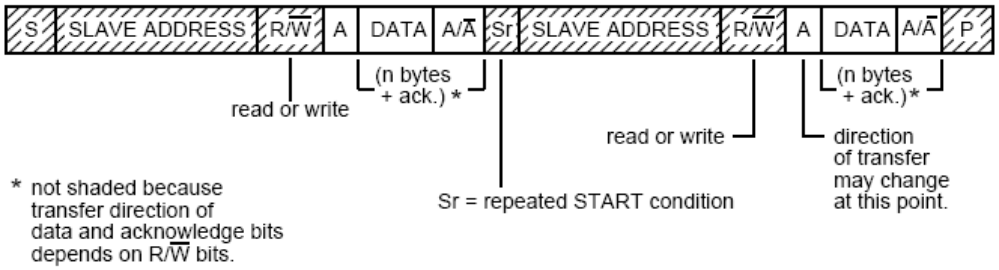


Figure 3-58 Data Format of Single Read Mode using Repeated START

3.4.3 Single Write Mode

Single Write Mode is not as complicated as Single Read Mode because data direction is always from host to ATA2508. The data format is as follows;

START -> CHIP ID (7) -> WRITE (1) -> ACK (1) -> Register Address (8) -> ACK (1) -> DATA (8) -> ACK (1) -> STOP.

Figure 3-59 describes Single Write Mode.

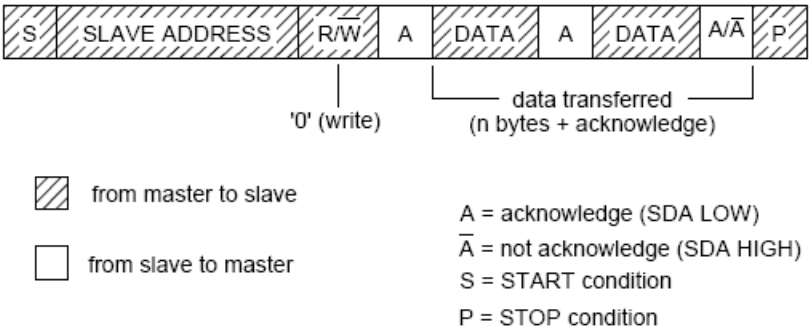


Figure 3-59 Data Format of Single Write Mode

3.4.4 Burst Read Mode

In order to accelerate data transition speed, Burst Mode operation is used. Important thing regarding Burst Mode is the addresses of registers to access must be consecutive. Register address is automatically incremented. In the case of Burst Read operation data transition format is as follow;

START -> CHIP ID (7) -> WRITE (1) -> ACK (1) -> Start Address of register (8) -> ACK (1) -> Repeated START -> CHIP ID (7) -> READ (1) -> ACK (1) -> DATA1 (8) -> ACK (1) -> DATA2 (8) -> ACK (1) .....-> DATAn (8) -> ACK (1) -> STOP

3.4.5 Burst Write Mode

Burst Write Mode is almost same as Burst Read Mode except data direction. The data format of Burst Write Mode is as follow;

START -> CHIP ID (7) -> WRITE (1) -> ACK (1) ->Start Address of Register (8) -> ACK (1) -> DATA1 (8) -> ACK (1) -> DATA2 (8) -> ACK (1) -> DATAn (8) -> ACK (1) -> STOP

3.4.6 Configuring SLAVE ADDRESS

ATA2508 has 2 pins to configure I^2C address (CHIP ID) in 4 ways. The configuration table is described as Table 2.

B6	B5	B4	B3	B2	B1	B0	HEX
1	0	1	1	0	0	0	58

Table 2. I^2C SLAVE ADDRESS Configuration Table

I^2C ID(Hex)	ID1	ID0
0x58	0	0
0x59	0	1
0x5A	1	0
0x5B	1	1

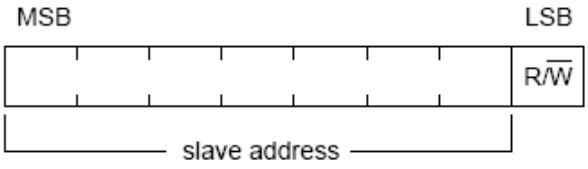


Figure 3-60 SLAVE ADDRESS with 7 bits

Since ATA2508 can be configured as 4 different CHIP IDs, host can control four ATA2508s at the same time which enables large number of sensor inputs for big screen applications.

3.4.7 Configuring flow while Host starts up

The typical flow is described below.

1. Power up. Disable Host's INT enable register that connected to ATA2508 TINT pin to prevent unexpected INT from ATA2508's TINT, because ATA2508 is not initialized yet as Host wants.
2. Initialize ATA2508 Configuration Registers through I^2C.
3. Wait for about 1 ms.
4. Transfer WARM_RESET (addr: 0xFF, data: dummy) to activate all configuration data.
5. Wait for about 1 ~ 10 ms until ATA2508 reset is completed.
6. Enable the Host's INT register again.

(*See the enclosed sample code- 5.2 Host startup sample code to initialize ATA2508)

3.4.8 Trouble shooting while I^2C interfacing

Table 3. Trouble Shooting of I^2C Interfacing

Problem	Typical Solution	Reference
ATA2508 doesn't respond to ACK	Ensure that CHIP ID is correct. Ensure that I^2C clock timing that Host reads ACK signal is correct.	
Once TINT is asserted, it is never asserted again.	TINT is self cleared by reading **TOUCH_BYTE_H**(0x76). Even though Host uses only **TOUCH_BYTE_L**(0x75), Host should read both of them.	3.5.34/35
Host can't catch TINT signal	Ensure that Host's INT edge type. ATA2508 can change the edge type of TINT such as falling or rising edge (default). See the **Control2** Register bit0 ("INT_POL")	3.5.22
Data Registers always read 0x00	Ensure that a pull-up resistor is connected to the SDA pin	2.2
Should I connect the unused Sensor Input pins to GND?	Just Open the unused sensor input pins.	
Should I connect the unused GPIO pins to GND?	If you set GPIO pins as output (default), just leave them open, otherwise, if you set them as input, then connect them to GND.	3.5.12-17

3.5 Register Table

3.5.1 Feature Select (Address: 0X00)

B7	B6	B5	B4	B3	B2	B1	B0
				APIS3	APIS2	APIS1	HYS On
				R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register for selecting the format of touch output.
- The bits of B3, B2, B1 and B0 are used. The rest are reserved.
- Caution: Just set only one bit to HIGH among B1, B2 and B3.

B0: If this bit is set to "H", the sensor output is from Hysteresis Delay Chain. If it is set to "L", the sensor output is from Non-Hysteresis Delay Chain. If you use Hysteresis Delay Chain, you can get de-bounced output hence data could be more stable.

B1: If this bit is set to "H", APIS mode 1 is activated. In APIS mode 1, the strongest sensor output is available among twelve Strength Registers (0x50~0x5B) in the give integration period. This mode is suitable for button applications to screen weakly touched buttons.

B2: If this bit is set to "H", APIS mode 2 is enabled. You need to preset Strength Threshold values in the registers (0x1C~0x27) to use APIS mode 2. The current strength values are stored in Strength Registers (0x50~0x5B). If the values of Strength Registers (0x50~0x5B) are greater than preset values of Strength Threshold values, all corresponding sensor outputs are available.

B3: If this bit is set to "H", Two strongest sensor outputs are available among twelve Strength Registers (0x50~0x5B) in the given integration period. This is called APIS mode 3. It is suitable for multi-touch applications.

※ The register' addresses are 0x1C~0x27 to set or change sensor's Strength Threshold values.
※ Current strength values can be read at Strength Registers (0x50~0x5B).

3.5.2 ALPHA 0~11 (Address: 0X01~0X0C)

B7	B6	B5	B4	B3	B2	B1	B0
	ALPHA Register						
	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	1	0	0	0

- The registers for setting sensitivity of each touch input.
- You can control sensitivity individually to set different values of S0 through S11.
- Bigger number will decrease the sensitivity and Small number will increase the sensitivity.
- The value starts from 0 and the maximum number is 99.

3.5.3 BETA (Address: 0X0D)

B7	B6	B5	B4	B3	B2	B1	B0
	BETA Register						
	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	1	0	0

- The register for setting AIC entering threshold value, BETA. It is used to detect non-touch conditions of all twelve inputs.
- Sensor checks current impedances and previous impedances of all twelve inputs before starting AIC operation. If all differences of them are lower than BETA, it starts AIC.

3.5.4 AIC_WAIT (WAIT before CALIBRATION Time) (Address: 0X0E)

B7	B6	B5	B4	B3	B2	B1	B0
AIC_WAIT Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	1	0	0	1	1	1

- The register to set AIC waiting time to stabilize AIC operation with BETA.
- AIC keeps blocked during AIC_WAIT time after all 12 inputs become non-touch condition.
- If at lease one input is touched during AIC_WAIT time, AIC_WAIT time is reloaded.
- If you assign bigger value of AIC_WAIT time, AIC needs more time to start.

AIC_WAIT Time = AIC_WAIT Register value * 64 * Clock Sensor Period

3.5.5 Reference Delay (Address: 0X0F)

B7	B6	B5	B4	B3	B2	B1	B0
	Reference Delay Register						
	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to change the value of Reference Delay Chain.
- Its range is 0 through99.
- One step is equivalent to 0.06pF. So, the maximum value of Reference delay is equal to attach 7.8pF of capacitance in the reference input.
- Warm reset is mandatory when you change this value.

3.5.6 Hysteresis Delay 0~11 (Address: 0X10~0X1B)

B7	B6	B5	B4	B3	B2	B1	B0
			Hysteresis Register				
			R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	1

- The registers for setting the values of Hysteresis Delays, S0 ~S11.
- The maximum value is 19 and always lower than corresponding ALPHA value.
- If it is set to higher than ALPHA value, the sensor output still remains in HIGH after un-touched.
- These registers are only applicable when B0 of Feature Select Register (0x00) is active.
- Since one step of delay chain is equivalent to 0.06pF, these registers can increase capacitance up to 1.2pF (20 steps).

3.5.7 Strength Threshold 0~11 (Address: 0X1C~0X27)

B7	B6	B5	B4	B3	B2	B1	B0
Strength Threshold Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	1

- The registers for setting Strength Threshold values of S0~S11.
- These registers are used for APIS mode to select touch outputs greater than Strength Threshold values in the given integration time.
- The Strength Threshold value must be set lower than Integration Time Register (0x29). Otherwise, No touch outputs are available.

3.5.8 Sampling Interval (Address: 0X28)

B7	B6	B5	B4	B3	B2	B1	B0
						Sampling Interval	
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
1	1	1	1	1	1	1	1

- The register to select Sampling Frequency
- This register is only applicable non-APIS mode (B1, B2 and B3 of Feature Select register are all zeros.).
- Only B0 and B1 are used. Others are reserved.
- System Clock Period is set by B4 and B5 of Control Register (0x33).

B[1:0] == 2'b00 -> Sampling Frequency = 1 / (System Clock Period × 500)

B[1:0] == 2'b01 -> Sampling Frequency = 1 / (System Clock Period × 1000)

B[1:0] == 2'b10 -> Sampling Frequency = 1 / (System Clock Period × 2000)

B[1:0] == 2'b11 -> Sampling Frequency = 1 / (System Clock Period × 4000)

	B[1:0] == 2'b00	B[1:0] == 2'b01	B[1:0] == 2'b10	B[1:0] == 2'b11
1.6MHz	3.2KHz	1.6KHz	800Hz	400Hz
800KHz	1.6KHz	800Hz	400Hz	200Hz
400KHz	800Hz	400Hz	200Hz	100Hz
200KHz	400Hz	200Hz	100Hz	50Hz

3.5.9 Integration Time (Address: 0X29)

B7	B6	B5	B4	B3	B2	B1	B0
Integration Time Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	1	1	1	1

- The register to set Integration Time for APIS mode.

- Period to update touch output in APIS mode is varied by the equation below.

Update Period of Touch Output = Sensor Clock Period × (Integration Time Register Value)

- The maximum value of Strength Registers (0x50~0x5B) are determined by this register. If this value is set to smaller, APIS filtering is getting worse. On the contrary, if this value is set to bigger, the response time of touch output is getting longer. For example, if sensor clock is set to 5KHz, and integration time is set to 255, the update period of touch output and Strength register is every 50msec.

3.5.10 IDLE Time (Address: 0X2A)

B7	B6	B5	B4	B3	B2	B1	B0
IDLE Time Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	1	1	1	1

- The register to determine entering time to IDLE mode after all inputs are in non-touch status..

- If sensor clock is 5KHz, IDLE time will be 1 sec.

IDLE Time = Register value X 5000 X Sensor CLK Period

SIF Setup (Address: 0X2B)

Reserved

3.5.11 MODE (Address: 0X2C)

B7	B6	B5	B4	B3	B2	B1	B0
				Mode Register			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	1

- The register to select Function Mode or TEST Modes.

- If only B0 is set to H, it is Function mode. Otherwise, it is Test Mode.

- You should not control this register.

3.5.12 GPIO REG L (Address: 0X2D)

B7	B6	B5	B4	B3	B2	B1	B0
GPIO REG L Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register for storing GPIO_LOW data

- The bits having HIGH in GPIO Configuration L Register (0x2F) outputs corresponding bits of GPIO REG L Register to DIO 0 ~ 7. (GPIO REG L: B0~B7 -> DIO 0~ DIO 7 port)

3.5.13 GPIO REG H (Address: 0X2E)

B7	B6	B5	B4	B3	B2	B1	B0
				GPIO REG H Register			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register for storing GPIO_HIGH data

- The bits having HIGH in GPIO Configuration H Register (0x30) outputs corresponding bits of GPIO REG H Register to DIO 8 ~ 11. (GPIO REG H: B0~B3 -> DIO 8~ DIO 11 port)

3.5.14 GPIO Configuration L (Address: 0X2F)

B7	B6	B5	B4	B3	B2	B1	B0
GPIO Configuration L Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine which DIO ports (DIO0~DIO7) are configured as GPIO.

- Only HIGH state bits in GPIO Configuration L Register outputs corresponding bits of GPIO REG L Register (0x2D).

3.5.15 GPIO Configuration H (Address: 0X30)

B7	B6	B5	B4	B3	B2	B1	B0
				GPIO Configuration H Register			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine which DIO ports (DIO8~DIO11) are configured as GPIO.

- Only HIGH state bits in GPIO Configuration H Register outputs corresponding bits of GPIO REG H Register (0x2E).

3.5.16 GPIO Direction L (Address: 0X31)

B7	B6	B5	B4	B3	B2	B1	B0
GPIO Direction L Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine GPIO_LOW (DIO0~DIO7) direction.

- If bit is set to HIGH, GPIO direction is set to Input, otherwise set to Output.

3.5.17 GPIO Direction H (Address: 0X32)

B7	B6	B5	B4	B3	B2	B1	B0
				GPIO Direction H Register			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine GPIO_HIGH (DIO8~DIO11) direction.

- If bit is set to HIGH, GPIO direction is set to Input, otherwise set to Output.

3.5.18 Control (Address: 0X33)

B7	B6	B5	B4	B3	B2	B1	B0
F2A	CLKSRC	P DIV[1]	P DIV[0]	N DIV	AIC	Force C	HOLD_C
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	1	0	0

B0: If B0 is set to "L", it is auto calibration mode. Otherwise, initial calibration is done during boot up and AIC is waiting until Force Cal (B1) is asserted.

B1: If the status of B0 is HIGH, and Force Cal bit is written by 1 by host, calibration is executed only once then B1 becomes LOW automatically. In other words, whenever Bi becomes HIGH, calibration is done.

B2: activates AIC function when this bit is HIGH. Otherwise AIC is OFF.

B3: changes Sensor Clock Frequency

"L": Sensor Clock Frequency is equal to System Clock /80.

"H": Sensor Clock Frequency is equal to System Clock/160.

Initial calibration time depends on Sensor Clock Period as below;

Initial Calibration Time = Sensor Clock Period X 150

Ex) Sensor Clock 5KHz setting시 = 200us X 150 = 30ms

B4~5: Internal Analog clock is 1.6MHz

B[5:4] = 2'b00 -> System Clock = Internal Analog Clock / 1 = 1.6MHz

B[5:4] = 2'b01 -> System Clock = Internal Analog Clock / 2 = 800KHz

B[5:4] = 2'b10 -> System Clock= Internal Analog Clock / 4 = 400KHz

B[5:4] = 2'b11 -> System Clock = Internal Analog Clock / 8 = 200KHz

B6: If this bit is set to "L", internal OSC is used. Otherwise, external OSC is used. Generally this bit is set to "L" in function mode.

B7: If this bit is set to HIGH, power state is always in ACTIVE mode. If power state is in IDLE mode, it becomes ACTIVE mode.

3.5.19 Interrupt Mask (Address: 0X34)

B7	B6	B5	B4	B3	B2	B1	B0
		Interrupt Mask Register					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
		0	0	0	0	0	0

- Interrupt Source Masking Register when using GINT.
- If each bit is set to HIGH, the corresponding interrupt is masked.
- Six interrupts are used as below;

B0: Touch Interrupt Source Mask Bit

B1: Active TO IDLE Interrupt Source Mask Bit

B2: IDLE TO Active Interrupt Source Mask Bit

B3: Serial Communication Error Interrupt Source Mask Bit

B4: I^2C Procedure Error Interrupt Source Mask Bit

B5: EINT Interrupt Source Mask Bit.

3.5.20 Interrupt Clear (Address: 0X35)

B7	B6	B5	B4	B3	B2	B1	B0
		Interrupt Clear Register					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- Interrupt Clear Register when using GINT
- This register is used to clear interrupt after finishing interrupt service when GINT occurred.

3.5.21 Interrupt Edge (Address: 0X36)

B7	B6	B5	B4	B3	B2	B1	B0
		Interrupt Edge Enable Register					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
1	1	1	1	1	1	1	1

- The register how to detect interrupt by level triggered or edge triggered.
- **All bits are reserved as HIGH for edge triggered.**

3.5.22 Control2 (Address: 0X37)

B7	B6	B5	B4	B3	B2	B1	B0
M RST	I RST	BGEB	LDEB	Filter En	SCD	Beep En	INT POL
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

B0: The bit to set Interrupt Polarity

If "H", GINT and TINT are falling edge, otherwise rising edge (Default).

The interrupt polarity depends on the polarity of MCU interrupt inputs.

B1: To enable Beep (TOUT Beep Frequency output) Default is "L" and beep is disabled.

B2: To disable twelve Sensor Clocks and Reference Sensor Clock for test purpose only. Even if this bit is active, Default Sensor clock and Ref Clock are still alive. Default is "L" and all clocks are enabled.

B3: To control FILTER On /Off to get more stable touch outputs. Default Setting is "L", Filter is OFF.

B4: Internal LDO On/Off Bit, Default is "L", and Internal LDO is On.

B5: To control On/Off of internal BIAS Block. Default is "L" and BIAS Block is ON. This bit must be "L" when using Internal LDO.

B6: If "L" (Default), internal Integration Count is always running in Active mode (one of power sate) during APIS mode. Otherwise, Integration Count is reset by Sampler output.

B7: To synchronize Sensor clock frequencies When using multi ATA2508s and external clock. If this bit is set to "H", you can assert RESET signal through DIO0 port but register values are not changed. Default is "L".

3.5.23 Beep Period (Address: 0X38)

B7	B6	B5	B4	B3	B2	B1	B0
Beep Period Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine Beep duration output to TOUT port when detecting the change of touch status.

Beep Period(ms) = System CLK Period × (Beep Time REG × 8)

3.5.24 Beep Frequency (Address: 0X39)

B7	B6	B5	B4	B3	B2	B1	B0
Beep Frequency Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine the frequency of Beep sound when detecting the change of touch status.

Beep Frequency (Hz) = System CLK/(Beep Frequency REG × 2)

3.5.25 Calibration Interval (Address: 0X3A)

B7	B6	B5	B4	B3	B2	B1	B0
Calibration Interval Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	1	1	0	0	0	0

- The register to set Auto Calibration Interval
- If Sensor Clock is 5KHz, calibration is done every Calibration Interval register value x 10ms.

Calibration interval = Sensor CLK Period X Register value X 50

- During APIS mode, if Calibration Interval is set by less than Integration Time, touch output is never generated because touch output is hold during Calibration.

3.5.26 EINT Enable (Address: 0X3B)

B7	B6	B5	B4	B3	B2	B1	B0
EINT Enable Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to activate interrupts when detecting data changes in DIO0~7(B0 ~B7) ports
- The interrupt is active when the corresponding bit is "H". Otherwise, disabled (Default).

3.5.27 EINT Polarity (Address: 0X3C)

B7	B6	B5	B4	B3	B2	B1	B0
EINT Polarity Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine DIO interrupts polarity.
- Rising Edge is Default ("L"), otherwise Falling Edge.

3.5.28 FILTER Period (Address: 0X3D)

B7	B6	B5	B4	B3	B2	B1	B0
FILTER Count Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- To set additional filter period to ensure more stable touch output during APIS mode and expand touch output period (Sampling period) when using high speed sensor clock.
- For example, if this register is set to 10, APIS touch output is monitored 10 times every Integration Time. And if touch state is HIGH, the corresponding Filter Counter is increased by 1 and the counter operation is done by 10 times, it will be reset and count up again.

3.5.29 FILTER Threshold (Address: 0X3E)

B7	B6	B5	B4	B3	B2	B1	B0
FILTER Threshold Register							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
0	0	0	0	0	0	0	0

- The register to determine threshold level of filtered touch output in APIS mode.
- Touch output is available only when Filter Counter value of each touch input is greater than FILTER Threshold.

3.5.30 Strength 0~11 (Address: 0X50~0X5B)

B7	B6	B5	B4	B3	B2	B1	B0
Strength Register							
R	R	R	R	R	R	R	R

- The registers to store Strength values of SO~S11 during every Integration Time in APIS Mode
- These registers are applicable only in APIS mode.
- The maximum value of Strength Registers depends on Integration Time Register (0x29).
- For example, if Integration Time is set to 10, the max value of Strength is also 10.
- To enlarge Strength range, you need to change Integration Time first.

3.5.31 Calibrated Impedance 0~11 (Address: 0X5C~0X67) - RED BAR

B7	B6	B5	B4	B3	B2	B1	B0
	Calibrated Impedance Register						
R	R	R	R	R	R	R	R

- The registers to store reference impedances of each touch input after AIC calibration.
- The range of Calibrated Impedance of S0~S11 is 0~99.
- Calibrated Impedances are available when all touch inputs become non-touch status.
- The value is equivalent to Impedance Value + ALPHA in the time of Touch OFF.

3.5.32 Impedance 0~11 (Address: 0X68~0X73) - BLUE BAR

B7	B6	B5	B4	B3	B2	B1	B0
	Impedance						
R	R	R	R	R	R	R	R

- The registers to store current sensor impedance of S0 ~S11.
- You can monitor capacitance variation by Touch On and Off.

3.5.33 Status (Address: 0X74)

B7	B6	B5	B4	B3	B2	B1	B0
					Status Register		
R	R	R	R	R	R	R	R

- The read only register to store current Power State. B2~B0 are used and others are reserved.

B[2:0] = 3'b000 : Reset State

B[2:0] = 3'b001 : Active State

B[2:0] = 3'b010 : IDLE State

B[2:0] = 3'b011 : SLEEP State

3.5.34 Touch Byte L (Address: 0X75)

B7	B6	B5	B4	B3	B2	B1	B0
S7	S6	S5	S4	S3	S2	S1	S0
R	R	R	R	R	R	R	R

- Read only Register to read sensor output of S7~S0 when TINT occurs
- B7~B0 represent sensor output of S7~S0

3.5.35 Touch Byte H (Address: 0X76)

B7	B6	B5	B4	B3	B2	B1	B0
				S11	S10	S9	S8
R	R	R	R	R	R	R	R

- The read only Register to read sensor output of S11~S8 when TINT occurs
- B3~B0 represent sensor output of S11~S8, other upper bits are reserved.

3.5.36 Interrupt Pending (Address: 0X79)

B7	B6	B5	B4	B3	B2	B1	B0
		EINT	Pro_E	Com_E	I2A	A2I	TINT
R	R	R	R	R	R	R	R

- The read only register to determine which interrupt should be handled among six interrupt sources of GINT.

B0: Touch Interrupt Pending

B1: Active TO IDLE Pending

B2: IDLE To Active Pending

B3: serial communication Error Pending

B4: Procedure Error Pending (Interrupt for I^2C Communication Error)

B5: EINT Pending Register

3.5.37 GPIO IN L (Address: 0X7A)

B7	B6	B5	B4	B3	B2	B1	B0
DIO IN L							
R	R	R	R	R	R	R	R

- The read only register to read data from DIO7~DIO0 when GPIO Direction H Register (0x31)is set to input.

3.5.38 GPIO IN H (Address: 0X7B)

B7	B6	B5	B4	B3	B2	B1	B0
DIO IN H							
R	R	R	R	R	R	R	R

- The read only register to read data from DIO11~DIO8 when GPIO Direction H Register (0x32)is set to input.

3.5.39 BIAS OFF (Address: 0XFA)

B7	B6	B5	B4	B3	B2	B1	B0
BIAS OFF Register							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- Only register address is meaningful. Analog BIAS Block Power Off control Register.

BIAS OFF is used to reduce power consumption, when you are using external core power source and ATA2508 is in the Sleep Mode. Because of the Internal LDO drives some current by itself.

3.5.40 BIAS ON (Address: 0XFB)

B7	B6	B5	B4	B3	B2	B1	B0
BIAS ON Register							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- Only register address is meaningful. Analog BIAS Block Power On Control Register

When using external core power, power consumption can be minimized by turning off BIAS in sleep mode. On the other hand, to wake up from sleep mode BIAS block should be turned on by executing 'BIOS ON register'.

3.5.41 Wakeup SLEEP (Address: 0XFC)

B7	B6	B5	B4	B3	B2	B1	B0
Wakeup SLEEP Register							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- The write only register to wake up from Sleep mode by writing any value to this register. Only register address is meaningful.

3.5.42 Enter SLEEP (Address: 0XFD)

B7	B6	B5	B4	B3	B2	B1	B0
Enter SLEEP							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- The write only register to enter Sleep mode by writing any value to this register. Only register address is meaningful.

3.5.43 Cold Reset (Address: 0XFE)

B7	B6	B5	B4	B3	B2	B1	B0
Cold Reset							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- The write only register for Cold Reset. Cold Reset initializes all blocks of ATA2508 including Register block. So, all register values are reset to Default. Only register address is meaningful.

3.5.44 Warm Reset (Address: 0XFF)

B7	B6	B5	B4	B3	B2	B1	B0
Warm Reset							
W	W	W	W	W	W	W	W
X	X	X	X	X	X	X	X

- The write only register for Warm Reset. Warm Reset initializes all blocks of ATA2508 except Register block. Therefore, the register values keep remaining. Only register address is meaningful.

4. Part Ⅲ: Register Map Summary

Ads	Register Name	Ads	Register Name	Ads	Register Name	Ads	Register Name
00	Feature	1C	Strength Threshold 0	39	Beep Frequency	66	Calibrated Impedance 10
01	ALPHA 0	1D	Strength Threshold 1	3A	Calibration Interval	67	Calibrated Impedance 11
02	ALPHA 1	1E	Strength Threshold 2	3B	EINT Enable	68	Impedance 0
03	ALPHA 2	1F	Strength Threshold 3	3C	EINT Polarity	69	Impedance 1
04	ALPHA 3	20	Strength Threshold 4	3D	FILTER Period	6A	Impedance 2
05	ALPHA 4	21	Strength Threshold 5	3E	FILTER Threshold	6B	Impedance 3
06	ALPHA 5	22	Strength Threshold 6	50	Strength 0	6C	Impedance 4
07	ALPHA 6	23	Strength Threshold 7	51	Strength 1	6D	Impedance 5
08	ALPHA 7	24	Strength Threshold 8	52	Strength 2	6E	Impedance 6
09	ALPHA 8	25	Strength Threshold 9	53	Strength 3	6F	Impedance 7
0A	ALPHA 9	26	Strength Threshold 10	54	Strength 4	70	Impedance 8
0B	ALPHA 10	27	Strength Threshold 11	55	Strength 5	71	Impedance 9
0C	ALPHA 11	28	Sampling Interval	56	Strength 6	72	Impedance 10
0D	BETA	29	Integration Time	57	Strength 7	73	Impedance 11
0E	COT	2A	IDLE Time	58	Strength 8	74	Status
0F	Reference Delay	2C	MODE	59	Strength 9	75	Touch Byte L
10	Hysteresis Delay 0	2D	GPIO REG L	5A	Strength 10	76	Touch Byte H
11	Hysteresis Delay 1	2E	GPIO REG H	5B	Strength 11	79	Interrupt Pending
12	Hysteresis Delay 2	2F	GPIO Configuration L	5C	Calibrated Impedance 0	7A	GPIO IN L
13	Hysteresis Delay 3	30	GPIO Configuration H	5D	Calibrated Impedance 1	7B	GPIO IN H
14	Hysteresis Delay 4	31	GPIO Direction L	5E	Calibrated Impedance 2	FA	BIAS OFF
15	Hysteresis Delay 5	32	GPIO Direction H	5F	Calibrated Impedance 3	FB	BIAS ON
16	Hysteresis Delay 6	33	Control	60	Calibrated Impedance 4	FC	Wakeup SLEEP
17	Hysteresis Delay 7	34	Interrupt Mask	61	Calibrated Impedance 5	FD	Enter SLEEP
18	Hysteresis Delay 8	35	Interrupt Clear	62	Calibrated Impedance 6	FE	Cold Reset
19	Hysteresis Delay 9	36	Interrupt Edge	63	Calibrated Impedance 7	FF	Warm Reset
1A	Hysteresis Delay 10	37	Control 2	64	Calibrated Impedance 8		
1B	Hysteresis Delay 11	38	Beep Period	65	Calibrated Impedance 9		

5. Part Ⅴ: APPENDIX A

5.1 ATA2508 Terminology

Table. 3. Terminology

TERM	DESCRIPTION	NOTE
AIC	Automatic Impedance Calibration	
APIS	Adjacent Pattern Interference Suppression	See 3.5.1
ALPHA	Sensitivity Threshold Value for AIC	See 3.5.2
BETA	AIC threshold value to determine AIC ON	See 3.5.3
Ref Delay	Reference impedance to determine sensor input impedance	See 3.5.5

5.2 Host startup sample code to initialize ATA2508

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////
// This is an example C-code to implement initializing ATA2508 registers.
// Most of register values are set as default value.
// And this is just a reference code, thus maybe not match to your touch module.
// You need to set again with AIC parameters, ALPHA, BETA, Reference Delay, etc, and
// timing registers.
//
// Released by ATLab Korea, 20070705, Leo.
// All rights are reserved.
//
//

////////////////////////////////////////////////////////////////////////////////////////////////////////////
// Init register set arrays...

const BYTE code init_data_alpha[] = {
                                    0x03,    // ALPHA0          , default value = 8
                                    0x03,    // ALPHA1          , default value = 8
```

```
                                0x03,   // ALPHA2       , default value = 8
                                0x03,   // ALPHA3       , default value = 8
                                0x03,   // ALPHA4       , default value = 8
                                0x03,   // ALPHA5       , default value = 8
                                0x03,   // ALPHA6       , default value = 8
                                0x04,   // ALPHA7       , default value = 8
                                0x04,   // ALPHA8       , default value = 8
                                0x04,   // ALPHA9       , default value = 8
                                0x05,   // ALPHA10      , default value = 8
                                0x03    // ALPHA11      , default value = 8
                                };

const BYTE code init_data_burst[] = {
                                0x04,   // BETA         , default value = 4
                                0x27,   // AIC_WAIT     , default value = 39
                                0x3C,   // REF_DELAY    , default value = 0
                                0x01,   // HYSTERESIS01  , default value = 1
                                0x01,   // HYSTERESIS1   , default value = 1
                                0x01,   // HYSTERESIS2   , default value = 1
                                0x01,   // HYSTERESIS3   , default value = 1
                                0x01,   // HYSTERESIS4   , default value = 1
                                0x01,   // HYSTERESIS51  , default value = 1
                                0x01,   // HYSTERESIS61  , default value = 1
                                0x01,   // HYSTERESIS7   , default value = 1
                                0x01,   // HYSTERESIS8   , default value = 1
                                0x01,   // HYSTERESIS9   , default value = 1
                                0x01,   // HYSTERESIS10  , default value = 1
                                0x01,   // HYSTERESIS11  , default value = 1
                                0x20,   // STRENGTH_THRESHOLD0, default value = 1
                                0x20,   // STRENGTH_THRESHOLD1, default value = 1
                                0x20,   // STRENGTH_THRESHOLD2, default value = 1
                                0x20,   // STRENGTH_THRESHOLD3, default value = 1
                                0x20,   // STRENGTH_THRESHOLD4, default value = 1
                                0x20,   // STRENGTH_THRESHOLD5, default value = 1
                                0x20,   // STRENGTH_THRESHOLD6, default value = 1
                                0x20,   // STRENGTH_THRESHOLD7, default value = 1
```

```
0x20,   // STRENGTH_THRESHOLD8, default value = 1
0x20,   // STRENGTH_THRESHOLD9, default value = 1
0x20,   // STRENGTH_THRESHOLD10, default value = 1
0x20,   // STRENGTH_THRESHOLD11, default value = 1
0x03,   // Sampling Interval,       default value = 3
0x64,   // INTEGRATION TIME,        default value = 15
0x0A,   // IDLE TIME,               default value = 15
0x00,   // SIF_SETUP(RESERVED), default value = 0
0x01,   // MODE                     , default value = 1
0x00,   // GPIO_REG_L               , default value = 0
0x00,   // GPIO_REG_H               , default value = 0
0x00,   // GPIO_CONFIGURATION_L, default value = 0
0x00,   // GPIO_CONFIGURATION_H, default value = 0
0x00,   // GPIO_DIR_L               , default value = 0
0x00,   // GPIO_DIR_H               , default value = 0
0x04,   // CONTROL                  , default value = 4
0x38,   // INT_MASK                 , default value = 0
0x00,   // INT_CLEAR                , default value = 0
0xFF,   // INT_edge                 , default value = 0xff
0x02,   // CONTROL_2                , default value = 0
0xAF,   // BEEP_TIME                , default value = 0,
0x7F,   // BEEP_FREQUENCY           , default value = 0,
0x45,   // CALIBRATION INTERVAL, default value = 0x3C
0x00,   // GPIO_DIR_L               , default value = 0
0x00,   // GPIO_DIR_H               , default value = 0
0x00,   // EINT_ENABLE              , default value = 0
0x00,   // EINT_POL                 , default value = 0
0x00,   // FILTER_PERIOD            , default value = 0
0x00,   // FILTER_THRESHOLD         , default value = 0
};
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////
// Init Function...

void ATA2508_Init()
```

```
{
        char i;
        BYTE data1[2];
        // ATA2508 INIT process....
        // Disable INT that connected to ATA2508's TINT.
        EX0 = 0;
        // Set APIS Mode...
        data1[0] = 0x04;            // APIS-2 mode. do not set bit1,2,3 simultaneoulsy.

        // usage : I2C_WRITE( address, data_size, data);
        I2C_WRITE(0, 1, data1);

        for(i=1; i<13; i++)
        {
                data1[0] =          init_data_alpha[i-1];
                I2C_WRITE(i, 1, data1);
        }

        for(i=13; i<59; i++)
        {
                data1[0] =          init_data_burst[i-13];
                I2C_WRITE(i, 1, data1);
        }

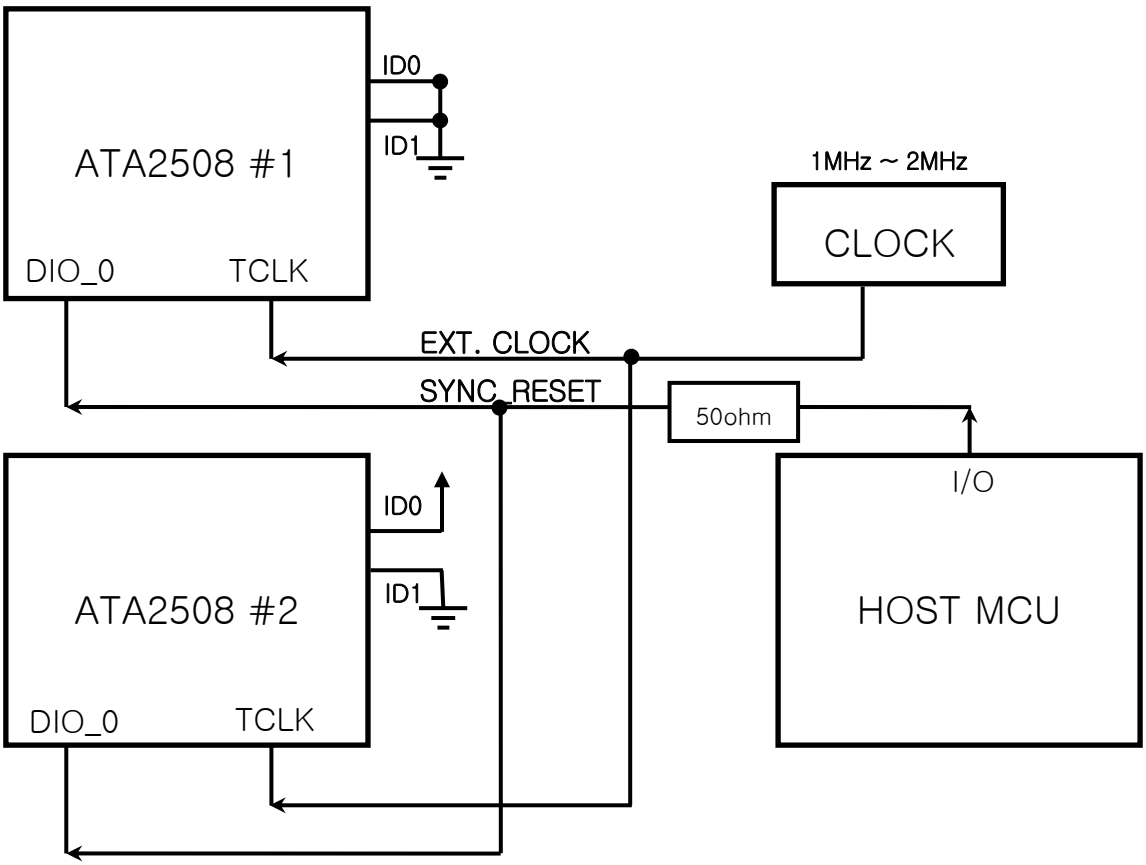
        // wait for 1 ms
        // to activate all the new settings, give a WARM RESET.
        I2C_WRITE(ADDR_WARM_RESET, 1, 0x00);

        // wait for 1 ~ 10 ms.

        // Enable INT that connected to ATA2508's TINT.
        EX0 = 1;

}
```

5.3 Multiple connection of ATA2508



Multiple Chip using Diagram

If you want to use more sensor channels, then you can connect same chips with other CHIP ID up to 4 chips. The above diagram shows the example. This feature enables multiple connections and can make more than 12 channel sensor pads as one. But, you need to set some configurations like below. They are 3 factors with this function. And when MCU accesses touch data and other registers, MCU should access all of the chips with its own ID.

1. EXTERNAL CLOCK SOURCE USE CONTROL

=> set CONTROL Register (0x33) bit 6 "CLKSRC" as '1.'

2. SYNC RESET USE CONTROL

=> set CONTROL2 Register (0x37) bit 7 "MRST" as '1.'

3. GPIO DIRECTION as INPUT

=> to receive SYNC RESET signal from HOST MCU with DIO_0

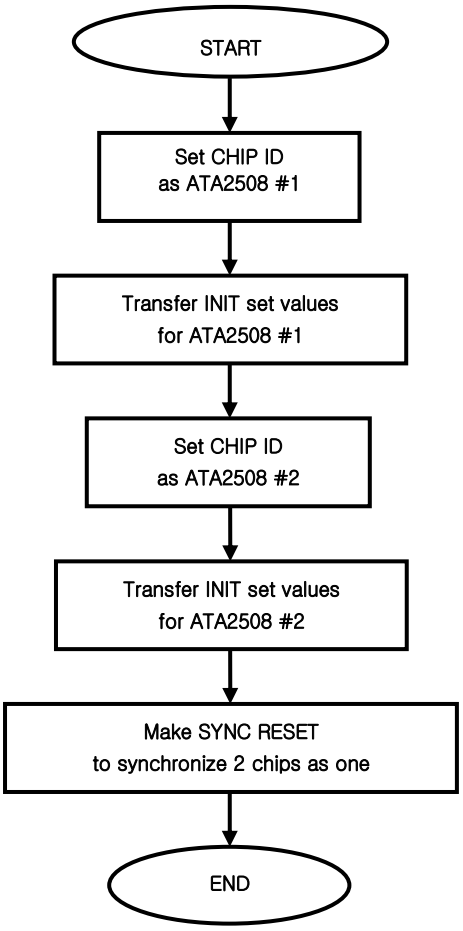
=> set GPIO CONFIGURATION_L Register (0x2f) bit 0 as '1.'

=> set GPIO DIRECTION_L Register (0x31) bit 0 as '1'

to make DIO_0 as input direction.

4. When starts up the HOST MCU, MCU must set the above configurations to all of the chips. And ATA2508 can connect on the same I2C connection up to 4 chips with separate chip ID.

The initializing flow is like below.



6. Revision History

V1.1

-1.7 CONFIG, ID Pin descriptions added.

-1.7 32QFN Package Type added.

-1.7 Package Outline dimension descriptions added.

-2.2.2 32QFN Mobile Application Circuit added.

-3.39/3.40 Analog BIAS On/Off Control Register added.

-4 Register Summary tables modified.

- Function Characteristic added.

V1.2

-2.2.2 Case by power consumption table added.

-1.7.2 32QFN Pin descriptions added.

-1.7. Package Dimension modified.

V1.3

- 3.4.7 Configuring flow while Host start up added
- 3.4.8 Trouble shooting while I^2C interfacing added
- 5.2 Host startup sample code to initialize ATA2508 added
- 24SSOP Pin description and application Circuit added
- 32QFN Package Dimension Modified
- Modify 3.2 and split to sub chapter and add newly update information
- 3.2.1 Main Window of Tuning Viewer added
- 3.2.2 Reading and Writing Registers added
- 3.2.3 Monitoring and Tuning Touch Module added
- 3.2.4 MCU Configuration Window
- 3.2.5 Data Saving Window
- Emphasis of uniformity in the scroll input pattern removed

V1.4

- 5.3 Multiple connection of ATA2508 added
- 5.2 Host startup sample code to initialize ATA2508 modified
- 24QFN,30SSOP,20SSOP package information and application circuits added